PART III

ODOUR THRESHOLD VALUES IN OTHER MEDIA

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

A

acetal⇒1,1-DIETHOXYETHANE
acetaldehyde⇒ETHANAL
acetaldehyde diethyl acetal ⇒1,1-DIETHOXYETHANE

ACETIC ACID (ethanoic acid) *[64-19-7]*

Salo *et al.* (1972)	d	26.0	9.4 % (w/w) grain spirit
Yoshida (1972)	d	6.9	nujiol (a mineral oil)
Yoshida (1972)	r	100	nujiol (a mineral oil)
Yoshida (1984);Takagi (1989)	d	100	nujol
Yoshida (1984);Takagi (1989)	r	100	nujol
Williams & Ismail (1981)		35	sugar-acid base
Guth & Grosch (1994)		32.3	0.03 M Na phosphate; pH 5.7
Reiners & Grosch (1998)	d	0.124	refined sunflower oil
Zehentbauer & Grosch (1998)		31.14	starch
Stephan & Steinhart (1999)	r	0.75	refined vegetable oil
Lee *et al.* (2000)	d	24-28	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	102-233	3 year old grain whisky: 23 % ethanol
Escudero *et al.* (2004)		300	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Utsunomiya *et al.* (2004a)	d	36 - 37	base sake
Utsunomiya *et al.* (2004a)	r	59	base sake
Morales *et al.* (2005)	d	0.50	deodorised olive oil
Cliff & Pickering (2006)	d	3,185	base ice wine
Averbeck & Schierberle (2011)	d	12,3	citrate buffer (pH 3.8)

acetoin⇒3-HYDROXY-2-BUTANONE
acetone⇒PROPANONE

ACETOPHENONE *[98-86-2]*

Schirack *et al.* (2006)	d	5.629	vegetable oil

acetovanillon⇒1-(4-HYDROXY-3-METHOXYPHENYL)ETHANONE

4-ACETYL-6-*tert*-BUTYL-1,1-DIMETHYLINDANE (celestolide) *[13171-00-1]*

D'Andrea (1975)	d	500 - 510	diethyl phthalate

2-acetyl-3,4-dihydro-2H-azole⇒2-ACETYL-1-PYRROLINE

7-ACETYL-1,1,3,4,4,6-HEXAMETHYL-1,2,3,4-TETRAHYDRONAPHTHALENE (fixolide, tonalide) *[1506-02-1]*

D'Andrea (1975)	d	1.2 - 160	diethyl phthalate
Jeffcoat & Willis (1986)		100	diethyl phthalate

ACETYLPYRAZINE *[22047-25-2]*

Schieberle (1993,1996)	d	0.01	sunflower oil

2-ACETYL-1-PYRROLINE (2-acetyl-3,4-dihydro-2H-azole) *[85213-22-5]*

Schieberle (1993,1996);			
Kubickova & Grosch (1998)	d	0.000 1	sunflower oil
Rychlik & Grosch (1996); Rychlik (1998)	d	0.000 0073	starch

2-ACETYLTETRAHYDROPYRIDINE

Chetschik *et al.* (2010)		0.0012	sunflower oil

6-ACETYL-1,2,3,4/2,3,4,5-TETRAHYDROPYRIDINE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Rychlik & Grosch (1996)	d	0.000 054	starch

2-ACETYL-2-THIAZOLINE *[29926-41-8]*

Ong & Acree (1998)		0.000 5	artificial saliva and fruit juice matrix
Kerscher (2000b)		0.0018	sunflower oil
Karagül-Yüceer *et al.* (2004)	d	0.002	skim milk
Limpawattana (2007)	d	0.0157	starch
Limpawattana (2007)	d	0.0046	partially deodorized rice at 70° C

acrolein⇒2-PROPENAL
acrylic acid⇒PROPENOIC ACID
active amyl acetate⇒2-METHYLBUTYL ACETATE
active amyl alcolhol⇒2-METHYL-1-BUTANOL
adamantane⇒TRICYCLO[3.3.1.1(3.7)]DECANE
allyl caproate⇒2-PROPENYL HEXANOATE

4-ALLYL-1,2-DIMETHOXYBENZENE (methyleugenol, eugenol methyl ether, 4-allylveratrole) *[93-15-2]*

Zea *et al.* (2001)		10	14 % (v/v) ethanol; tartaric acid to pH3.5

1-ALLYL-4-METHOXYBENZENE (estragole) *[140-67-0]*

Zeller & Rychlik (2007)	r	6.3	cellulose

4-ALLYL-2-METHOXYPHENOL (eugenol) *[97-53-0]*

Proetz (1924)		0.53	liq. petrolatum
Rosenthal (1927)		88	diethyl phthalate
Swan & Burtles (1978); Swan *et al.* (1981)		0.05	20 % ethanol-water
Jounela-Eriksson & Lehtonen (1981)		0.011	10 vol. % grain spirit
Etiévant *et al.* (1983)		6.4	base white sweet wine
Ophir *et al.* (1986)	d	42-81	diethyl phthalate
Gross-Isseroff & Lancet (1988)	d	0.032-320	diethyl phthalate
Ophir *et al.* (1988)	d	10.5-33.3	diethyl phthalate
Gross-Isseroff *et al.* (1992)	d	10,000	diethyl phthalate
Ferreira *et al.* (1998)		0.01	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Jagella & Grosch (1999b)	d	0.000 98	starch
Zimmermann (2001)	r	0.0098	starch
Moyano *et al.* (2002)		0.01	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005);Chaves *et al.* (2007);			
Moyano *et al.* (2010)		0.005	14 % ethanolic solution
Campo *et al.* (2006); Escudero *et al.* (2007)		0.006	10 % water/ethanol. 5 g/l tartaric acid, pH 3.2
Ferreira *et al.* (2006)		0.158	ethanol 10% v/v, tartaric acid 2 g/l, pH 3.4
Callejón *et al.* (2008)	d	0.000 17	7 % acetic acid solution
Poisson & Schieberle (2008)		0.0071	water/ethanol 6:4 by vol.
Söllner & Schieberle (2009)		0.512	odorless sunflower oil

4-ALLYL-1,2-(METHYLENEDIOXY)BENZENE (safrole) *[94-59-7]*

Rosenthal (1927)	177	diethyl phthalate
Kendall & Neilson (1964)	51	hexadecane
Kendall & Neilson (1964)	0.09	aqueous 0.1 % ethanol
Koelega & Köster (1974)	177-221	benzyl benzoate

allyl methyl sulphide⇒3-METHYLTHIO-1-PROPENE

ALLYL SULPHOCYANATE

Proetz (1924)	3.05	liq. Petrolatum

4-allylveratrole⇒4-ALLYL-1,2-DIMETHOXYBENZENE
ambrettolide⇒16-HYDROXY-7-HEXADECENOIC ACID LACTONE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

2'-AMINOACETOPHENONE *[551-93-9]*

Karagül-Yüceer *et al.* (2004)		d	0.000 4	skim milk

(*R*)-(+)-2-AMINO-3-MERCAPTOPROPANOIC ACID (L-cysteine) *[52-90-4]*

Naim *et al.* (1997)			220	orange juice

amyl acetate⇒PENTYL ACETATE
amyl alcohol⇒1-PENTANOL
amyl salicylate⇒PENTYL SALICYLATE

5,16-ANDROSTADIEN-3β-OL *[1224-94-8]*

Brooks & Pearson (1989)		d	8.9	oil/water

4,16-ANDROSTADIEN-3-ONE ((8*S*,9*S*,10*R*,13*R*,14*S*)-10,13-dimethyl-1,2,6,7,8,9,11,12,14,15-decahydrocyclopenta[a]phenanthren-3-one) *[4075-07-4]*

Brooks & Pearson (1989)		d	7.8	oil/water
Lundström *et al.* (2003)		d	57	propylene glycol
Hummel *et al.* (2005)		d	13.6 -1,360	white mineral oil
Jacob *et al.* (2006)		d	0.0018- 1,300	silicone oil
Lundström *et al.* (2006)		d	1.6 - 12.7	propylene glycol
Chopra *et al.* (2008)	[female pre-puberty]	d	0.058	propylene glycol
Chopra *et al.* (2008)	[female puberty]	d	0.029	propylene glycol
Chopra *et al.* (2008)	[female post-puberty]	d	0.059	propylene glycol
Chopra *et al.* (2008)	[male pre-puberty]	d	0.11	propylene glycol
Chopra *et al.* (2008)	[male puberty]	d	0.98	propylene glycol
Chopra *et al.* (2008)	[male post-puberty]	d	0.013	propylene glycol

5α-ANDROST-16-EN-3α-OL *[1153-51-1]*

Brooks & Pearson (1989)		d	0.9	oil/water
Morofushi *et al.* (2000)			3-30	mineral oil

5α-ANDROST-16-EN-3β-OL *[7148-51-8]*

Brooks & Pearson (1989)		d	1.2	oil/water

5α-ANDROST-16-EN-3-ONE *[18339-16-7]*

Pollack *et al.* (1982)			0.5 - 1	light mineral oil
Thomson (1984)			0.0013	aqueous ethanol
Wysocki & Beauchamp (1984)			62.5-125	mineral oil
Gross-Isseroff *et al.* (1987)		d	2.4->1,250	heavy mineral oil
Brooks & Pearson (1989)		d	0.6	oil/water
Dorries *et al.* (1989)			31 - 250	mineral oil
Wysocki *et al.* (1989)	*anosmic*	d	125-2,000	light mineral oil
Hummel *et al.* (1991)		d	1.5-6.3	propylene glycol
Gross-Isseroff *et al.* (1992)		d	2.4->1,250	heavy mineral oil
Pierce *et al.* (1993)		r	1.95	light mineral oil
Sather (1995)			1.9	sunflower oil
Annor-Frempong *et al.* (1997)		d	0.152	sunflower oil and vegetable oil
Font I Furnols *et al.* (2000)		d	0.10	refined sunflower oil
Morofushi *et al.* (2000)			13-63	mineral oil
Lübke *et al.* (2009)		d	4.9 - 78	1,2-propanediol
Lunde *et al.* (2010)			6	minced meat

anethole⇒1-METHOXY-4-(1-PROPENYL)BENZENE
anisaldehyde⇒4-METHOXYBENZALDEHYDE
m-anisaldehyde⇒3-METHOXYBENZALDEHYDE
o-anisaldehyde⇒2-METHOXYBENZALDEHYDE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

p-anisaldehyde⇒4-METHOXYBENZALDEHYDE
azine⇒PYRIDINE

B

BENZALDEHYDE *[100-52-7]*

Proetz (1924)		0.33	liq. petrolatum
Rosenthal (1927)		44	diethyl phthalate
Moncrieff (1957)		500	propylene glycol
Williams & Ismail (1981)		0.19	sugar-acid base
Delfini (1987)		3 - 3.5	white wine
Gross-Isseroff & Lancet (1988)	d	0.000 1-1,000	diethyl phthalate
Gross-Isseroff & Lancet (1988)	d	1,550	diethyl phthalate
Utsunomiya *et al.* (2004a)	d	0.97 - 0.99	base sake
Utsunomiya *et al.* (2004a)	r	1.9	base sake
Moreno *et al.* (2005)		5	14 % ethanolic solution
García-González *et al.* (2008)	d	0.06	deodorised edible oil
Tesfaye *et al.* (2008)		0.158	7 % acetic acid solution

BENZOIC ACID *[65-85-0]*

Culleré *et al.* (2004); Gómez-Míguez *et al.* (2007); Campo *et al.* (2006); Escudero *et al.* (2007)		1	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2

BENZYL ACETATE *[140-11-4]*

Rosenthal (1927)		88	diethyl phthalate
Williams & Ismail (1981)		0.3	sugar-acid base
Aceña *et al.* (2011)	d	2.3	7 % (w/v) acetic acid water volume

BENZYL ALCOHOL (phenylmethanol, α-hydroxytoluene) *[100-51-6]*

Rosenthal (1927)		590	diethyl phthalate
Harder (1975)		10,000	diethyl phthalate
Lindeman *et al.* (1982)	d	159	10 % ethanol
Delfini (1987)		2,300 - 2,500	white wine
Culleré *et al.* (2004); Gómez-Míguez *et al.* (2007); Escudero *et al.* (2007)		200	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Moreno *et al.* (2005)		900	14 % ethanolic solution
Callejón *et al.* (2008)	d	16.9	7 % acetic acid solution
Aceña *et al.* (2011)	d	5.1	7 % (w/v) acetic acid water volume

benzylbenzene⇒DIPHENYLMETHANE

BENZYL BENZOATE *[120-51-4]*

Rosenthal (1927)		885	diethyl phthalate

BENZYL FORMATE *[104-57-4]*

Rosenthal (1927)		177	diethyl phthalate

benzyl methyl sulphide⇒α-(METHYLTHIO)TOLUENE
bis(2-methyl-3-furyl) disulphide⇒3-(2-METHYL)FURYLDITHIO-3-(2-METHYL)FURAN

BIS(METHYLTHIO)METHANE (2,4-dithiapentane, methylene bis(methyl sulphide)) *[1618-26-4]*

Adda *et al.* (1978); Cuer *et al.* (1979)	d	0.060	liquid cheese
Kubickova & Grosch (1998)		0.03	sunflower oil

borneol⇒*endo*-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL
(l)-borneol⇒(1*S*)-(-)-*endo*-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

bornyl acetate⇒*endo*-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTYL ACETATE
bourgeonal⇒3-(4-*tert*-BUTYLPHENYL)PROPANAL
(1'*R*,2*R*)-brahmanol⇒(1'*R*,2*R*)-2-METHYL-4-(2,2,3-TRIMETHYL-1-CYCLOPENTEN-3-YL)-1-BUTANOL
(1'*R*,2*S*)-brahmanol⇒(1'*R*,2*S*)-2-METHYL-4-(2,2,3-TRIMETHYL-1-CYCLOPENTEN-3-YL)-1-BUTANOL
(1'*S*,2*R*)-brahmanol⇒(1'*S*,2*R*)-2-METHYL-4-(2,2,3-TRIMETHYL-1-CYCLOPENTEN-3-YL)-1-BUTANOL
(1'*S*,2*S*)-brahmanol⇒(1'*S*,2*S*)-2-METHYL-4-(2,2,3-TRIMETHYL-1-CYCLOPENTEN-3-YL)-1-BUTANOL
bromoform⇒TRIBROMOMETHANE

1-BROMONAPHTHALENE *[90-11-9]*			
Dhont *et al.* (1972)	d	0.25	'liquid sugar'
2-BROMOPHENOL *[95-56-7]*			
Bemelmans & Den Braber (1983)	d	0.018	peanut oil
3-BROMOPHENOL *[591-20-8]*			
Bemelmans & Den Braber (1983)	d	18	peanut oil
4-BROMOPHENOL *[106-41-2]*			
Bemelmans & Den Braber (1983)	d	27	peanut oil
BUTANAL (butyraldehyde) *[123-72-8]*			
Meijboom (1964)		0.15	paraffin oil
Salo *et al.* (1972)	d	0.028	9.4 % (w/w) grain spirit
2,3-BUTANEDIOL *[513-85-9]*			
Moreno *et al.* (2005)		668	14 % ethanolic solution
2,3-BUTANEDIONE (diacetyl) *[431-03-8]*			
Van Niel *et al.* (1929)		>2 - 4	butter
Sega *et al.* (1967)	d	0.162	lager beer
Salo (1970a); Salo *et al.* (1972)	d	0.0025-0.020	9.4 % (w/w) grain spirit
Selfridge & Amerine (1978)	d	0.05	artificial wine
Koch *et al.* (1978)		0.2 - 0.3	apple wine
Herrmann & Abd El Salam (1980b)		0.003	sunflower oil
Rothe (1991)		0.4	neutral margarine mass
Schieberle *et al.* (1993)		0.0045	sunflower oil
Preininger & Grosch (1994); Kubickova & Grosch (1998); Burdack-Freitag (2007); Buhr *et al.* (2010)		0.010	sunflower oil
Mayer (1996)		0.025	cellulose
Rychlik & Grosch (1996)	d	0.0065	starch
Buettner *et al.* (1998)		0.004	2 mg/l furaneol (=200 OAV) in water
Buettner *et al.* (1998)		0.003	0.05 mg/l (*Z*)-3-hexenal (=200 OAV) in water
Lee *et al.* (2000)	d	0.002-0.02	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	0.006-0.04	3 year old grain whisky: 23 % ethanol
Rogerson *et al.* (2001)	r (buttery)	0.0195	20 % ethanol, tartaric acid 5g/l, pH 3.5
Rogerson *et al.* (2001)	r (caramel)	1.25	20 % ethanol, tartaric acid 5g/l, pH 3.5
Utsunomiya *et al.* (2004a)	d	0.08 - 0.083	base sake
Utsunomiya *et al.* (2004a)	r	0.14	base sake
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		0.1	14 % ethanolic solution
Callejón *et al.* (2008)	d	0.040	7 % acetic acid solution
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.0408	water with 1 % caseinate, pH 7.0
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.0449	water with 1 % caseinate, pH 5.5
Leksrisompong (2008);			

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Leksrisompong *et al.* (2010)	d	0.0056	emulsion with 10 % fat, pH 7.0
Leksrisompong (2008);			
Leksrisompong *et al.* (2010)	d	0.0092	emulsion with 10 % fat, pH 5.5
Leksrisompong (2008);			
Leksrisompong *et al.* (2010)	d	0.0218	emulsion with 20 % fat, pH 7.0
Leksrisompong (2008);			
Leksrisompong *et al.* (2010)	d	0.0086	emulsion with 20 % fat, pH 5.5
Leksrisompong (2008);			
Leksrisompong *et al.* (2010)	d	0.0995	oil
Poisson & Schieberle (2008)		0.0028	water/ethanol 6:4 by vol.
Buhr *et al.* (2010)	d	0.050	whole milk
Aceña *et al.* (2011)	d	0.0953	7 % (w/v) acetic acid water volume
BUTANOIC ACID (butyric acid) *[107-92-6]*			
Salo (1970a,1970b); Salo *et al.* (1972)	d	3.4/4	9.4 % (w/w) grain spirit
Baldwin *et al.* (1973)	r	0.3	buffered solution pH 3.2
Baldwin *et al.* (1973)	r	0.9	buffered solution pH 4.5
Baldwin *et al.* (1973)	r	4.8	buffered solution pH 6.0
Amoore & Buttery (1978)	d	0.109	safflower oil
Haslbeck *et al.* (1986)		44 - 56	cream
Haslbeck *et al.* (1986)		17 - 49	30 % coconut oil
Haslbeck *et al.* (1986)		53 - 210	38 % coconut oil aromatized
Schieberle *et al.* (1993)		0.135	sunflower oil
Guth & Grosch (1994);			
Kerscher & Grosch (2000)		2.73	0.03 M Na phosphate; pH 5.7
Rychlik & Grosch (1996)	d	0.1	starch
Stephan & Steinhart (1999)	r	0.205	refined vegetable oil
Moyano *et al.* (2002)		2.5	14 % (v/v) ethanol; tartaric acid to pH 3.5
Karagül-Yüceer *et al.* (2004)	d	3.09	skim milk
Utsunomiya *et al.* (2004a)	d	4.3	base sake
Utsunomiya *et al.* (2004a)	r	7.9 - 8	base sake
Morales *et al.* (2005);			
García-González *et al.* (2008)	d	0.14/0.65	deodorised olive/edible oil
Moreno *et al.* (2005); Moyano *et al.* (2010)		10	14 % ethanolic solution
Steinhaus & Schieberle (2005)	r	7.5	starch with 12 white pepper odorants
Rychlik *et al.* (2006)		3.54	fresh yoghurt with 3.04 mg/kg butanoic acid
Rychlik *et al.* (2006)		1.73	three-month-old yoghurt with 4.89 mg/kg butanoic acid
Averbeck & Schierberle (2011)	d	0.145	citrate buffer (pH 3.8)
1-BUTANOL (butyl alcohol) *[71-36-3]*			
Furchtgott & Friedman (1960)		3,112-4,076	mineral oil
Kimbrell & Furchtgott (1963)	d	1,000-60,000	mineral oil
Cain (1966)	r	2,380	diethyl phthalate
Semb (1968)		100-5,000	diethyl phthalate
Salo *et al.* (1972)	d	>5.0	9.4 % (w/w) grain spirit
Granzer *et al.* (1986)		10 - 200	coconut oil
Moyano *et al.* (2002)		160	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		820	14 % ethanolic solution
García-González *et al.* (2008)	d	0.038	deodorised edible oil
2-BUTANOL (*sec*-butyl alcohol) *[78-92-2]*			
Salo *et al.* (1972)	d	>10.0	9.4 % (w/w) grain spirit
Zea *et al.* (2001); Moyano *et al.* (2002)		50	14 % (v/v) ethanol; tartaric acid to pH 3.5
Morales *et al.* (2005)	d	0.10/0.15	deodorised olive oil
Moreno *et al.* (2005)		1,000	14 % ethanolic solution
García-González *et al.* (2008)	d	0.50	deodorised edible oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

BUTANONE (methyl ethyl ketone) *[78-93-3]*

Forrai *et al.* (1970)	*trimodal distribution*	r	2.4-156	silicone oil
Doty (1975)		d	98	propylene glycol
Granzer *et al.* (1986)			10-100	coconut oil
Granzer *et al.* (1986)			100-1,000	liquid cheese
Granzer *et al.* (1986)			5-100	biscuits
Granzer *et al.* (1986)			100-500	marmelade
Rüter (1992)		d	24	potato crisps
Rüter (1992)		r	110	potato crisps
Rüter (1992)		d	86	coffee powder
Rüter (1992)		r	440	coffee powder
Rüter (1992)		d	32	'Gummibären'
Rüter (1992)		r	70	'Gummibären'
Rüter (1992)		d	44	milk chocolate
Rüter (1992)		r	90	milk chocolate
Morales *et al.* (2005); Luna *et al.* (2006); García-González *et al.* (2008)		d	40	deodorised/refined olive/edible oil

1-BUTEN-3-ONE (3-buten-2-one, methyl vinyl ketone) *[78-94-4]*

Evans *et al.* (1971)	d	0.2	cottonseed oil

3-buten-2-one⇒1-BUTEN-3-ONE

BUTYL ACETATE *[123-86-4]*

Moncrieff (1957)		300	propylene glycol
Sega *et al.* (1967)	d	2.63	lager beer
De Wet (1978)	d	1.83	wine residue medium
Granzer *et al.* (1986)		0.05 - 2	coconut oil
Granzer *et al.* (1986)		2 - 20	liquid cheese
Granzer *et al.* (1986)		0.5 - 5	biscuits
Granzer *et al.* (1986)		0.1 - 2	marmelade
Morales *et al.* (2005)	d	0.3	deodorised olive oil
Luna *et al.* (2006)	d	0.1	refined olive oil
Callejón *et al.* (2008)	d	0.453	7 % acetic acid solution
García-González *et al.* (2008)	d	0.30	deodorised edible oil
Pineau *et al.* (2009)	d	1.83	dearomatized red wine

4'-*tert*-BUTYLACETOPHENONE *[943-27-1]*

Cerveny *et al.* (1993)	224.0	paraffin oil

butyl acrylate⇒BUTYL 2-PROPENOATE
butyl alcohol⇒1-BUTANOL
sec-butyl alcohol⇒2-BUTANOL

4-*tert*-BUTYL-2,6-DIMETHYL-3,5-DINITROACETOPHENONE (musk ketone) *[81-14-1]*

Kalmus & Seedburgh (1975)	20	paraffin oil

1-*tert*-BUTYL-3,5-DIMETHYL-2,4,6-TRINITROBENZENE (musk xylene) *[81-15-2]*

Jeffcoat & Willis (1986)	3,600	diethyl phthalate

5-BUTYL-2-ETHOXY-3-METHYLPYRAZINE

Masuda *et al.* (1981)	0.2	aqueous ethanol solution

2-BUTYLFURAN *[4466-24-4]*

Evans *et al.* (1971)	d	10	cottonseed oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

BUTYL 2-HYDROXYPROPANOATE (butyl lactate) *[138-22-7]*

Chaves *et al.* (2007)		10	14 % v/v ethanol/water

butyl lactate⇒BUTYL 2-HYDROXYPROPANOATE
butyl methacrylate⇒BUTYL 2-METHYL-2-PROPENOATE

1-*tert*-BUTYL-2-METHOXY-4-METHYL-3,5-DINITROBENZENE (musk ambrette) *[83-66-9]*

Rosenthal (1927)		44	diethyl phthalate
Kalmus & Seedburgh (1975)		49	paraffin oil
D'Andrea (1975)	d	500 - 10,000	diethyl phthalate

2-BUTYL-3-METHOXYPYRAZINE *[32737-11-4]*

Maga (1990)	r	0.000 002	model wine
Maga (1990)	r	0.000 008	red wine

2-*sec*-BUTYL-3-METHOXYPYRAZINE *[24168-70-5]*

Preininger & Grosch (1994)		0.000 5	sunflower oil
Neta *et al.* (2008)	d	0.000 016	whole milk

BUTYL 2-METHYL-2-PROPENOATE (butyl methacrylate) *[97-88-1]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.1 - 4	sunflower oil
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.05 - 0.5	10 vol. % ethanol
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.05 - 0.5	3 vol. % acetic acid

1-(4-*tert*-BUTYLPHENYL)ETHANOL *[34386-42-0]*

Cerveny *et al.* (1993)		219.4	paraffin oil

1-(4-*tert*-BUTYLPHENYL)ETHYL ACETATE

Cerveny *et al.* (1993)		465.0	paraffin oil

3-(4-*tert*-BUTYLPHENYL)PROPANAL (bourgeonal) *[18127-01-0]*

Olsson & Laska (2010)	d	412 - 1.235	diethyl phthalate

BUTYL 2-PROPENOATE (butyl acrylate) *[141-32-2]*

Boidron & Pons (1978)	d	0.012	ethanol 12 % + acid
Boidron & Pons (1978)	d	0.018	ethanol 12 % + sucrose
Boidron & Pons (1978)	d	0.029	white wine
Boidron & Pons (1978)	d	0.015	red wine
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.1 - 1	sunflower oil
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.005 - 0.1	10 vol. % ethanol
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.005 - 0.2	3 vol. % acetic acid

butyraldehyde⇒BUTANAL
butyric acid⇒BUTANOIC ACID
γ-butyrolactone⇒4-HYDROXYBUTANOIC ACID LACTONE

C

camphor⇒1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
capric acid⇒DECANOIC ACID
caproic acid⇒HEXANOIC ACID
γ-caprolactone⇒4-HYDROXYHEXANOIC ACID LACTONE
caprylaldehyde⇒OCTANAL
caprylic acid⇒OCTANOIC ACID

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

CARBON DISULPHIDE *[75-15-0]*

Kleinschmidt (1983)		r	1	paraffin oil
Chopra *et al.* (2008)	[female pre-puberty]	d	0.0039	propylene glycol
Chopra *et al.* (2008)	[female puberty]	d	0.0035	propylene glycol
Chopra *et al.* (2008)	[female post-puberty]	d	0.0030	propylene glycol
Chopra *et al.* (2008)	[male pre-puberty]	d	0.024	propylene glycol
Chopra *et al.* (2008)	[male puberty]	d	0.058	propylene glycol
Chopra *et al.* (2008)	[male post-puberty]	d	0.014	propylene glycol

carvacrol⇒5-ISOPROPYL-2-METHYLPHENOL
carvone⇒5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
(*d*)-carvone⇒(*S*)-(+)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
(*l*)-carvone⇒(*R*)-(-)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
R-(-)-carvone⇒(*R*)-(-)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
S-(+)-carvone⇒(*S*)-(+)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
celestolide⇒4-ACETYL-6-*tert*-BUTYL-1,1-DIMETHYLINDANE
cellosolve⇒2-ETHOXYETHANOL
cetyl alcohol⇒1-HEXADECANOL

CHLORINE *[7782-50-5]*

McDonald *et al.* (2009)	0.05	water with 0.3 mg/L bromine

6-chloro-*o*-cresol⇒6-CHLORO-2-METHYLPHENOL
2-chloro-6-methylphenol⇒6-CHLORO-2-METHYLPHENOL

6-CHLORO-2-METHYLPHENOL (2-chloro-6-methylphenol, 6-chloro-*o*-cresol) *[87-64-9]*

Coulter *et al.* (2008)	d	0.000 070	neutral, dry white wine

2-CHLOROPHENOL *[95-57-8]*

Coulter *et al.* (2008)	d	0.000 437	neutral, dry white wine

4-CHLOROPHENOL *[106-48-9]*

Coulter *et al.* (2008)	d	> 0.000 64	neutral, dry white wine

1,4-cineole⇒1-ISOPROPYL-4-METHYL-7-OXABICYCLO[2.2.1]HEPTANE
1,8-cineole⇒1,3,3-TRIMETHYL-2-OXABICYCLO[2.2.2]OCTANE
cinnamaldehyde⇒3-PHENYL-2-PROPENAL
cinnamic alcohol⇒3-PHENYL-2-PROPEN-1-OL
cinnamyl alcohol⇒3-PHENYL-2-PROPEN-1-OL
(*E*)-cinnamyl acetate⇒(*E*)-3-PHENYL-2-PROPEN-1-YL ACETATE
cinnamyl butyrate ⇒ 3-PHENYL-2-PROPEN-1-YL BUTANOATE
citral⇒3,7-DIMETHYL-2,6-OCTADIENAL
citronellal⇒3,7-DIMETHYL-6-OCTENAL
(3*R*)-(+)-citronellal⇒(*R*)-(+)-3,7-DIMETHYL-6-OCTENAL
(3*S*)-(-)-citronellal⇒(*S*)-(-)-3,7-DIMETHYL-6-OCTENAL
citronellol⇒3,7-DIMETHYL-6-OCTEN-1-OL
(3*R*)-(+)-citronellol⇒(*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-OL
(3*S*)-(-)-citronellol⇒(*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-OL
β-citronellol⇒3,7-DIMETHYL-6-OCTEN-1-OL
(3*R*)-citronellyl acetate⇒(*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-YL ACETATE
(3*S*)-citronellyl acetate⇒(*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-YL ACETATE
(3*R*)-citronellylethylether⇒(*R*)-(+)-1-ETHOXY-3,7-DIMETHYL-6-OCTENE
(3*S*)-citronellylethylether⇒(*S*)-(-)-1-ETHOXY-3,7-DIMETHYL-6-OCTENE
(3*R*)-citronellyl ethyl oxalate⇒ETHYL (*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-YL OXALATE
(3*S*)-citronellyl ethyl oxalate⇒ETHYL (*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-YL OXALATE
(3*R*)-citronellylnitrile⇒(*R*)-(-)-2,6-DIMETHYL-5-HEPTENYL CYANIDE
(3*S*)-citronellylnitrile⇒(*S*)-(+)-2,6-DIMETHYL-5-HEPTENYL CYANIDE
civetone⇒9-*cis*-CYCLOHEPTADECEN-1-ONE

COUMARIN *[91-64-5]*

Proetz (1924)	0.77	liq. petrolatum

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Rosenthal (1927)		6	diethyl phthalate
Mair *et al.* (1978)		< 140	dibutyl phthalate

m-cresol⇒3-METHYLPHENOL
o-cresol⇒2-METHYLPHENOL
p-cresol⇒4-METHYLPHENOL
creosol⇒2-METHOXY-4-METHYLPHENOL

CUBENOL ([(1*S*)-(1α,4β,4aβ,7α,8aα)]-1,3,4,5,6,8a-hexahydro-4,7-dimethyl-1-(1-methylethyl)-4a(2*H*)-naphthalenol) *[21284-22-0]*

Kajiwara *et al.* (1988)		100 - 250	ethanol

cuminaldehyde⇒4-ISOPROPYLBENZALDEHYDE

9-*cis*-CYCLOHEPTADECEN-1-ONE (civetone) *[542-46-1]*

Jeffcoat & Willis (1986)		7,900	diethyl phthalate
Hiraoka *et al.* (1995)	d	1-10 10^3	liquid paraffin

CYCLOHEPTANONE *[502-42-1]*

Matsushita *et al.* (1967)		130	liquid paraffin

CYCLOHEXANE *[110-82-7]*

Rüter (1992)	d	200	potato crisps
Rüter (1992)	r	650	potato crisps
Rüter (1992)	d	200	coffee powder
Rüter (1992)	r	620	coffee powder
Rüter (1992)	d	240	'Gummibären'
Rüter (1992)	r	2,400	'Gummibären'
Rüter (1992)	d	70	milk chocolate
Rüter (1992)	r	260	milk chocolate

CYCLOHEXANOL *[108-93-0]*

Matsushita *et al.* (1967)		160	liquid paraffin

cyclopentadecanolide⇒15-HYDROXYPENTADECANOIC ACID LACTONE
cyclotene⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE
p-cymene⇒1-ISOPROPYL-4-METHYLBENZENE
p-cymene-2,3-diol ⇒2,3-DIHYDROXY-4-ISOPROPYL-1-METHYLBENZENE
L-cysteine⇒(*R*)-(+)-2-AMINO-3-MERCAPTOPROPANOIC ACID

D

β-damascenone⇒1-(2,6,6-TRIMETHYL-1,3-CYCLOHEXADIEN-1-YL)-(*E*)-2-BUTEN-1-ONE
(*E*)-damascenone⇒1-(2,6,6-TRIMETHYL-1,3-CYCLOHEXADIEN-1-YL)-(*E*)-2-BUTEN-1-ONE

2,4-DECADIENAL *[2363-88-4]*

Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	0.135	vegetable oil
Morales *et al.* (2005)	d	2.15	deodorised olive oil

(*E*,*E*)-2,4-DECADIENAL *[25152-84-5]*

Meijboom (1964)		2.15	paraffin oil
Guth & Grosch (1990a); Schieberle (1996)	d	0.18	sunflower oil
Guth & Grosch (1993)		0.0075	cellulose
Brewer & Vega (1995)	d	0.47	lean ground beef
Rychlik & Grosch (1996)	d	0.0027	starch
Jensen *et al.* (1999)	d	0.063	mashed potatoes

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

García-González *et al.* (2008)	d	2.50	deodorised edible oil
Poisson & Schieberle (2008)		0.0011	water/ethanol 6:4 by vol.

(*E*,*Z*)-2,4-DECADIENAL *[25152-83-4]*

Gassenmeier & Schieberle (1994)		0.010	sunflower oil
Wagner & Grosch (1998)		0.004	sunflower oil
Zehentbauer & Grosch (1998)		0.00181	starch

(*E*,*E*)-2,7-DECADIENAL

Meijboom (1964); Meijboom & Jongenotter (1981)		1.0	paraffin oil

(*E*,*Z*)-2,7-DECADIENAL

Meijboom (1964)		0.048	paraffin oil

γ-decalactone⇒4-HYDROXYDECANOIC ACID LACTONE
(*R*)-γ-decalactone⇒(*R*)-4-HYDROXYDECANOIC ACID LACTONE
(*S*)-γ-decalactone⇒(*S*)-4-HYDROXYDECANOIC ACID LACTONE
δ-decalactone⇒5-HYDROXYDECANOIC ACID LACTONE

DECANAL (decyl aldehyde) *[112-31-2]*

Meijboom (1964); Belitz & Grosch (1992)		6.7	paraffin oil
Rosenthal (1927)		9	diethyl phthalate
Pokorný *et al.* (1978)		3	orange beverage
Tajima *et al.* (1990)	d	10.0	limonene in water?
Plotto *et al.* (2004)		0.197 - 0.204	deodorized orange juice matrix
Culleré *et al.* (2004)		0.01	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Morales *et al.* (2005); García-González *et al.* (2008)	d	0.65	deodorised olive/edible oil
Ferreira *et al.* (2006)		0.0017	ethanol 10% v/v, tartaric acid 2 g/l, pH 3.4
Chaves *et al.* (2007)		1	14 % v/v ethanol/water
Culleré *et al.* (2011)	d	0.00125	hydroethanolic solution (13 % v/v), tartaric acid 5 g/L, pH 3.2
Culleré *et al.* (2011)	d	0.0075	citrus-smelling white wine
Culleré *et al.* (2011)	d	0.00375	neutral white wine
Culleré *et al.* (2011)	d	0.00187	floral white wine

DECANE *[124-18-5]*

Champagne & Nawar (1969)		4,000	mineral oil

DECANOIC ACID (capric acid) *[334-48-5]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	8.0/9.4	9.4 % (w/w) grain spirit
Baldwin *et al.* (1973)	r	1.7	buffered solution pH 3.2
Baldwin *et al.* (1973)	r	2.3	buffered solution pH 4.5
Baldwin *et al.* (1973)	r	16.0	buffered solution pH 6.0
Haslbeck *et al.* (1986)		> 400	cream
Haslbeck *et al.* (1986)		> 1,000	30 % coconut oil
Haslbeck *et al.* (1986)		> 1,000	38 % coconut oil aromatized
Moreno *et al.* (2005); Chaves *et al.* (2007)		15	14 % ethanolic solution
Aceña *et al.* (2011)	d	1.1	7 % (w/v) acetic acid water volume

1-DECANOL (decyl alcohol) *[112-30-1]*

Rosenthal (1927)		446	diethyl phthalate
Salo *et al.* (1972)	d	0.21	9.4 % (w/w) grain spirit
De Wet (1978)	d	0.36	wine residue medium
Moyano *et al.* (2002)		0.5	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		5	14 % ethanolic solution

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

4-decanolide⇒4-HYDROXYDECANOIC ACID LACTONE
5-decanolide⇒5-HYDROXYDECANOIC ACID LACTONE

2-DECENAL *[3913-71-1]*

Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	2.1	vegetable oil

(*E*)-2-DECENAL *[3913-81-3]*

Meijboom (1964); Keppler (1977);			
Belitz & Grosch (1992)		33.8 - 34	paraffin oil
Morales *et al.* (2005)	d	0.01	deodorised olive oil
Poisson & Schieberle (2008)		0.0052	water/ethanol 6:4 by vol.
Burdack-Freitag (2007)		3.22	sunflower oil

(*Z*)-2-DECENAL *[2497-25-8]*

Burdack-Freitag (2007)		0.05	sunflower oil

(*E*)-5-DECENAL *[60671-72-9]*

Keppler (1977); Meijboom & Jongenotter (1981)		5.0	paraffin oil

(*E*)-7-DECENAL

Keppler (1977); Meijboom & Jongenotter (1981)		1.55	paraffin oil

(*Z*)-7-DECENAL *[21661-97-2]*

Meijboom & Jongenotter (1981)		0.45	paraffin oil

1-DECENE *[872-05-9]*

Champagne & Nawar (1969)		150	mineral oil
Evans *et al.* (1971)	d	7	cottonseed oil

9-DECENOIC ACID *[14436-32-9]*

Brennand *et al.* (1989)	r	4.3	pH 2.0 aqueous solution

δ-decenolactone⇒5-HYDROXY-2-DECENOIC ACID LACTONE
decyl alcohol⇒1-DECANOL
decyl aldehyde⇒DECANAL

1-DECYNE *[764-93-2]*

Champagne & Nawar (1969)		60	mineral oil
Evans *et al.* (1971)	d	4	cottonseed oil

diacetone alcohol⇒4-HYDROXY-4-METHYLPENTAN-2-ONE
diacetyl⇒2,3-BUTANEDIONE
diallyl disulphide⇒2-PROPENYLDITHIO-2-PROPENE
diallyl sulphide⇒2-PROPENYLTHIO-2-PROPENE

2,6-DIBROMOPHENOL *[608-33-3]*

Siegmund & Pöllinger-Zierler (2006)	d	0.000 009	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.000 085	apple juice
Coulter *et al.* (2008)	d	0.000 077	neutral, dry white wine

1,2-DICHLOROETHANE (ethylene chloride, ethylene dichloride) *[107-06-2]*

Rosen *et al.* (1979)	d	9.9	mineral oil
Kleinschmidt (1983)	r	31.5	paraffin oil

2,4-DICHLOROPHENOL *[120-83-2]*

Van Gemert (1984)	d	0.006	cassis soft drink

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Van Gemert (1984)	d	0.006	orange soft drink
Coulter *et al.* (2008)	d	> 0.000 896	neutral, dry white wine
Coulter *et al.* (2008)	d (aroma and taste)	0.000 739	neutral, dry white wine

2,6-DICHLOROPHENOL *[87-65-0]*

Coulter *et al.* (2008)	d	0.000 032	neutral, dry white wine

3,5-DICHLOROPHENOL *[591-35-5]*

Van Gemert (1984)	d	2.2	cassis soft drink
Van Gemert (1984)	d	1.1	orange soft drink

1,1-DIETHOXYETHANE (acetaldehyde diethyl acetal, acetal) *[105-57-7]*

Perry (1989)		1	23 % v. ethanol
Moyano *et al.* (2002)		1.4	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		1	14 % ethanolic solution
Callejón *et al.* (2008)	d	0.133	7 % acetic acid solution
Poisson & Schieberle (2008)		0.719	water/ethanol 6:4 by vol.

diethyl disulphide⇒ETHYLDITHIOETHANE

DIETHYL MALATE *[7554-12-3]*

Zea *et al.* (2001); Moyano *et al.* (2002)		10	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		760	14 % ethanolic solution

2,3-DIETHYL-5-METHYLPYRAZINE *[18138-04-0]*

Czerny (1996)		0.0012	
Wagner & Grosch (1998)		0.000 5	sunflower oil
Zehentbauer & Grosch (1998)		0.000 14	starch
Stephan & Steinhart (1999)	r	0.000 5	refined vegetable oil

2,5-DIETHYLPYRAZINE *[13238-84-1]*

Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	0.27	vegetable oil

DIETHYL SUCCINATE *[123-25-1]*

Shinohara & Watanabe (1981)		75	wine
Moyano *et al.* (2002)		1,250	14 % (v/v) ethanol; tartaric acid to pH 3.5
Isogai *et al.* (2005)	d	100	base sake
Moreno *et al.* (2005); Chaves *et al.* (2007)		100	14 % ethanolic solution

diethyl sulphide⇒ETHYLTHIOETHANE
(+)-dihydrocarveol⇒(+)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANOL
(-)-dihydrocarveol⇒(-)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANOL
(+)-dihydrocarvone⇒(+)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE
(-)-dihydrocarvone⇒(-)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE
(+)-dihydrocarvyl acetate⇒(+)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXYL ACETATE
(-)-dihydrocarvyl acetate⇒(-)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXYL ACETATE
(3*R*)-dihydrocitronellal⇒(*R*)-(+)-3,7-DIMETHYLOCTANAL
(3*S*)-dihydrocitronellal⇒(*S*)-(-)-3,7-DIMETHYLOCTANAL
(3*R*)-dihydrocitronellol⇒(*R*)-(+)-3,7-DIMETHYL-1-OCTANOL
(3*S*)-dihydrocitronellol⇒(*S*)-(-)-3,7-DIMETHYL-1-OCTANOL
(3*R*)-dihydrocitronellyl acetate⇒(*R*)-(+)-3,7-DIMETHYLOCTYL ACETATE
(3*S*)-dihydrocitronellyl acetate⇒(*S*)-(-)-3,7-DIMETHYLOCTYL ACETATE
(3*R*)-dihydrocitronellylethylether⇒(*R*)-(+)-1-ETHOXY-3,7-DIMETHYLOCTANE
(3*S*)-dihydrocitronellylethylether⇒(*S*)-(-)-1-ETHOXY-3,7-DIMETHYLOCTANE
(3*R*)-dihydrocitronellylnitrile⇒(*R*)-(-)-2,6-DIMETHYLHEPTYL CYANIDE
(3*S*)-dihydrocitronellylnitrile⇒(*S*)-(+)-2,6-DIMETHYLHEPTYL CYANIDE
(1*R*,6*S*)-(*E*)-dihydrodamascone ⇒(*E*)-(1*R*,6*S*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-2-BUTEN-1-ONE
(1*S*,6*R*)-(*E*)-dihydrodamascone ⇒(*E*)-(1*S*,6*R*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-2-BUTEN-1-ONE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

5,6-DIHYDRO-6-PENTYL-2*H*-PYRAN-2-ONE (6-pentyl-5,6-dihydro-2*H*-pyran-2-one, massoia lactone) *[54814-64-1]*

Cadwallader *et al.* (2010)	d	1.1	canola oil

(4*R*)-*cis*-dihydrorose oxide⇒(4*R*)-*cis*-2-ISOBUTYL-4-METHYLTETRAHYDROPYRAN
(4*R*)-*trans*-dihydrorose oxide⇒(4*R*)-*trans*-2-ISOBUTYL-4-METHYLTETRAHYDROPYRAN
(4*S*)-*cis*-dihydrorose oxide⇒(4*S*)-*cis*-2-ISOBUTYL-4-METHYLTETRAHYDROPYRAN
(4*S*)-*trans*-dihydrorose oxide⇒(4*S*)-*trans*-2-ISOBUTYL-4-METHYLTETRAHYDROPYRAN

3,4-DIHYDROXY-3-HEXEN-2,5-DIONE

Engel *et al.* (2001)		0.01	sunflower oil

2,3-DIHYDROXY-4-ISOPROPYL-1-METHYLBENZENE (*p*-cymene-2,3-diol, 3-isopropyl-6-methylpyrocatechol) *[490-062]*

Bitar *et al.* (2008)	d	794.33	sunflower oil

1,3-DIMETHOXYBENZENE (dimethyl ether resorcinol) *[151-10-0]*

Rogerson *et al.* (2002)	d	0.0214	12 % ethanol, tartaric acid 5 g/l, pH 3.5
Rogerson *et al.* (2002)	d	0.0468	20 % ethanol, tartaric acid 5 g/l, pH 3.5

2,6-DIMETHOXYPHENOL (syringol) *[91-10-1]*

Wasserman (1966)		0.34	mineral oil
Ferreira *et al.* (2002) ; Lopez *et al.* (2002)		0.57	10 % hydroalcoholic solution, pH 3.2
Söllner & Schieberle (2009)		0.263	odorless sunflower oil

1,3-DIMETHYLBENZENE (*m*-xylene) *[108-38-3]*

Koelega & Köster (1974)		2,300-5,100	benzyl benzoate

4-(4',4'-DIMETHYLCYCLOHEXYL)-2-METHYLCYCLOHEXANONE

Pierce *et al.* (1993)	r	0.98-1.95	light mineral oil

(8*S*,9*S*,10*R*,13*R*,14*S*)-10,13-dimethyl-1,2,6,7,8,9,11,12,14,15-decahydrocyclopenta[a]phenanthren-3-one⇒4,16-ANDROSTADIEN-3-ONE

trans-1,10-DIMETHYL-*trans*-9-DECALOL (geosmin) *[19700-21-1]*

Tyler *et al.* (1979)	d	0.0058	beet juice
Persson (1980)	r	0.000 90	cooked bream
Persson (1980)	r	0.000 59	cooked pike
Persson (1980)	r	0.0065	cooked rainbow trout
Lovell *et al.* (1986)	d	0.0085	channel catfish
Ito *et al.* (1988)	d	3.5-5.5 10^{-5}	water with 0.6 mg/l chlorine
Ito *et al.* (1988)	d	2.9-16.4 10^{-5}	water with 1.5 mg/l chlorine
Ito *et al.* (1988)	r	8.2-17.5 10^{-5}	water with 0.6 mg/l chlorine
Ito *et al.* (1988)	r	6.7-34.3 10^{-5}	water with 1.5 mg/l chlorine
Tomita *et al.* (1988)		1-10 10^{-5}	water with 2 mg/l chlorine
Amon *et al.* (1989)	d	0.000 025	non-aromatic dry white wine
Siegmund & Pöllinger-Zierler (2006)	d	0.000 023	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.000 027	apple juice

(+)-*trans*-1,10-DIMETHYL-*trans*-9-DECALOL ((+)-geosmin)

Polak & Provasi (1992)		0.000 078	water and dipropylene glycol

(-)-*trans*-1,10-DIMETHYL-*trans*-9-DECALOL ((-)-geosmin)

Polak & Provasi (1992)		0.000 0095	water and dipropylene glycol

dimethyl disulphide⇒METHYLDITHIOMETHANE
dimethyl ether resorcinol⇒1,3-DIMETHOXYBENZENE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

(*R*)-(-)-2,6-DIMETHYL-5-HEPTENYL CYANIDE ((3*R*)-citronellylnitrile)

Yamamoto *et al.* (2004)		0.1	500 ppm ethanol in water

(*S*)-(+)-2,6-DIMETHYL-5-HEPTENYL CYANIDE ((3*S*)-citronellylnitrile)

Yamamoto *et al.* (2004)		0.075	500 ppm ethanol in water

(*R*)-(-)-2,6-DIMETHYLHEPTYL CYANIDE ((3*R*)-dihydrocitronellylnitrile)

Yamamoto *et al.* (2004)		0.075	500 ppm ethanol in water

(*S*)-(+)-2,6-DIMETHYLHEPTYL CYANIDE ((3*S*)-dihydrocitronellylnitrile)

Yamamoto *et al.* (2004)		0.05	500 ppm ethanol in water

2,5-DIMETHYL-4-METHOXY-3(2*H*)-FURANONE *[4077-47-8]*

Pino & Queris (2011)		0.016	11 % ethanol solution

6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE (β-pinene) *[127-91-3]*

Pelosi & Pisanelli (1981)	d	430	mineral oil
Plotto *et al.* (2004)		37.2 - 38.7	deodorized orange juice matrix

(1*S*)-(-)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE ((-)-β-pinene) *[18172-67-3]*

Jagella & Grosch (1999a)	d	2.9	starch

(1*R*)-(+)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE ((+)-β-pinene) *[19902-08-0]*

Jagella & Grosch (1999a)	d	2.1	starch

(*R*)-(-)-3,7-DIMETHYL-2-METHYLENE-6-OCTENAL ((3*R*)-α-methylene-3,7-dimethyl-6-octen-1-al)

Yamamoto *et al.* (2004)		0.25	500 ppm ethanol in water

(*S*)-(+)-3,7-DIMETHYL-2-METHYLENE-6-OCTENAL ((3*S*)-α-methylene-3,7-dimethyl-6-octen-1-al)

Yamamoto *et al.* (2004)		0.25	500 ppm ethanol in water

(*R*)-(-)-3,7-DIMETHYL-2-METHYLENE-6-OCTEN-1-OL ((3*R*)-α-methylene-3,7-dimethyl-6-octen-1-ol)

Yamamoto *et al.* (2004)		1	500 ppm ethanol in water

(*S*)-(+)-3,7-DIMETHYL-2-METHYLENE-6-OCTEN-1-OL ((3*S*)-α-methylene-3,7-dimethyl-6-octen-1-ol)

Yamamoto *et al.* (2004)		0.5	500 ppm ethanol in water

3,7-DIMETHYL-2,6-OCTADIENAL (citral) *[5392-40-5]*

Rosenthal (1927)		295	diethyl phthalate
Kendall & Neilson (1964)		9	hexadecane
Kendall & Neilson (1964)		0.03	aqueous 0.1 % ethanol
Pokorný *et al.* (1978)		11	orange beverage
Knupfer & Spiegel (1986)	d	100	propylene glycol
Ophir *et al.* (1986)	d	17-20	diethyl phthalate
Gross-Isseroff & Lancet (1988)	d	0.001-3,200	diethyl phthalate
Gross-Isseroff & Lancet (1988)	d	2,040	diethyl phthalate
Gross-Isseroff *et al.* (1992)	d	1,000	diethyl phthalate
Plotto *et al.* (2004)		0.656 - 1.23	deodorized orange juice matrix

3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-OL (geraniol) *[106-24-1]*

Rosenthal (1927)		35	diethyl phthalate
Terrier *et al.* (1972);			
Ribéreau-Gayon *et al.* (1975)	d	0.132	water and sugar
Gross-Isseroff & Lancet (1988)	d	1,000	diethyl phthalate
Ong & Acree (1998)		0.005	artificial saliva and fruit juice matrix

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Lee *et al.* (2000)	d	0.2 - 5.6	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	1 - 16	3 year old grain whisky: 23 % ethanol
Escudero *et al.* (2004)		0.036	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Escudero *et al.* (2007)		0.02	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2

3,7-DIMETHYL-(*Z*)-2,6-OCTADIEN-1-OL (nerol) *[106-25-2]*

Terrier *et al.* (1972);			
Ribéreau-Gayon *et al.* (1975)	d	0.4	water and sugar
Ferreira *et al.* (1998)		0.7	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Zea *et al.* (2001); Moyano *et al.* (2002)		0.5	14 % (v/v) ethanol; tartaric acid to pH 3.5

3,7-DIMETHYL-1,6-OCTADIEN-3-OL (linalool) *[78-70-6]*

Rosenthal (1927)		4	diethyl phthalate
Terrier *et al.* (1972);			
Ribéreau-Gayon *et al.* (1975)	d	0.1	water and sugar
Pokorný *et al.* (1978)		8	orange beverage
Williams & Ismail (1981)		0.10	sugar-acid base
Guth & Grosch (1993)		0.07	cellulose
Ferreira *et al.* (1998)		0.03	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Ong & Acree (1998)		0.0015	artificial saliva and fruit juice matrix
Jagella & Grosch (1998,1999a)	d	0.069	starch
Plotto *et al.* (2004)		0.067 - 0.113	deodorized orange juice matrix
Frasnelli & Hummel (2005)	r	947 – 60,625	propylene glycol (“trigeminal”)
Ferreira *et al.* (2006)		0.132	ethanol 10% v/v, tartaric acid 2 g/l, pH 3.4
Frauendorfer & Schieberle (2006)	d	0.037	sunflower oil
Chaves *et al.* (2007)		0.015	14 % v/v ethanol/water

(*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL ((*l*)-linalool, *R*-(-)-linalool) *[126-91-0]*

Fritsch (2001)		0.0022	beer

3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE (linalyl acetate) *[115-95-7]*

Rosenthal (1927)		88	diethyl phthalate
Matsushita *et al.* (1967)		10	liquid paraffin

(*R*)-(+)-3,7-DIMETHYLOCTANAL ((3*R*)-dihydrocitronellal)

Yamamoto *et al.* (2004)	d	0.005	500 ppm ethanol in water

(*S*)-(-)-3,7-DIMETHYLOCTANAL ((3*S*)-dihydrocitronellal)

Yamamoto *et al.* (2004)	d	0.005	500 ppm ethanol in water

(*R*)-(+)-3,7-DIMETHYL-1-OCTANOL ((3*R*)-dihydrocitronellol)

Yamamoto *et al.* (2004)	d	0.1	500 ppm ethanol in water

(*S*)-(-)-3,7-DIMETHYL-1-OCTANOL ((3*S*)-dihydrocitronellol)

Yamamoto *et al.* (2004)	d	0.25	500 ppm ethanol in water

3,7-DIMETHYL-6-OCTENAL (citronellal) *[106-23-0]*

Rosenthal (1927)		35	diethyl phthalate

(*R*)-(+)-3,7-DIMETHYL-6-OCTENAL ((3*R*)-(+)-citronellal) *[2385-77-5]*

Yamamoto *et al.* (2004)	d	0.025	500 ppm ethanol in water

(*S*)-(-)-3,7-DIMETHYL-6-OCTENAL ((3*S*)-(-)-citronellal) *[5949-05-3]*

Yamamoto *et al.* (2004)	d	0.025	500 ppm ethanol in water

3,7-DIMETHYL-6-OCTEN-1-OL (β-citronellol, citronellol) *[106-22-9]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Rosenthal (1927)		44	diethyl phthalate
Pokorný *et al.* (1978)		12	orange beverage
Ferreira *et al.* (1998)		0.7	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Moyano *et al.* (2002)		0.1	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		0.1	14 % ethanolic solution

(*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-OL ((3*R*)-(+)-citronellol) *[1117-61-9]*

Yamamoto *et al.* (2004)	d	0.05	500 ppm ethanol in water

(*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-OL ((3*S*)-(-)-citronellol) *[7540-51-4]*

Yamamoto *et al.* (2004)	d	0.05	500 ppm ethanol in water

(*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-YL ACETATE ((3*R*)-citronellyl acetate)

Yamamoto *et al.* (2004)	d	0.25	500 ppm ethanol in water

(*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-YL ACETATE ((3*S*)-citronellyl acetate)

Yamamoto *et al.* (2004)	d	0.25	500 ppm ethanol in water

(*R*)-(+)-3,7-DIMETHYLOCTYL ACETATE ((3*R*)-dihydrocitronellyl acetate)

Yamamoto *et al.* (2004)	d	0.25	500 ppm ethanol in water

(*S*)-(-)-3,7-DIMETHYLOCTYL ACETATE ((3*S*)-dihydrocitronellyl acetate)

Yamamoto *et al.* (2004)	d	0.25	500 ppm ethanol in water

2,3-DIMETHYLPHENOL (2,3-xylenol) *[526-75-0]*

Swan & Burtles (1978); Swan *et al.* (1981)		0.5	20 % ethanol-water

2,3-DIMETHYLPYRAZINE *[123-32-0]*

Siegmund & Pöllinger-Zierler (2006)	d	0.123	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	> 0.14	apple juice

2,5-DIMETHYLPYRAZINE *[123-32-0]*

Koehler *et al.* (1971)		17	paraffin oil
Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	2.6	vegetable oil

2,6-DIMETHYLPYRAZINE *[108-50-9]*

Koehler *et al.* (1971)		8	paraffin oil
Schirack *et al.* (2006)	d	1.021	vegetable oil

dimethyl sulphide⇒METHYLTHIOMETHANE
dimethyl tetrasulphide ⇒METHYLTETRATHIOMETHANE
dimethyl trisulphide⇒METHYLTRITHIOMETHANE

6,10-DIMETHYLUNDECA-5,9-DIEN-2-ONE (geranylacetone) *[3796-70-1]*

Tandon *et al.* (2001)	d	0.621	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	0.681	deodorized tomato homogenate

diphenyl ether⇒PHENOXYBENZENE

DIPHENYLMETHANE (benzylbenzene, ditan) *[101-81-5]*

Gross-Isseroff & Lancet (1988)	d	0.000 32-3,200	diethyl phthalate

ditan⇒DIPHENYLMETHANE
3,4-dithiahexane⇒ETHYLDITHIOETHANE

2,4-DITHIA-6-HEXANOL (2-[[(methylthio)methyl]thio]ethanol)

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Rapp *et al.* (1985)		0.1 - 1	15 vol. % ethanol

2,3-dithiapentane⇒METHYLDITHIOETHANE
2,4-dithiapentane⇒BIS(METHYLTHIO)METHANE
3,3'-dithiobis(2-methylfuran) ⇒3-(2-METHYL)FURYLDITHIO-3-(2-METHYL)FURAN
dithioglycolic acid⇒HYDROXYDITHIOACETIC ACID
DMDS⇒METHYLDITHIOMETHANE
γ-dodecalactone⇒4-HYDROXYDODECANOIC ACID LACTONE
(*R*)-γ-dodecalactone⇒(*R*)-4-HYDROXYDODECANOIC ACID LACTONE
(*S*)-γ-dodecalactone⇒(*S*)-4-HYDROXYDODECANOIC ACID LACTONE
δ-dodecalactone⇒5-HYDROXYDODECANOIC ACID LACTONE

DODECANAL (lauraldehyde) *[112-54-9]*

Meijboom (1964)		3.0	paraffin oil

DODECANE *[112-40-3]*

Champagne & Nawar (1969)		> 13,000	mineral oil

DODECANOIC ACID (lauric acid) *[143-07-7]*

Salo *et al.* (1972)	d	7.2	9.4 % (w/w) grain spirit
Haslbeck *et al.* (1986)		> 400	cream
Haslbeck *et al.* (1986)		> 1,000	30 % coconut oil
Haslbeck *et al.* (1986)		> 1,000	38 % coconut oil aromatized
Moreno *et al.* (2005)		10	14 % ethanolic solution

1-DODECANOL *[112-53-8]*

Salo *et al.* (1972)	d	1.0	9.4 % (w/w) grain spirit
De Wet (1978)	d	1.001	wine residue medium

4-dodecanolide⇒4-HYDROXYDODECANOIC ACID LACTONE
5-dodecanolide⇒5-HYDROXYDODECANOIC ACID LACTONE

(*E*)-2-DODECENAL *[20407-84-5]*

Meijboom (1964)		36.0	paraffin oil

1-DODECENE *[112-41-4]*

Champagne & Nawar (1969)		1,280	mineral oil

(*Z*)-6-dodeceno-γ-lactone⇒(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE

E

γ-eicosalactone⇒4-HYDROXYEICOSANOIC ACID LACTONE
δ-eicosalactone⇒5-HYDROXYEICOSANOIC ACID LACTONE
4-eicosanolide⇒4-HYDROXYEICOSANOIC ACID LACTONE
5-eicosanolide⇒5-HYDROXYEICOSANOIC ACID LACTONE

4,5-EPOXY-(*E*)-2-DECENAL *[73528-38-8]*

Heiler & Schieberle (1996,1997)	d	0.000 8	buttermilk
Ong & Acree (1998)		0.005	artificial saliva and fruit juice matrix

trans-4,5-EPOXY-(*E*)-2-DECENAL *[134454-31-2]*

Guth & Grosch (1990a,1990b)		5 - 20 10^{-4}	(sunflower) oil
Zehentbauer & Grosch (1998)		0.000 19	starch
Chetschik *et al.* (2010)		0.025	sunflower oil

ETHANAL (acetaldehyde) *[75-07-0]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Salo *et al.* (1972)	d	1.20	9.4 % (w/w) grain spirit
Czerny (1996)		0.062	cellulose
Kubickova & Grosch (1998); Reiners & Grosch (1998)	d	0.000 22	(refined) sunflower oil
Moyano *et al.* (2002)		110	14 % (v/v) ethanol; tartaric acid to pH 3.5
Utsunomiya *et al.* (2004a)	d	11	base sake
Utsunomiya *et al.* (2004a)	r	22	base sake
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		10	14 % ethanolic solution
Tesfaye *et al.* (2008)		0.402	7 % acetic acid solution

ETHANETHIOL (ethyl mercaptan) *[75-08-1]*

Matsushita *et al.* (1967)		0.0033	liquid paraffin
Goniak & Noble (1987)	d	0.0011	white wine
Utsunomiya *et al.* (2004a)	d	35 - 41 10^{-5}	base sake
Utsunomiya *et al.* (2004a)	r	81 - 82 10^{-5}	base sake

ethanoic acid ⇒ACETIC ACID

ETHANOL (ethyl alcohol) *[64-17-5]*

Adda & Jounela-Eriksson (1979)		120	9.4 % ethanol-water
Granzer *et al.* (1986)		50 - 500	coconut oil
Granzer *et al.* (1986)		5-50 10^3	liquid cheese
Granzer *et al.* (1986)		1-50 10^3	marmelade
Rüter (1992)	d	150	potato crisps
Rüter (1992)	r	330	potato crisps
Rüter (1992)	d	64	coffee powder
Rüter (1992)	r	1,000	coffee powder
Rüter (1992)	d	680	'Gummibären'
Rüter (1992)	r	2,000	'Gummibären'
Rüter (1992)	d	600	milk chocolate
Rüter (1992)	r	1,400	milk chocolate
Morales *et al.* (2005); Luna *et al.* (2006); García-González *et al.* (2008)	d	30	deodorised/refined olive/edible oil
Yu & Pickering (2008)	d	5,000	Chardonnay wine, ethanol 11.6 %
Yu & Pickering (2008)	d	5,800	Chardonnay wine, ethanol 13.4 %
Yu & Pickering (2008)	d	10,800	Zinfandel wine, ethanol 11.5 %
Yu & Pickering (2008)	d	11,400	Zinfandel wine, ethanol 13.4 %

ETHENYLBENZENE (styrene, vinylbenzene) *[100-42-5]*

Moiroud & Berger (1977)		0.10 - 0.15	wine
Boidron & Pons (1978)	d	0.125	ethanol 12 % + acid
Boidron & Pons (1978)	d	0.16	ethanol 12 % + sucrose
Boidron & Pons (1978)	d	0.25	white wine
Boidron & Pons (1978)	d	0.25	red wine
Dalton *et al.* (2003)		3.1 - 12.2	light, white mineral oil

2-ETHENYL-3,5-DIMETHYLPYRAZINE *[157615-33-3]*

Czerny (1996)		0.000 3	cellulose

2-ETHENYL-3-ETHYL-5-METHYLPYRAZINE *[181589-32-2]*

Czerny (1996)		0.000 2	cellulose
Wagner & Grosch (1998)		0.000 5	sunflower oil

2-ETHENYLFURAN (2-vinylfuran) *[1487-18-9]*

Evans *et al.* (1971)	d	1	cottonseed oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

4-ETHENYL-2-METHOXYPHENOL (*p*-vinylguaiacol, 4-vinylguaiacol, 2-methoxy-4-vinylphenol) *[7786-61-0]*

Schieberle (1991)		0.5 - 0.8	pale lager beer
Schieberle (1993,1996)	d	0.05	sunflower oil
Mayer (1996)		1.042	cellulose
Rychlik & Grosch (1996)	d	0.018	starch
Lee *et al.* (2000)	d	0.2-0.7	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	2-5	3 year old grain whisky: 23 % ethanol
Mayer (2000)		0.08	cellulose
Vanbeneden *et al.* (2008)	d	0.294	pilsner beer
Vanbeneden *et al.* (2008)	d	0.400	white beer
Vanbeneden *et al.* (2008)	d	0.516	blond specialty beer
Vichi *et al.* (2008)	d	0.2	refined sunflower oil
Averbeck & Schierberle (2011)	d	0.039	freshly reconstituted orange juice

5-ETHENYL-2-METHOXYPHENOL (2-methoxy-5-vinylphenol) *[621-58-9]*

Scheidig *et al.* (2007)	d	0.0012	oil
Scheidig *et al.* (2007)	r	0.0052	oil
Scheidig *et al.* (2007)	d	0.0022	cellulose
Scheidig *et al.* (2007)	r	0.0058	cellulose

4-ETHENYLPHENOL (4-vinylphenol) *[2628-17-3]*

Vichi *et al.* (2008)	d	0.4	refined sunflower oil
Chetschik *et al.* (2010)		14.72	sunflower oil

trans-5-ETHENYLTETRAHYDRO-α,α,5-TRIMETHYL-2-FURANMETHANOL (*trans*-furan linalool oxide, linalool oxide A) *[34995-77-2]*

Terrier *et al.* (1972); Ribéreau-Gayon *et al.* (1975); Ribéreau-Gayon (1978)	d	> 6 - 7	water and sugar
Ong & Acree (1998)		0.06	artificial saliva and fruit juice matrix

cis-5-ETHENYLTETRAHYDRO-α,α,5-TRIMETHYL-2-FURANMETHANOL (*cis*-furan linalool oxide, linalool oxide B) *[5989-33-3]*

Terrier *et al.* (1972); Ribéreau-Gayon *et al.* (1975); Ribéreau-Gayon (1978)	d	> 6 - 65	water and sugar

trans-6-ETHENYLTETRAHYDRO-2,2,6-TRIMETHYL-2*H*-PYRAN-3-OL (*trans*-pyran linalool oxide, linalool oxide C) *[39028-58-5]*

Terrier *et al.* (1972); Ribéreau-Gayon *et al.* (1975); Ribéreau-Gayon (1978)	d	3 - 3.6	water and sugar

cis-6-ETHENYLTETRAHYDRO-2,2,6-TRIMETHYL-2*H*-PYRAN-3-OL (*cis*-pyran linalool oxide, linalool oxide D) *[14009-71-3]*

Terrier *et al.* (1972); Ribéreau-Gayon *et al.* (1975); Ribéreau-Gayon (1978)	d	5 - 5.4	water and sugar

(*R*)-(+)-1-ETHOXY-3,7-DIMETHYLOCTANE ((3*R*)-dihydrocitronellylethylether)

Yamamoto *et al.* (2004)		0.5	500 ppm ethanol in water

(*S*)-(-)-1-ETHOXY-3,7-DIMETHYLOCTANE ((3*S*)-dihydrocitronellylethylether)

Yamamoto *et al.* (2004)		1	500 ppm ethanol in water

(*R*)-(+)-1-ETHOXY-3,7-DIMETHYL-6-OCTENE ((3*R*)-citronellylethylether)

Yamamoto *et al.* (2004)		0.25	500 ppm ethanol in water

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

(*S*)-(-)-1-ETHOXY-3,7-DIMETHYL-6-OCTENE ((3*S*)-citronellylethylether)

Yamamoto *et al.* (2004)		0.5	500 ppm ethanol in water

2-ETHOXYETHANOL (cellosolve, ethyl glycol, ethylene glycol ethyl ether) *[110-80-5]*

Granzer *et al.* (1986)		100 - 1,500	coconut oil
Granzer *et al.* (1986)		1-50 10^3	liquid cheese
Granzer *et al.* (1986)		100 - 1,000	biscuits
Granzer *et al.* (1986)		500 - 5,000	marmelade
Granzer *et al.* (1986)		400 - 1,500	fresh cheese

2-ETHOXYETHYL ACETATE (ethyl glycol acetate) *[111-15-9]*

Granzer *et al.* (1986)		2 - 100	coconut oil
Granzer *et al.* (1986)		2 - 50	liquid cheese

4-(1-ethoxyethyl)guaiacol⇒4-(1-ETHOXYETHYL)-2-METHOXYPHENOL

4-(1-ETHOXYETHYL)-2-METHOXYPHENOL (4-(1-ethoxyethyl)guaiacol)

Dugelay *et al.* (1995)	d	2.5	water-ethanol 998:2 v:v

4-(1-ETHOXYETHYL)PHENOL

Dugelay *et al.* (1995)	d	12.8	water-ethanol 998:2 v:v

2-ETHOXY-3,5-HEXADIENE *[56752-55-7]*

Chisholm & Samuels (1992)	d	< 0.000 001	wine

2-ETHOXYNAPHTHALENE (ethyl β-naphthyl ether) *[93-18-5]*

Rosenthal (1927)		446	diethyl phthalate

3-ETHOXY-1-PROPANOL *[111-35-3]*

Moyano *et al.* (2002)		0.1	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		50	14 % ethanolic solution

1-ETHOXY-2-PROPANOL *[1569-02-4]*

Rüter (1992)	d	88	potato crisps
Rüter (1992)	r	440	potato crisps
Rüter (1992)	d	300	coffee powder
Rüter (1992)	r	750	coffee powder
Rüter (1992)	d	70	'Gummibären'
Rüter (1992)	r	500	'Gummibären'
Rüter (1992)	d	75	milk chocolate
Rüter (1992)	r	600	milk chocolate

ETHYL ACETATE *[141-78-6]*

Sega *et al.* (1967)	d	93.5	lager beer
Salo (1970a,1970b); Salo *et al.* (1972)	d	17	9.4 % (w/w) grain spirit
Boidron (1978)		40	ethanol 12 % + acid
Boidron (1978)		160	dry white wine
Selfridge & Amerine (1978)	d	18	artificial wine
De Wet (1978)	d	12.27	wine residue medium
Corison *et al.* (1979)	d	170	white table wine
Corison *et al.* (1979)	d	160	red table wine
Granzer *et al.* (1986)		10 - 100	coconut oil
Granzer *et al.* (1986)		50 - 500	liquid cheese
Granzer *et al.* (1986)		10 - 100	biscuits
Granzer *et al.* (1986)		10 - 100	marmelade
Granzer *et al.* (1986)		10 - 60	fresh cheese

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Perry (1989)		74	23 % v. ethanol
Rüter (1992)	d	9	potato crisps
Rüter (1992)	r	27	potato crisps
Rüter (1992)	d	58	coffee powder
Rüter (1992)	r	470	coffee powder
Rüter (1992)	d	17	'Gummibären'
Rüter (1992)	r	40	'Gummibären'
Rüter (1992)	d	22	milk chocolate
Rüter (1992)	r	57	milk chocolate
Moyano *et al.* (2002)		15	14 % (v/v) ethanol; tartaric acid to pH 3.5
Escudero *et al.* (2004)		12.3	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Utsunomiya *et al.* (2004b)	d	24	base sake
Utsunomiya *et al.* (2004b)	r	45 - 46	base sake
Morales *et al.* (2005); Luna *et al.* (2006)	d	0.94	deodorised/refined olive oil
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		7.5	14 % ethanolic solution
Cliff & Pickering (2006)	d	160	base ice wine
Tesfaye *et al.* (2008)		91	7 % acetic acid solution

ethyl acid succinate⇒MONOETHYL SUCCINATE
ethyl acrylate⇒ETHYL 2-PROPENOATE
ethyl alcohol⇒ETHANOL
ethyl amyl ketone⇒3-OCTANONE

ETHYLBENZENE *[100-41-4]*

Boidron & Pons (1978)	d	1.2	ethanol 12 % + acid
Boidron & Pons (1978)	d	3.3	ethanol 12 % + sucrose
Boidron & Pons (1978)	d	7 - 8	white wine
Boidron & Pons (1978)	d	9	red wine

ETHYL BENZOATE *[93-89-0]*

Zea *et al.* (2001)		0.5	14 % (v/v) ethanol; tartaric acid to pH 3.5
Escudero *et al.* (2004)		0.74	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Chaves *et al.* (2007)		5	14 % v/v ethanol/water
Callejón *et al.* (2008)	d	0.210	7 % acetic acid solution

ETHYL BUTANOATE (ethyl butyrate) *[105-54-4]*

Salo *et al.* (1972)	d	0.15	9.4 % (w/w) grain spirit
Preininger & Grosch (1994)		0.028	sunflower oil
Moyano *et al.* (2002)		0.6	14 % (v/v) ethanol; tartaric acid to pH 3.5
Morales *et al.* (2005)	d	0.03	deodorised olive oil
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		0.02	14 % ethanolic solution
Poisson & Schieberle (2008)		0.0095	water/ethanol 6:4 by vol.
Pineau *et al.* (2009)	d	0.6	dearomatized red wine
Aceña *et al.* (2011)	d	0.073	7 % (w/v) acetic acid water volume

2-ETHYLBUTANOIC ACID *[88-09-5]*

Brennand *et al.* (1989)	r	5.7	pH 2.0 aqueous solution

ethyl butyrate⇒ETHYL BUTANOATE
ethyl caprate⇒ETHYL DECANOATE
ethyl caproate⇒ETHYL HEXANOATE
ethyl caprylate⇒ETHYL OCTANOATE
ethyl cinnamate⇒ETHYL 3-PHENYLPROPENOATE
ethyl (*E*)-cinnamate⇒ETHYL (*E*)-3-PHENYLPROPENOATE
trans-ethyl cinnamate⇒ETHYL (*E*)-3-PHENYLPROPENOATE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

ETHYL CYCLOHEXANECARBOXYLATE *[3289-28-9]*

Reiners & Grosch (1998)	d	0.000 16	refined sunflower oil

ETHYL DECANOATE (ethyl caprate) *[110-38-3]*

Salo *et al.* (1972)	d	1.10	9.4 % (w/w) grain spirit
De Wet (1978)	d	0.51	wine residue medium
Ferreira *et al.* (1998)		0.2	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Chaves *et al.* (2007)		0.51	14 % v/v ethanol/water

2-ETHYLDECANOIC ACID *2874-76-2]*

Brennand *et al.* (1989)	r	0.02	pH 2.0 aqueous solution

ETHYL (*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-YL OXALATE ((3*R*)-citronellyl ethyl oxalate)

Yamamoto *et al.* (2004)	d	0.2	500 ppm ethanol in water

ETHYL (*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-YL OXALATE ((3*S*)-citronellyl ethyl oxalate)

Yamamoto *et al.* (2004)	d	0.5	500 ppm ethanol in water

2-ETHYL-3,5-DIMETHYLPYRAZINE *[13925-07-0]*

Koehler *et al.* (1971)		24	paraffin oil
Schieberle (1993,1996)	d	0.003	sunflower oil
Czerny (1996)		0.000 5	cellulose
Rychlik & Grosch (1996)	d	0.000 17	starch
Stephan & Steinhart (1999)	r	0.0018	refined vegetable oil
Wagner & Grosch (1998)		0.0022	sunflower oil
Chetschik *et al.* (2010)		0.0075	sunflower oil

3-ETHYL-2,5-DIMETHYLPYRAZINE (2-ethyl-3,6-dimethylpyrazine) *[13360-65-1]*

Koehler *et al.* (1971)		24	paraffin oil
Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	0.024	vegetable oil
Wagner & Grosch (1998)		0.057	sunflower oil
Chetschik *et al.* (2010)		0.166	sunflower oil

ETHYLDITHIOETHANE (diethyl disulphide, 3,4-dithiahexane) *[110-81-6]*

Adda & Jounela-Eriksson (1979)		0.01	9.4 % ethanol-water
Goniak & Noble (1987)	d	0.0043	white wine
Stephan & Steinhart (1999)	r	0.078	refined vegetable oil

ETHYL DODECANOATE (ethyl laurate) *[106-33-2]*

Salo *et al.* (1972)	d	0.64	9.4 % (w/w) grain spirit
Lee *et al.* (2000)	d	0.7-0.8	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	2-3	3 year old grain whisky: 23 % ethanol
Zea *et al.* (2001); Moyano *et al.* (2002)		0.5	14 % (v/v) ethanol; tartaric acid to pH 3.5

ethylene chloride⇒1,2-DICHLOROETHANE
ethylene dichloride⇒1,2-DICHLOROETHANE
ethylene glycol ethyl ether⇒2-ETHOXYETHANOL

ETHYL FORMATE *[109-94-4]*

De Wet (1978)	d	155.2	wine residue medium

2-ETHYLFURAN *[3208-16-0]*

Evans *et al.* (1971)	d	8	cottonseed oil

ETHYL FURANCARBOXYLATE (ethyl 2-furoate) *[1335-40-6]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Moreno *et al.* (2005); Chaves *et al.* (2007)		1	14 % ethanolic solution

ethyl 2-furoate⇒ETHYL FURANCARBOXYLATE
ethyl glycol⇒2-ETHOXYETHANOL
ethyl glycol acetate⇒2-ETHOXYETHYL ACETATE
p-ethylguaiacol⇒4-ETHYL-2-METHOXYPHENOL
4-ethylguaiacol⇒4-ETHYL-2-METHOXYPHENOL

ETHYL HEPTANOATE *[106-30-9]*

De Wet (1978)	d	0.22	wine residue medium
Ferreira *et al.* (1998)		0.018	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Moyano *et al.* (2002)		0.3	14 % (v/v) ethanol; tartaric acid to pH 3.5

ETHYL HEXADECANOATE (ethyl palmitate) *[628-97-7]*

Salo *et al.* (1972)	d	> 14	9.4 % (w/w) grain spirit
Zea *et al.* (2001); Moyano *et al.* (2002)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5

ETHYL (*Z*)-9-HEXADECENOATE (ethyl palmitoleate) *[56219-10-4]*

Salo *et al.* (1972)	d	10.0	9.4 % (w/w) grain spirit

ETHYL 3,5-HEXADIENOATE (ethyl sorbate) *[2396-84-1]*

De Rosa *et al.* (1983)		0.3	10 % ethanol-water
Chisholm & Samuels (1992)	d	0.56	wine

ETHYL HEXANOATE (ethyl caproate) *[123-66-0]*

Salo *et al.* (1972)	d	0.076	9.4 % (w/w) grain spirit
Boidron (1978)		0.037	ethanol 12 % + acid
Boidron (1978)		0.056	ethanol 12 % + acid + sugar
Boidron (1978)		0.85	dry white wine
De Wet (1978)	d	0.08	wine residue medium
Preininger & Grosch (1994)		0.040	sunflower oil
Ferreira *et al.* (1998)		0.014	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Lee *et al.* (2000)	d	0.07	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	0.4-0.8	3 year old grain whisky: 23 % ethanol
Moyano *et al.* (2002)		0.1	14 % (v/v) ethanol; tartaric acid to pH 3.5
Utsunomiya *et al.* (2004a)	d	0.12	base sake
Utsunomiya *et al.* (2004a)	r	0.27	base sake
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		0.005	14 % ethanolic solution
Pineau *et al.* (2007)	d	0.000 0345	12 % ethanoll/water, 4 g/l tartaric aicd, pH 3.5
Pineau *et al.* (2007)	d	0.000 025	12 % ethanoll/water, 4 g/l tartaric aicd, pH 3.5 + 50 ng/L β-damascenone
Poisson & Schieberle (2008)		0.03	water/ethanol 6:4 by vol.
Pineau *et al.* (2009)	d	0.44	dearomatized red wine
Buhr *et al.* (2010)	d	0.3	whole milk
Buhr *et al.* (2010)	d	0.04	sunflower oil

2-ETHYLHEXANOIC ACID *[149-57-5]*

Brennand *et al.* (1989)	r	82.4	pH 2.0 aqueous solution

2-ethylhexyl acrylate⇒2-ETHYLHEXYL 2-PROPENOATE
2-ethylhexyl methacrylate⇒2-ETHYLHEXYL 2-METHYL-2-PROPENOATE

2-ETHYLHEXYL 2-METHYL-2-PROPENOATE (2-ethylhexyl methacrylate) *[688-84-6]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.5 - 10	sunflower oil
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.05 - 0.5	10 vol. % ethanol
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.05 - 0.5	3 vol. % acetic acid

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

2-ETHYLHEXYL 2-PROPENOATE (2-ethylhexyl acrylate) *[103-11-7]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.2 - 4	sunflower oil
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.01 - 0.2	10 vol. % ethanol
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.01 - 0.1	3 vol. % acetic acid

ETHYL 3-HYDROXYBUTANOATE *[5405-41-4]*

Moyano *et al.* (2002)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
Culleré *et al.* (2004); Gómez-Míguez *et al.* (2007); Campo *et al.* (2006); Escudero *et al.* (2007)		20	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Moreno *et al.* (2005); Chaves *et al.* (2007)		67	14 % ethanolic solution
Pineau *et al.* (2009)	d	1.2	dearomatized red wine

ETHYL 4-HYDROXYBUTANOATE *[999-10-0]*

Akita *et al.* (1988)		40 - 50	15 % ethanol

ETHYL 3-HYDROXYHEXANOATE *[2305-25-1]*

Moyano *et al.* (2010)		0.045	14 % v/v ethanol/water

ETHYL 6-HYDROXYHEXANOATE *[5299-60-5]*

Pineau *et al.* (2009)	d	1.8	dearomatized red wine

ETHYL 4-HYDROXY-3-METHOXYBENZOATE (ethyl vanillate) *[617-05-0]*

Lopez *et al.* (2002)		0.99	10 % hydroalcoholic solution at pH 3.2

2-ETHYL-4-HYDROXY-5-METHYL-3(2*H*)-FURANONE (sugarone, homofuraneol) *[27538-10-9]*

Ferreira *et al.* (2002)		0.125	10 % hydroalcoholic solution, pH 3.2

5-ETHYL-4-HYDROXY-2-METHYL-3(2*H*)-FURANONE *[27538-09-6]*

Preininger & Grosch (1994)		0.006	sunflower oil

ETHYL 2-HYDROXYPROPANOATE (ethyl lactate) *[97-64-3]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	14	9.4 % (w/w) grain spirit
Moyano *et al.* (2002)		146	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		100	14 % ethanolic solution

ETHYL *(R)*-(+)-2-HYDROXYPROPANOATE (ethyl D-lactate) *[7699-00-5]*

Lee *et al.* (2000)	d	> 1,400	3 year old grain whisky: 23 % ethanol

ethyl isobutyrate⇒ETHYL 2-METHYLPROPANOATE

ETHYL 2-ISOPROPYL-2-THIAZOLINE-4(*R*)-CARBOXYLATE

Fernandez *et al.* (2002)	d	< 10	ethanol

ETHYL 2-ISOPROPYL-3-THIAZOLINE-4-CARBOXYLATE

Fernandez *et al.* (2002)	d	< 10	ethanol

ethyl isovalerate⇒ETHYL 3-METHYLBUTANOATE
ethyl lactate⇒ETHYL 2-HYDROXYPROPANOATE
ethyl D-lactate⇒ETHYL (*R*)-(+)-2-HYDROXYPROPANOATE
ethyl laurate⇒ETHYL DODECANOATE
ethyl levulinate⇒ETHYL 4-OXOPENTANOATE
ethyl linoleate⇒ETHYL (*Z,Z*)-9,12-OCTADECADIENOATE
ethyl mercaptan⇒ETHANETHIOL

ETHYL 2-MERCAPTOACETATE (ethyl thioglycolate, ethyl 2-thiolacetate) *[623-51-8]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Brenner *et al.* (1954)		0.07 - 0.1	beer

ETHYL 3-MERCAPTOPROPANOATE *[5466-06-8]*

Sourabié *et al.* (2010)		0.000 723	fresh creame

ethyl methacrylate⇒ETHYL 2-METHYL-2-PROPENOATE

4-ETHYL-2-METHOXYPHENOL (*p*-ethylguaiacol, 4-ethylguaiacol) *[2785-89-9]*

Williams (1953)		0.05	ethanol solution
Swan & Burtles (1978); Swan *et al.* (1981)		0.05	20 % ethanol-water
Jounela-Eriksson & Lehtonen (1981)		0.033	10 vol. % grain spirit
Mayer (2000)		0.035	cellulose
Moyano *et al.* (2002)		0.02	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Moyano *et al.* (2010)		0.046	14 % ethanolic solution
Schmitt (2005)		0.972	oil
Poisson & Schieberle (2008)		0.0069	water/ethanol 6:4 by vol.
Vichi *et al.* (2008)	d	0.05	refined sunflower oil
Aceña *et al.* (2011)	d	0.0695	7 % (w/v) acetic acid water volume

ETHYL 2-METHYLBUTANOATE (ethyl 2-methylbutyrate) *[7452-79-1]*

Reiners & Grosch (1998)	d	0.000 72	refined sunflower oil
Czerny & Grosch (2000)	d	0.000 5	cellulose
Ferreira *et al.* (2006)		0.074	ethanol 10% v/v, tartaric acid 2 g/l, pH 3.4
Elss *et al.* (2007)		2.5 - 5 10^{-4}	10.5 g sucrose + 1.5 g citric acid.H_2O/L tap water
Elss *et al.* (2007)		3 - 16 10^{-4}	orange juice substitute without aroma
Elss *et al.* (2007)		3.5 - 40 10^{-4}	apple juice substitute without aroma
Elss *et al.* (2007)		0.003 - 0.007	commercial orange juice
Barbe *et al.* (2008); Pineau *et al.* (2009)	d	1.83 or 5.6	dearomatized red wine
Aceña *et al.* (2011)	d	0.0165	7 % (w/v) acetic acid water volume

ETHYL (*S*)-2-METHYLBUTANOATE *[10307-61-6]*

Reiners & Grosch (1998)	d	0.000 26	refined sunflower oil
Poisson & Schieberle (2008)		0.000 2	water/ethanol 6:4 by vol.

ETHYL 3-METHYLBUTANOATE (ethyl isovalerate, ethyl 3-methylbutyrate) *[108-64-5]*

Preininger & Grosch (1994)		0.000 1	sunflower oil
Ferreira *et al.* (1998)		0.003	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Reiners & Grosch (1998)	d	0.000 62	refined sunflower oil
Czerny & Grosch (2000)	d	0.000 6	cellulose
Callejón *et al.* (2008)	d	0.0044	7 % acetic acid solution
Poisson & Schieberle (2008)		0.0016	water/ethanol 6:4 by vol.
Aceña *et al.* (2011)	d	0.0022	7 % (w/v) acetic acid water volume

ethyl 2-methylbutyrate⇒ETHYL 2-METHYLBUTANOATE
ethyl 3-methylbutyrate⇒ETHYL 3-METHYLBUTANOATE

ETHYL 2-METHYLPROPANOATE (ethyl isobutyrate) *[97-62-1]*

Reiners & Grosch (1998)	d	0.0012	refined sunflower oil
Moyano *et al.* (2002)		5.6	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Moyano *et al.* (2010)		0.015	14 % ethanolic solution
Callejón *et al.* (2008)	d	0.00366	7 % acetic acid solution
Poisson & Schieberle (2008)		0.0045	water/ethanol 6:4 by vol.
Barbe *et al.* (2008); Pineau *et al.* (2009)	d	1.83 or 5.6	dearomatized red wine
Aceña *et al.* (2011)	d	0.0052	7 % (w/v) acetic acid water volume

ETHYL 2-METHYL-2-PROPENOATE (ethyl methacrylate) *[97-63-2]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.05 - 1	sunflower oil
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.02 - 0.2	10 vol. % ethanol
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.01 - 0.1	3 vol. % acetic acid

2-ETHYL-5-METHYLPYRAZINE *[13360-64-0]*

Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	0.32	vegetable oil

ethyl myristate⇒ETHYL TETRADECANOATE
ethyl β-naphthyl ether⇒2-ETHOXYNAPHTHALENE
ethyl nicotinate⇒ETHYL 3-PYRIDINECARBOXYLATE

ETHYL NONANOATE (ethyl pelargonate) *[123-29-5]*

De Wet (1978)	d	0.85	wine residue medium
Williams & Ismail (1981)		0.39	sugar-acid base

ETHYL (*Z*,*Z*)-9,12-OCTADECADIENOATE (ethyl linoleate) *[544-35-4]*

Salo *et al.* (1972)	d	0.45	9.4 % (w/w) grain spirit

ETHYL OCTADECANOATE (ethyl stearate) *[111-61-5]*

Salo *et al.* (1972)	d	> 0.5	9.4 % (w/w) grain spirit

ETHYL (*Z*)-9-OCTADECENOATE (ethyl oleate) *[111-62-6]*

Salo *et al.* (1972)	d	0.87	9.4 % (w/w) grain spirit

ETHYL OCTANOATE (ethyl caprylate) *[106-32-1]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	0.24/0.25	9.4 % (w/w) grain spirit
De Wet (1978)	d	0.58	wine residue medium
Rychlik & Grosch (1996)	d	0.038	starch
Ferreira *et al.* (1998)		0.005	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Moyano *et al.* (2002)		0.6	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		0.002	14 % ethanolic solution
Callejón *et al.* (2008)	d	0.0015	7 % acetic acid solution
Poisson & Schieberle (2008)		0.147	water/ethanol 6:4 by vol.
Pineau *et al.* (2009)	d	0.96	dearomatized red wine

4-ETHYLOCTANOIC ACID *[16493-80-4]*

Brennand *et al.* (1989)	r	0.006	pH 2.0 aqueous solution
Salles *et al.* (2002)	d	0.000 43	citrate buffer pH 5

ethyl oleate⇒ETHYL (*Z*)-9-OCTADECENOATE

ETHYL 4-OXOPENTANOATE (ethyl levulinate) *[539-88-8]*

Pineau *et al.* (2009)	d	0.35	dearomatized red wine

ethyl 2-oxopropanoate⇒ETHYL PYRUVATE
ethyl palmitate⇒ETHYL HEXADECANOATE
ethyl palmitoleate⇒ETHYL (*Z*)-9-HEXADECENOATE
ethyl pelargonate⇒ETHYL NONANOATE

ETHYL PENTANOATE (ethyl valerate) *[539-82-2]*

De Wet (1978)	d	0.01	wine residue medium

3-ETHYLPHENOL *[620-17-7]*

Söllner & Schieberle (2009)		0.044	odorless sunflower oil

4-ETHYLPHENOL *[123-07-9]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Williams (1953)		1.0	ethanol solution
Swan & Burtles (1978); Swan *et al.* (1981)		0.6	20 % ethanol-water
Jounela-Eriksson & Lehtonen (1981)		0.14	10 vol. % grain spirit
Moreno *et al.* (2005)		140	14 % ethanolic solution
Callejón *et al.* (2008)	d	0.004	7 % acetic acid solution
Vichi *et al.* (2008)	d	0.2	refined sunflower oil
Söllner & Schieberle (2009)		0.627	odorless sunflower oil
Aceña *et al.* (2011)	d	0.051	7 % (w/v) acetic acid water volume

ETHYL PHENYLACETATE *[101-97-3]*

Moio *et al.* (2004)		0.25	'wine'
Isogai *et al.* (2005)	d	0.1	base sake
Tat *et al.* (2007)	d	0.073	basic red wine
Aceña *et al.* (2011)	d	0.1482	7 % (w/v) acetic acid water volume

ETHYL 3-PHENYLPROPENOATE (ethyl cinnamate) *[103-36-6]*

Rosenthal (1927)		885	ethyl phthalate
Etiévant *et al.* (1983)	d	0.048	base white sweet wine
Pineau *et al.* (2007)	d	>0.0015	12 % ethanoll/water, 4 g/l tartaric aicd, pH 3.5
Pineau *et al.* (2007)	d	0.00145	12 % ethanoll/water, 4 g/l tartaric aicd, pH 3.5 + 50 ng/L β-damascenone

ETHYL (*E*)-3-PHENYLPROPENOATE (ethyl (*E*)-cinnamate, *trans*-ethyl cinnamate) *[4192-77-2]*

Reiners & Grosch (1998)	d	0.15	refined sunflower oil
Poisson & Schieberle (2008)		0.000 7	water/ethanol 6:4 by vol.
Aceña *et al.* (2011)	d	0.000 480	7 % (w/v) acetic acid water volume

ETHYL PROPANOATE *[105-37-3]*

Salo *et al.* (1972)	d	> 4.0	9.4 % (w/w) grain spirit
De Wet (1978)	d	1.84	wine residue medium
Moyano *et al.* (2002)		2.1	14 % (v/v) ethanol; tartaric acid to pH 3.5
Morales *et al.* (2005); Luna *et al.* (2006)	d	0.1	deodorised/refined olive oil
Moreno *et al.* (2005)		5	14 % ethanolic solution
Callejón *et al.* (2008)	d	0.516	7 % acetic acid solution
Poisson & Schieberle (2008)		3.452	water/ethanol 6:4 by vol.
Barbe *et al.* (2008); Pineau *et al.* (2009)	d	2.1	dearomatized red wine

ETHYL 2-PROPENOATE (ethyl acrylate) *[140-88-5]*

Piringer & Granzer (1984); Granzer *et al.* (1986)	0.001 - 0.05	sunflower oil
Piringer & Granzer (1984); Granzer *et al.* (1986)	0.001 - 0.01	10 vol. % ethanol
Piringer & Granzer (1984); Granzer *et al.* (1986)	5-50 10^{-4}	3 vol. % acetic acid

ETHYLPYRAZINE *[13925-00-3]*

Koehler *et al.* (1971)	17	paraffin oil

ETHYL 3-PYRIDINECARBOXYLATE (ethyl nicotinate) *[614-18-6]*

Saison *et al.* (2009)	d	4.555	carbonated beer served at 7-8 °C

ETHYL PYRUVATE (ethyl 2-oxopropanoate) *[617-35-6]*

Zea *et al.* (2001); Moyano *et al.* (2002)	5.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)	100	14 % ethanolic solution

ETHYL SALICYLATE *[118-61-6]*

Lindeman *et al.* (1982)	d	0.115	10 % ethanol

ethyl sorbate⇒ETHYL 3,5-HEXADIENOATE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

ethyl stearate⇒ETHYL OCTADECANOATE

ETHYL TETRADECANOATE (ethyl myristate) *[124-06-1]*

Salo *et al.* (1972)	d	> 5.7	9.4 % (w/w) grain spirit
Zea *et al.* (2001); Moyano *et al.* (2002)		0.5	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		494	14 % ethanolic solution

3'-ETHYL-5',6',7',8'-TETRAHYDRO-5',5',8',8'-TETRAMETHYL-2'-ACETONAPHTHONE (musk 36A) *[88-29-9]*

D'Andrea (1975)	d	50 - 1,000	diethyl phthalate

ETHYLTHIOETHANE (diethyl sulphide) *[352-93-2]*

Goniak & Noble (1987)	d	0.000 93	white wine

ethyl thioglycolate⇒ETHYL 2-MERCAPTOACETATE
ethyl 2-thiolacetate⇒ETHYL 2-MERCAPTOACETATE

(1*R*,6*S*)-ETHYL 2,2,6-TRIMETHYLCYCLOHEXYLCARBOXYLATE

Yamamoto *et al.* (2004)		0.02	500 ppm ethanol in water

(1*S*,6*R*)-ETHYL 2,2,6-TRIMETHYLCYCLOHEXYLCARBOXYLATE

Yamamoto *et al.* (2004)		0.1	500 ppm ethanol in water

ethyl valerate⇒ETHYL PENTANOATE
ethyl vanillate⇒ETHYL 4-HYDROXY-3-METHOXYBENZOATE
ethyl vinyl ketone⇒1-PENTEN-3-ONE
eucalyptol⇒1,3,3-TRIMETHYL-2-OXABICYCLO[2.2.2]OCTANE
eugenol⇒4-ALLYL-2-METHOXYPHENOL
eugenol methyl ether⇒4-ALLYL-1,2-DIMETHOXYBENZENE

EUGENYL ACETATE *[93-28-7]*

Harder (1975)		100	diethyl phthalate

exaltolide⇒15-HYDROXYPENTADECANOIC ACID LACTONE

F

farnesol⇒3,7,11-TRIMETHYL-2,6,10-DODECTRIEN-1-OL
fenchyl alcohol⇒1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL
fixolide⇒7-ACETYL-1,1,3,4,4,6-HEXAMETHYL-1,2,3,4-TETRAHYDRONAPHTHALENE

FORMIC ACID (methanoic acid) *[64-18-6]*

Kleinschmidt (1983)	r	141	paraffin oil

4-formyl-2-methoxyphenol⇒4-HYDROXY-3-METHOXYBENZALDEHYDE
2-furaldehyde⇒FURFURAL
furaneol⇒4-HYDROXY-2,5-DIMETHYL-3(2*H*)-FURANONE
trans-furan linalool oxide⇒*trans*-5-ETHENYLTETRAHYDRO-α,α,5-TRIMETHYL-2-FURANMETHANOL
cis-furan linalool oxide⇒*cis*-5-ETHENYLTETRAHYDRO-α,α,5-TRIMETHYL-2-FURANMETHANOL
2-furanmethanethiol⇒2-FURYLMETHANETHIOL

2-FURANMETHANOL (furfuryl alcohol, furfurylic alcohol) *[98-00-0]*

Moreno *et al.* (2005); Chaves *et al.* (2007)		15	14 % ethanolic solution
Callejón *et al.* (2008)	d	1.415	7 % acetic acid solution

FURFURAL (2-furaldehyde) *[98-01-1]*

Salo *et al.* (1972)	d	5.80	9.4 % (w/w) grain spirit

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Doty (1975)	d	155	propylene glycol
Doty *et al.* (1981)	r	< 25	propylene glycol
Lee *et al.* (2000)	d	39	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	210	3 year old grain whisky: 23 % ethanol
Utsunomiya *et al.* (2004a)	d	11	base sake
Utsunomiya *et al.* (2004a)	r	21	base sake
Moreno *et al.* (2005); Chaves *et al.* (2007)		15	14 % ethanolic solution
Tesfaye *et al.* (2008)		6.2	7 % acetic acid solution
Saison *et al.* (2009)	d	15.157	carbonated beer served at 7-8 °C

furfuryl alcohol⇒2-FURANMETHANOL
furfurylic alcohol⇒2-FURANMETHANOL
furfuryl ethyl ether⇒2-FUROYLETHANE
furfuryl mercaptan⇒2-FURYLMETHANETHIOL
furfurylthiol⇒2-FURYLMETHANETHIOL

2-FUROYLETHANE (furfuryl ethyl ether) *[6270-56-0]*

Spillman *et al.* (1998)		0.43	neutral dry whit wine
Saison *et al.* (2009)	d	0.011	carbonated beer served at 7-8 °C

2-FURYLMETHANETHIOL (furfuryl mercaptan, furfurylthiol, 2-furanmethanethiol) *[98-02-2]*

Rychlik (1991)		0.000 1	starch
Schieberle (1993,1996)	d	0.000 4	sunflower oil
Mayer (1996)		0.000 3	cellulose

G

galaxolide⇒1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA(g)-2-BENZOPYRAN
geosmin⇒*trans*-1,10-DIMETHYL-*trans*-9-DECALOL
(+)-geosmin⇒(+)-*trans*-1,10-DIMETHYL-*trans*-9-DECALOL
(-)-geosmin⇒(-)-*trans*-1,10-DIMETHYL-*trans*-9-DECALOL
geraniol⇒3,7-DIMETHYL-*trans*-2,6-OCTADIEN-1-OL
geranylacetone⇒ 6,10-DIMETHYLUNDECA-5,9-DIEN-2-ONE
guaiacol⇒2-METHOXYPHENOL

H

helional⇒2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL
heliotropine⇒3,4-METHYLENEDIOXYBENZALDEHYDE

2,4-HEPTADIENAL *[5910-85-0]*

Morales *et al.* (2005)	d	0.36/3.62	deodorised olive oil

(*E*,*E*)-2,4-HEPTADIENAL *[4313-03-5]*

Meijboom (1964); Keppler (1977); Meijboom & Jongenotter (1981); Belitz & Grosch (1992)		10.0	paraffin oil

(*E*,*Z*)-2,4-HEPTADIENAL *[4313-02-4]*

Meijboom (1964); Keppler (1977); Belitz & Grosch (1992)		3.6 - 4	paraffin oil

1,4-HEPTADIENE *[592-45-0]*

Evans *et al.* (1971)	d	9	cottonseed oil

d-γ-heptalactone⇒(*R*)-(+)-4-HYDROXYHEPTANOIC ACID LACTONE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

l-γ-heptalactone⇒(*S*)-(-)-4-HYDROXYHEPTANOIC ACID LACTONE
heptaldehyde⇒HEPTANAL

HEPTANAL (heptaldehyde) *[111-71-7]*

Engen (1960)		760-6,100	benzyl benzoate
Meijboom (1964); Belitz & Grosch (1992)		3.2	paraffin oil
Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	0.25	vegetable oil
Brewer & Vega (1995)	d	0.23	lean ground beef
Morales *et al.* (2005); García-González *et al.* (2008)	d	0.5	deodorised olive/edible oil
Limpawattana (2007)	d	0.1809	starch
Limpawattana (2007)	d	0.0648	partially deodorized rice at 70° C

HEPTANE *[142-82-5]*

Engen (1960)		24-10,000	benzyl benzoate
Champagne & Nawar (1969)		250	mineral oil

HEPTANOIC ACID (oenanthic acid) *[111-14-8]*

Salo *et al.* (1972)	d	> 0.5	9.4 % (w/w) grain spirit
Brennand *et al.* (1989)	r	0.28	pH 2.0 aqueous solution
Morales *et al.* (2005)	d	0.1	deodorised olive oil

1-HEPTANOL *[111-70-6]*

Matsushita *et al.* (1967)		20	liquid paraffin
De Wet (1978)	d	2.45	wine residue medium
Chaves *et al.* (2007)		2.5	14 % v/v ethanol/water

2-HEPTANOL *[543-49-7]*

Morales *et al.* (2005); García-González *et al.* (2008)	d	0.01	deodorised olive/edible oil

2-HEPTANONE (methyl amyl ketone) *[110-43-0]*

Barylko-Pikielna *et al.* (1983)	d	62 - 98	sunflower oil
Barylko-Pikielna *et al.* (1983)	r	243 - 392	sunflower oil
Rothe (1991)		24	cheese 20 % fat dry matter
Rothe (1991,1992)		27	cheese 45 % fat dry matter
Rothe (1991)		35	cheese 70 % fat dry matter
Rothe (1991)		34	vegetable oil
Rothe (1991)		83	vegetable oil
Preininger & Grosch (1994)		1.5	sunflower oil
Morales *et al.* (2005); Luna *et al.* (2006); García-González *et al.* (2008)	d	0.3	deodorised/refined olive/edible oil

2-HEPTENAL *[2463-63-0]*

Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	1.5	vegetable oil

(*E*)-2-HEPTENAL *[18829-55-5]*

Meijboom (1964); Meiboom & Jongenotter (1981)		14.0	paraffin oil
Guth (1991)		3.75	oil
Morales *et al.* (2005)	d	0.001/0.005	deodorised olive oil
Culleré *et al.* (2007)	d	0.0046	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2
García-González *et al.* (2008)	d	0.05	deodorised edible oil

(*E*)-3-HEPTENAL

Meijboom & Jongenotter (1981)		0.31	paraffin oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

(*Z*)-3-HEPTENAL *[21662-18-0]*

Meijboom & Jongenotter (1981)		0.025	paraffin oil

(*E*)-4-HEPTENAL *[929-22-6]*

Hoffmann & Meijboom (1968);			
Meijboom & Jongenotter (1981)		2.3	paraffin oil

(*Z*)-4-HEPTENAL *[6728-31-0]*

Hoffmann & Meijboom (1968);			
Meijboom & Jongenotter (1981)		0.01	paraffin oil
Belitz & Grosch (1992)		0.002	oil
Guth & Grosch (1993)		0.003	cellulose
Grosch *et al.* (1994)		0.0035	butter oil

(*E*)-5-HEPTENAL

Meijboom & Jongenotter (1981)		0.16	paraffin oil

(*Z*)-5-HEPTENAL

Meijboom & Jongenotter (1981)		1.0	paraffin oil

6-HEPTENAL

Meijboom & Jongenotter (1981)		0.66	paraffin oil

1-HEPTENE *[592-76-7]*

Champagne & Nawar (1969)		1	mineral oil

1-HEPTEN-3-OL *[4938-52-7]*

Evans *et al.* (1971)	d	3.0	cottonseed oil

1-HEPTEN-3-ONE *[2918-13-0]*

Evans *et al.* (1971)	d	0.007	cottonseed oil

HEPTYL ACETATE *[112-06-1]*

De Wet (1978)	d	0.83	wine residue medium

1-HEPTYNE *[628-71-7]*

Champagne & Nawar (1969)		100	mineral oil

γ-hexadecalactone⇒4-HYDROXYHEXADECANOIC ACID LACTONE
δ-hexadecalactone⇒5-HYDROXYHEXADECANOIC ACID LACTONE
ω-hexadecalactone⇒16-HYDROXYHEXADECANOIC ACID LACTONE

HEXADECANE *[544-76-3]*

Champagne & Nawar (1969)		> 13,000	mineral oil

HEXADECANOIC ACID (palmitic acid) *[57-10-3]*

Salo *et al.* (1972)	d	> 10	9.4 % (w/w) grain spirit

1-HEXADECANOL (cetyl alcohol) *[36653-82-4]*

Salo *et al.* (1972)	d	> 1.1	9.4 % (w/w) grain spirit

4-hexadecanolide⇒4-HYDROXYHEXADECANOIC ACID LACTONE
5-hexadecanolide⇒5-HYDROXYHEXADECANOIC ACID LACTONE
hexadecanolide⇒16-HYDROXYHEXADECANOIC ACID LACTONE

1-HEXADECENE *[629-73-2*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Champagne & Nawar (1969)		3,200	mineral oil

(*Z*)-9-HEXADECENOIC ACID (palmitoleic acid) *[373-49-9]*

Salo *et al.* (1972)	d	> 11	9.4 % (w/w) grain spirit

2,4-HEXADIENAL *[142-83-6]*

Luna *et al.* (2006)	d	2	refined olive oil

(*E*,*E*)-2,4-HEXADIENAL (sorbic aldehyde) *[80466-34-8]*

Meijboom (1964)		0.27	paraffin oil

1,3-HEXADIENE *[592-48-3]*

Evans *et al.* (1971)	d	2	cottonseed oil

1,5-HEXADIENE *[592-42-7]*

Evans *et al.* (1971)	d	0.5	cottonseed oil

(*Z*,*Z*)-2,4-HEXADIENE *[6108-61-8]*

Evans *et al.* (1971)	d	3	cottonseed oil

(*Z*,*E*)-2,4-HEXADIENE *[5194-50-3]*

Evans *et al.* (1971)	d	30	cottonseed oil

(*E*,*E*)-2,4-HEXADIENE *[5194-51-4]*

Evans *et al.* (1971)	d	38	cottonseed oil

[(1*S*)-(1α,4β,4aβ,7α,8aα)]-1,3,4,5,6,8a-hexahydro-4,7-dimethyl-1-(1-methylethyl)-4a(2*H*)-naphthalenol⇒CUBENOL

1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA(g)-2-BENZOPYRAN (galaxolide) *[1222-05-5]*

D'Andrea (1975)	d	50 - 100	diethyl phthalate
Jeffcoat & Willis (1986)		700	diethyl phthalate
Pierce *et al.* (1993)	r	380-759	light mineral oil

γ-hexalactone⇒4-HYDROXYHEXANOIC ACID LACTONE

HEXANAL *[66-25-1]*

Meijboom (1964); Keppler (1977)		0.3 - 0.32	paraffin oil
Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	0.12	vegetable oil
Guth & Grosch (1990a)		0.3	sunflower oil
Brewer & Vega (1995)	d	5.87	lean ground beef
Kubickova & Grosch (1998)		0.12	sunflower oil
Zehentbauer & Grosch (1998)		0.03	starch
Jensen *et al.* (1999)	d	0.047	mashed potatoes
Lee *et al.* (2000)	d	0.07-0.1	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	0.7-2	3 year old grain whisky: 23 % ethanol
Tandon *et al.* (2001)	d	1.047	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	2.427	deodorized tomato homogenate
Aparicio & Luna (2002)		0.075	refined vegetable oil
Plotto *et al.* (2004)		0.135 - 0.151	deodorized orange juice matrix
Morales *et al.* (2005)	d	0.08/0.32	deodorised olive oil
Luna *et al.* (2006); García-González *et al.* (2008)	d	0.075 - 0.08	deodorised/refined olive/edible oil

HEXANE *[110-54-3]*

Champagne & Nawar (1969)		90	mineral oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

García-González *et al.* (2008)	d	1.50	deodorised edible oil

HEXANOIC ACID (caproic acid) *[142-62-1]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	8.6/8.8	9.4 % (w/w) grain spirit
Amoore & Buttery (1978)	d	29.7	safflower oil
Williams & Ismail (1981)		4	sugar-acid base
Haslbeck *et al.* (1986)		80 - 95	cream
Haslbeck *et al.* (1986)		21 - 27	30 % coconut oil
Haslbeck *et al.* (1986)		108 - 187	38 % coconut oil aromatized
Schieberle *et al.* (1993)		5.4	sunflower oil
Rychlik & Grosch (1996)	d	11	starch
Moyano *et al.* (2002)		8.5	14 % (v/v) ethanol; tartaric acid to pH 3.5
Karagül-Yüceer *et al.* (2004)	d	0.065	skim milk
Utsunomiya *et al.* (2004a)	d	2.2 - 2.3	base sake
Utsunomiya *et al.* (2004a)	r	4.3	base sake
Morales *et al.* (2005)	d	0.7	deodorised olive oil
Moreno *et al.* (2005); Chaves *et al.* (2007)		3	14 % ethanolic solution
Rychlik *et al.* (2006)		11.98	fresh yoghurt with 1.65 mg/kg hexanoic acid
Callejón *et al.* (2008)	d	2.6	7 % acetic acid solution

1-HEXANOL (hexyl alcohol) *[111-27-3]*

Williams (1953)		10	ethanol solution
Rosano & Scheps (1964)		102 - 511	mineral oil
Cain (1966)	r	1,890	diethyl phthalate
Salo (1970a,1970b); Salo *et al.* (1972)	d	5.2	9.4 % (w/w) grain spirit
De Wet (1978)	d	5.28	wine residue medium
Williams & Ismail (1981)		4.0	sugar-acid base
Morales *et al.* (1996);			
Aparicio & Morales (1998)		0.4	sunflower oil
Ferreira *et al.* (1998)		8	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Tandon *et al.* (2001)	d	1.067	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	1.102	deodorized tomato homogenate
Moyano *et al.* (2002)		1.3	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007)		8	14 % ethanolic solution
Aparicio & Luna (2002); Luna *et al.* (2006);			
García-González *et al.* (2008)	d	0.4	refined vegetable/olive/edible oil
Limpawattana (2007)	d	0.8321	starch
Limpawattana (2007)	d	0.3807	partially deodorized rice at 70° C

4-hexanolide⇒4-HYDROXYHEXANOIC ACID LACTONE

2-HEXENAL *[505-57-7]*

Buttery *et al.* (1973)	d	0.85	vegetable oil

(*E*)-2-HEXENAL (leaf aldehyde) *[6728-26-3]*

Meijboom (1964)		10.0	paraffin oil
Brewer & Vega (1995)	d	7.87	lean ground beef
Hatanaka *et al.* (1992)		0.01	dipropylene glycol
Reiners & Grosch (1998)	d	0.424	refined sunflower oil
Tandon *et al.* (2001)	d	0.284	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	0.592	deodorized tomato homogenate
Aparicio & Luna (2002)		1.125	refined vegetable oil
Morales *et al.* (2005); Luna *et al.* (2006)	d	0.42 - 0.424	deodorised/refined olive oil
Culleré *et al.* (2007)	d	0.004	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2

(*Z*)-2-HEXENAL *[16635-54-4]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Hatanaka *et al.* (1992)		0.01	dipropylene glycol
(*E*)-3-HEXENAL *[69112-21-6]*			
Meijboom (1964); Meijboom & Jongenotter (1981)		1.2	paraffin oil
Hatanaka *et al.* (1992)		0.001	dipropylene glycol
Luna *et al.* (2006)	d	0.003	refined olive oil
(*Z*)-3-HEXENAL *[6789-80-6]*			
Meijboom (1964); Keppler (1977); Meijboom & Jongenotter (1981)		0.11 - 0.15	paraffin oil
Guth & Grosch (1990a)		0.0135	sunflower oil
Hatanaka *et al.* (1992)		0.001	dipropylene glycol
Guth & Grosch (1993)		0.002	cellulose
Buettner *et al.* (1998)		0.000 4	2 mg/l furaneol (=200 OAV) in water
Buettner *et al.* (1998)		0.000 4	0.6 mg/l 2,3-butanedione (=200 OAV) in water
Reiners & Grosch (1998)	d	0.0017	refined sunflower oil
Tandon *et al.* (2001)	d	0.17	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	0.469	deodorized tomato homogenate
Aparicio & Luna (2002)		0.003	refined vegetable oil
Luna *et al.* (2006)	d	0.0017	refined olive oil
(*E*)-4-HEXENAL *[25166-87-4]*			
Hatanaka *et al.* (1992)		0.001	dipropylene glycol
(*Z*)-4-HEXENAL *[4684-89-3]*			
Keppler (1977); Meijboom & Jongenotter (1981)		0.002	paraffin oil
Hatanaka *et al.* (1992)		0.001	dipropylene glycol
5-HEXENAL *[764-59-0]*			
Keppler (1977); Meijboom & Jongenotter (1981)		0.019 - 0.025	paraffin oil
Hatanaka *et al.* (1992)		0.001	dipropylene glycol
1-HEXENE *[592-41-6]*			
Champagne & Nawar (1969)		3	mineral oil
Evans *et al.* (1971)	d	0.02	cottonseed oil
1-HEXEN-3-OL *[4798-44-1]*			
Evans *et al.* (1971)	d	0.5	cottonseed oil
(*E*)-2-HEXEN-1-OL *[928-95-0]*			
Hatanaka *et al.* (1992)		10	dipropylene glycol
Aparicio & Morales (1998)		8	sunflower oil
Morales *et al.* (2005); Luna *et al.* (2006)	d	5	deodorised/refined olive oil
(*Z*)-2-HEXEN-1-OL *[928-94-9]*			
Hatanaka *et al.* (1992)		10	dipropylene glycol
Luna *et al.* (2006)	d	1	refined olive oil
(*E*)-3-HEXEN-1-OL *[928-97-2]*			
Hatanaka *et al.* (1992)		1	dipropylene glycol
Aparicio & Morales (1998)		1.5	sunflower oil
Zea *et al.* (2001); Moyano *et al.* (2002)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
(*Z*)-3-HEXEN-1-OL (leaf alcohol) *[928-96-1]*			
Rosenthal (1927)		18	diethyl phthalate
Williams & Ismail (1981)		0.22	sugar-acid base

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Hatanaka *et al.* (1992)		1	dipropylene glycol
Reiners & Grosch (1998)	d	1.1	refined sunflower oil
Tandon *et al.* (2001)	d	0.479	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	1.602	deodorized tomato homogenate
Aparicio & Luna (2002)		6	refined vegetable oil
Zea *et al.* (2001); Moyano *et al.* (2002)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
Luna *et al.* (2006)	d	1.1	refined olive oil

(*E*)-4-HEXEN-1-OL *[928-92-7]*

Hatanaka *et al.* (1992)		0.1	dipropylene glycol

(*Z*)-4-HEXEN-1-OL *[928-91-6]*

Hatanaka *et al.* (1992)		0.1	dipropylene glycol

5-HEXEN-1-OL *[821-41-0]*

Hatanaka *et al.* (1992)		1	dipropylene glycol

1-HEXEN-3-ONE *[1629-60-3]*

Evans *et al.* (1971)	d	0.005	cottonseed oil

(*Z*)-3-HEXENYL ACETATE *[3681-71-8]*

Reiners & Grosch (1998)	d	0.2	refined sunflower oil
Aparicio & Luna (2002)		0.75	refined vegetable oil
Luna *et al.* (2006)	d	0.2	refined olive oil

HEXYL ACETATE *[142-92-7]*

De Wet (1978)	d	0.67	wine residue medium
Aparicio & Morales (1998)	d	1.04	sunflower oil
Aparicio & Luna (2002); Luna *et al.* (2006)	d	1.04	refined vegetable/olive oil
Chaves *et al.* (2007)		1	14 % v/v ethanol/water
Pineau *et al.* (2009)	d	0.67	dearomatized red wine

hexyl alcohol⇒1-HEXANOL

HEXYL HEXANOATE *[6378-65-0]*

Zea *et al.* (2001)		0.5	14 % (v/v) ethanol; tartaric acid to pH 3.5

HEXYL 2-HYDROXYPROPANOATE (hexyl lactate) *[20279-51-0]*

Zea *et al.* (2001)		5	14 % (v/v) ethanol; tartaric acid to pH 3.5

hexyl lactate⇒HEXYL 2-HYDROXYPROPANOATE

1-HEXYNE *[693-02-7]*

Champagne & Nawar (1969)		160	mineral oil
Evans *et al.* (1971)	d	0.2	cottonseed oil

homofuraneol⇒2-ETHYL-4-HYDROXY-5-METHYL-3(2*H*)-FURANONE
hydrocinnamic acid⇒3-PHENYLPROPANOIC ACID

HYDROGEN SULPHIDE *[7783-06-4]*

Brenner *et al.* (1953)	d	0.005	beer
Brenner *et al.* (1953)	r	0.1	beer
Pippen & Mecchi (1969)		0.162	broth
Amerine & Roessler (1976)		0.001 - 0.01	wine
Lucas (2000)	d	0.005	aqueous-alcoholic distillate from herbs
Lucas (2000)	r	0.025 - 0.03	aqueous-alcoholic distillate from herbs

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Siebert *et al.* (2009)	d	0.0016	neutral white wine
Siebert *et al.* (2009)	d	0.0011	red wine

2-HYDROXYBENZALDEHYDE (salicylaldehyde) *[90-02-8]*

Lindeman *et al.* (1982)	d	0.028	10 % ethanol

4-HYDROXYBUTANOIC ACID LACTONE (γ-butyrolactone) *[96-48-0]*

Akita *et al.* (1988)		20 - 30	15 % ethanol
Moyano *et al.* (2002)		20	14 % (v/v) ethanol; tartaric acid to pH 3.5
Culleré *et al.* (2004)		0.035	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Moreno *et al.* (2005); Chaves *et al.* (2007)		100	14 % ethanolic solution
Campo *et al.* (2006); Escudero *et al.* (2007)		35	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2

3-HYDROXY-2-BUTANONE (acetoin) *[513-86-0]*

Moyano *et al.* (2002)		155	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007);			
Moyano *et al.* (2010)		30	14 % ethanolic solution
Callejón *et al.* (2008)	d	8.8	7 % acetic acid solution

4-HYDROXYDECANOIC ACID LACTONE (γ-decalactone, 4-decanolide) *[706-14-9]*

Williams & Ismail (1981)		0.17	sugar-acid base
Wagner & Grosch (1998)		0.32	sunflower oil
Zea *et al.* (2001; Moyano *et al.* (2002)		0.01	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		1	14 % ethanolic solution
Elss *et al.* (2007)		0.003 - 0.01	10.5 g sucrose + 1.5 g citric acid.H_2O/L tap water
Elss *et al.* (2007)		0.01 - 0.025	orange juice substitute without aroma
Elss *et al.* (2007)		0.02 - 0.025	apple juice substitute without aroma
Elss *et al.* (2007)		0.02 - 0.03	commercial orange juice
Schlutt *et al.* (2007)	r	0.051	oil/water emulsion
Poisson & Schieberle (2008)		0.021	water/ethanol 6:4 by vol.

(*R*)-4-HYDROXYDECANOIC ACID LACTONE ((*R*)-γ-decalactone)

Brown (2007); Cooke *et al.* (2009)	d	0.033 - 0.34	red wine

(*S*)-4-HYDROXYDECANOIC ACID LACTONE ((*S*)-γ-decalactone)

Brown (2007); Cooke *et al.* (2009)	d	0.047	red wine

5-HYDROXYDECANOIC ACID LACTONE (δ-decalactone, 5-decanolide) *[705-86-2]*

Etiévant *et al.* (1983)	d	4.2	base white sweet wine
Rothe (1991)		2.4	neutral margarine mass
Gassenmeier & Schieberle (1994)		0.12	sunflower oil
Preininger & Grosch (1994)		0.400	sunflower oil
Kubickova & Grosch (1998)		0.4	sunflower oil
Karagül-Yüceer *et al.* (2004)	d	0.603	skim milk
Schlutt *et al.* (2007)	r	0.077	oil/water emulsion
Leksrisompong (2008);			
Leksrisompong *et al.* (2010)	d	0.0435	water with 1 % caseinate, pH 7.0
Leksrisompong (2008);			
Leksrisompong *et al.* (2010)	d	0.0358	water with 1 % caseinate, pH 5.5
Leksrisompong (2008);			
Leksrisompong *et al.* (2010)	d	0.546	emulsion with 10 % fat, pH 7.0
Leksrisompong (2008);			
Leksrisompong *et al.* (2010)	d	0.294	emulsion with 10 % fat, pH 5.5
Leksrisompong (2008);			
Leksrisompong *et al.* (2010)	d	0.113	emulsion with 20 % fat, pH 7.0
Leksrisompong (2008);			

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Leksrisompong *et al.* (2010)	d	0.329	emulsion with 20 % fat, pH 5.5
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	1.55	oil

5-HYDROXY-2-DECENOIC ACID LACTONE (δ-decenolactone) *[51154-96-2]*

Frauendorfer & Schieberle (2006)	d	0.59	sunflower oil

3-HYDROXY-4,5-DIMETHYL-2(5*H*)-FURANONE (sotolon) *[28664-35-9]*

Martin (1989); Martin *et al.* (1992)	d	0.015	white wine
Guth & Grosch (1993)		0.0013	cellulose
Czerny (1996); Czerny & Grosch (2000)	d	0.0021	cellulose
Wagner & Grosch (1998)		0.000 2	sunflower oil
Utsunomiya *et al.* (2004a)	d	0.0023	base sake
Utsunomiya *et al.* (2004a)	r	0.0031	base sake
Moreno *et al.* (2005); Moyano *et al.* (2010)		0.005	14 % ethanolic solution
Campo *et al.* (2006)		0.009	10% water/ethanol, 5 g/l tartaric acid/pH 3.2
Callejón *et al.* (2008)	d	0.016	7 % acetic acid solution

4-HYDROXY-2,5-DIMETHYL-3(2*H*)-FURANONE (furaneol) *[3658-77-3]*

Wilson *et al.* (1990)	d	173	mango puree
Schieberle (1992)	r	0.5 - 1	pale lager beer
Schieberle (1993,1996); Gassenmeier & Schieberle (1994); Preininger & Grosch (1994)	d	0.025 - 0.05	sunflower oil
Guth & Grosch (1993)		0.075	cellulose
Buttery *et al.* (1995)	d	0.031	1 % NaH_2PO_4 (pH 4.5)
Buttery *et al.* (1995)	d	0.021	1 % NaH_2PO_4 and 0.5 % phosphoric acid (pH 3)
Czerny (1996)		0.032	cellulose
Rychlik & Grosch (1996)	d	0.013	starch
Buettner *et al.* (1998)		0.18	0.6 mg/l 2,3-butanedione (=200 OAV) in water
Buettner *et al.* (1998)		0.56	0.05 mg/l (*Z*)-3-hexenal (=200 OAV) in water
Ong & Acree (1998)		0.025	artificial saliva and fruit juice matrix
Ferreira *et al.* (2002)		0.005	10 % hydroalcoholic solution, pH 3.2
Karagül-Yüceer *et al.* (2004)	d	0.056	skim milk
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.0908	water with 1 % caseinate, pH 7.0
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.0464	water with 1 % caseinate, pH 5.5
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.0564	emulsion with 10 % fat, pH 7.0
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.0678	emulsion with 10 % fat, pH 5.5
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.0669	emulsion with 20 % fat, pH 7.0
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.134	emulsion with 20 % fat, pH 5.5
Leksrisompong (2008); Leksrisompong *et al.* (2010)	d	0.0274	oil
Averbeck & Schierberle (2009, 2011)	d	0.021	citrate buffer (pH 3.8)
Chetschik *et al.* (2010)		0.0016	sunflower oil

HYDROXYDITHIOACETIC ACID (dithioglycolic acid)

Brenner *et al.* (1954)		0.5	beer

4-HYDROXYDODECANOIC ACID LACTONE (γ-dodecalactone, 4-dodecanolide) *[2305-05-7]*

Schlutt *et al.* (2007)	r	0.063	oil/water emulsion

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

(*R*)-4-HYDROXYDODECANOIC ACID LACTONE ((*R*)-γ-dodecalactone)

Brown (2007); Cooke *et al.* (2009)	d	0.008	red wine

(*S*)-4-HYDROXYDODECANOIC ACID LACTONE ((*S*)-γ-dodecalactone)

Brown (2007); Cooke *et al.* (2009)	d	0.039 - 0.041	red wine

5-HYDROXYDODECANOIC ACID LACTONE (δ-dodecalactone, 5-dodecanolide) *[713-95-1]*

Rothe (1991)		72	neutral margarine mass
Schieberle *et al.* (1993)		0.12	sunflower oil
Karagül-Yüceer *et al.* (2004)	d	0.933	skim milk
Schlutt *et al.* (2007)	r	0.15	oil/water emulsion

(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE ((*Z*)-6-dodeceno-γ-lactone) *[18679-18-0]*

Schieberle *et al.* (1993)		0.25	sunflower oil
Guth (1996)		0.000 1	ethanol/water

4-HYDROXYEICOSANOIC ACID LACTONE (γ-eicosalactone, 4-eicosanolide)

Schlutt *et al.* (2007)	r	> 3,100	oil/water emulsion

5-HYDROXYEICOSANOIC ACID LACTONE (δ-eicosalactone, 5-eicosanolide)

Schlutt *et al.* (2007)	r	> 3,100	oil/water emulsion

5-(2-HYDROXYETHYL)-4-METHYLTHIAZOLE (sulfurol) *[137-00-8]*

Rapp *et al.* (1985)		0.1 - 1	15 vol. % ethanol

(*R*)-(+)-4-HYDROXYHEPTANOIC ACID LACTONE (*d*-γ-heptalactone)

Yamamoto (1999)		0.7	500 ppm ethanol in water?

(*S*)-(-)-4-HYDROXYHEPTANOIC ACID LACTONE (*l*-γ-heptalactone)

Yamamoto (1999)		0.89	500 ppm ethanol in water?

4-HYDROXYHEXADECANOIC ACID LACTONE (γ-hexadecalactone, 4-hexadecanolide) *[730-46-1]*

Schlutt *et al.* (2007)	r	89	oil/water emulsion

5-HYDROXYHEXADECANOIC ACID LACTONE (δ-hexadecalactone, 5-hexadecanolide) *[7370-44-7]*

Schlutt *et al.* (2007)	r	43	oil/water emulsion

16-HYDROXYHEXADECANOIC ACID LACTONE (ω-hexadecalactone, hexadecanolide) *[109-29-5]*

Jeffcoat & Willis (1986)		100	diethyl phthalate

16-HYDROXY-7-HEXADECENOIC ACID LACTONE (ambrettolide) *[123-69-3]*

Harder (1975)		0.1	diethyl phthalate

4-HYDROXYHEXANOIC ACID LACTONE (γ-hexalactone, γ-caprolactone, 4-hexanolide) *[695-06-7]*

Escudero *et al.* (2004)		16	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Ito & Kubota (2005)	r	12.5 - 200	jasmine tea infusion

4-HYDROXY(2-HYDROXY-3,3-DIMETHYL)BUTANOIC ACID LACTONE (pantolactone) *[599-04-2]*

Moyano *et al.* (2002)		2.2	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007)		500	14 % ethanolic solution

4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE (α-terpineol, α-terpinol) *[98-55-5]*

Terrier *et al.* (1972);			
Ribéreau-Gayon *et al.* (1975)	d	0.46	water and sugar
Bielig *et al.* (1974)		10	orange juice

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Harder (1975)		1,000	diethyl phthalate
Pokorný *et al.* (1978)		26	orange beverage
Ferreira *et al.* (1998)		0.25	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Zea *et al.* (2001); Moyano *et al.* (2002)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
Plotto *et al.* (2004)		16.6 - 25.9	deodorized orange juice matrix
Siegmund & Pöllinger-Zierler (2006)	d	0.314	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.483	apple juice
Elss *et al.* (2007)		3 - 7	10.5 g sucrose + 1.5 g citric acid.H_2O/L tap water
Elss *et al.* (2007)		5 - 11	orange juice substitute without aroma
Elss *et al.* (2007)		3 - 11	apple juice substitute without aroma
Elss *et al.* (2007)		3 - 7	commercial apple juice

(*R*)-(+)-4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE ((4*R*)-(+)-α-terpineol) *[7785-53-7]*

Averbeck & Schierberle (2011)	d	6.8	freshly reconstituted orange juice

4-HYDROXY-3-METHOXYBENZALDEHYDE (vanillin, 4-formyl-2-methoxyphenol) *[121-33-5]*

Proetz (1924)		0.41	liq. petrolatum
Rosenthal (1927)		18	diethyl phthalate
Engen (1960)		10-39 10^3	benzyl benzoate
Herrmann & Abd El Salam (1980a)		0.037	paraffin oil
Herrmann & Abd El Salam (1980b)		1	sunflower oil
Lehtonen & Jounela-Eriksson (1983)		2.2	Scotch whisky
Lehtonen & Jounela-Eriksson (1983)		4.5	Canadian whisky
Lehtonen & Jounela-Eriksson (1983)		14	Bourbon whiskey
Kleinschmidt (1983)	r	1	paraffin oil
Perry (1989)		0.1	23 % v. ethanol
Rychlik & Grosch (1996)	d	0.0046	starch
Ong & Acree (1998)		1.2	artificial saliva and fruit juice matrix
Pfnür (1998c)		0.181	sunflower oil
Lee *et al.* (2000)	d	0.6	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	3	3 year old grain whisky: 23 % ethanol
Mayer (2000)		0.1	cellulose
Culleré *et al.* (2004); Gómez-Míguez *et al.* (2007);			
Culleré *et al.* (2008)		0.06	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2
Karagül-Yüceer *et al.* (2004)	d	7.413	skim milk
Plotto *et al.* (2004)		0.74 - 0.764	deodorized orange juice matrix
Campo *et al.* (2006); Escudero *et al.* (2007)		0.995	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2
Poisson & Schieberle (2008)		0.022	water/ethanol 6:4 by vol.
Tesfaye *et al.* (2008)		0.094	7 % acetic acid solution

4-HYDROXY-3-METHOXYBENZOIC ACID (vanillic acid) *[121-34-6]*

Lehtonen & Jounela-Eriksson (1983)		1,300	Scotch whisky

1-(4-HYDROXY-3-METHOXYPHENYL)ETHANONE (acetovanillon) *[498-02-2]*

Lopez *et al.* (2002)		1	10 % hydroalcoholic solution at pH 3.2
Culleré *et al.* (2004);Gómez-Míguez *et al.* (2007);			
Campo *et al.* (2006); Escudero *et al.* (2007)		1	10 % ethanol/water, 5 g/l tartaric acid, pH 3.2

2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE (3-methyl-1,2-cyclopentanedione, cyclotene, methylcyclopentenolone, maple lactone) *[80-71-7, 765-70-8]*

Yoshida (1972)	d	2.8	nujiol (a mineral oil)
Yoshida (1972)	r	100	nujiol (a mineral oil)
Yoshida (1984);Takagi (1989)	d	10	nujol or propylene glycol
Yoshida (1984);Takagi (1989)	r	10	nujol or propylene glycol

4-HYDROXY-5-METHYL-3(2*H*)-FURANONE (norfuraneol) *[19322-27-1]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Buttery *et al.* (1995)	d	2.1	1 % NaH_2PO_4 (pH 4.5)
Buttery *et al.* (1995)	d	2.5	1 % NaH_2PO_4 and 0.5 % phosphoric acid (pH 3)

5-(HYDROXYMETHYL)FURFURAL (oxymethylfurfural) *[67-47-0]*

Saison *et al.* (2009)	d	35.784	carbonated beer served at 7-8 °C

4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE (3-methyl-4-hydroxyoctanoic acid lactone, β-methyl-γ-octalactone, oak lactone, whisky lactone) *[39212-23-2]*

Piggott *et al.* (1995)		0.075	12 % ethanol
Piggott *et al.* (1995)		0.241	white wine
Piggott *et al.* (1995)		0.853	red wine
Lee *et al.* (2000)	d	0.3-0.5	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	1-7	3 year old grain whisky: 23 % ethanol
Callejón *et al.* (2008)	d	0.078	7 % acetic acid solution

cis-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE (*cis*-β-methyl-γ-octalactone, Z-whisky lactone, Z-oak lactone) *[55013-32-6]*

Otsuka *et al.* (1974)		0.79	30 % ethanol
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		0.035	14 % ethanolic solution

cis-(4*R*,5*R*)-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

Wilkinson *et al.* (2004)	d	0.082	neutral white wine
Brown *et al.* (2006a,b); Brown (2007)	d	0.132	white wine
Brown *et al.* (2006a,b); Brown (2007)	d	0.170	red wine

cis-(4*S*,5*S*)-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

Wilkinson *et al.* (2004)	d	0.023	neutral white wine
Wilkinson *et al.* (2004)	d	0.046	red wine
Brown *et al.* (2006a,b); Brown (2007)	d	0.020	white wine
Brown *et al.* (2006a,b); Brown (2007)	d	0.054	red wine

trans-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE (*trans*-β-methyl-γ-octalactone, E-oak lactone) *[39638-67-0]*

Salo *et al.* (1972,1976)	d	0.05	9.4 % (w/w) grain spirit
Otsuka *et al.* (1974)		0.067	30 % ethanol
Chaves *et al.* (2007)		0.122	14 % v/v ethanol/water

trans-(4*R*,5*S*)-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

Brown *et al.* (2006a,b); Brown (2007)	d	0.330	white wine
Brown *et al.* (2006a,b); Brown (2007)	d	0.305	red wine

trans-(4*S*,5*R*)-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

Brown *et al.* (2006a,b); Brown (2007)	d	0.140	white wine
Brown *et al.* (2006a,b); Brown (2007)	d	0.370	red wine

4-HYDROXY-4-METHYLPENTAN-2-ONE (diacetone alcohol) *[123-42-2]*

Engen (1960)		39-156 10^3	benzyl benzoate

3-HYDROXY-2-METHYL-4-PYRONE (maltol) *[118-71-8]*

Harder (1975)		0.000 1	diethyl phthalate
Lee *et al.* (2000)	d	8-21	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	20-36	3 year old grain whisky: 23 % ethanol
Culleré *et al.* (2004)		5	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Karagül-Yüceer *et al.* (2004)	d	16.596	skim milk

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

4-HYDROXYNONANOIC ACID LACTONE (γ-nonalactone, 4-nonanolide, prunolide) *[104-61-0]*

Etiévant *et al.* (1983)	d	0.46	base white sweet wine
Knupfer & Spiegel (1986)	d	100	propylene glycol
Nakamura *et al.* (1988)	d	0.030	basic white wine
Rothe (1991)		3.4	neutral margarine mass
Ong & Acree (1998)		0.002	artificial saliva and fruit juice matrix
Wagner & Grosch (1998)		0.148	sunflower oil
Poisson & Schieberle (2008)		0.021	water/ethanol 6:4 by vol.
Saison *et al.* (2009)	d	0.607	carbonated beer served at 7-8 °C

(*R*)-(+)-4-HYDROXYNONANOIC ACID LACTONE ((*R*)-γ-nonalactone)

Yamamoto (1999)		0.22	500 ppm ethanol in water?
Brown (2007); Cooke *et al.* (2009)	d	0.284 - 0.285	red wine

(*S*)-(-)-4-HYDROXYNONANOIC ACID LACTONE ((*S*)-γ-nonalactone)

Yamamoto (1999)		0.15	500 ppm ethanol in water?
Brown (2007); Cooke *et al.* (2009)	d	0.091	red wine

5-HYDROXYNONANOIC ACID LACTONE (δ-nonalactone, 5-nonanolide) *[3301-94-8]*

Etiévant *et al.* (1983)	d	2.6	base white sweet wine

4-HYDROXYOCTADECANOIC ACID LACTONE (γ-octadecalactone, 4-octadecanolide) *[502-26-1]*

Schlutt *et al.* (2007)	r	217	oil/water emulsion

5-HYDROXYOCTADECANOIC ACID LACTONE (δ-octadecalactone, 5-octadecanolide) *[1227-51-6]*

Schlutt *et al.* (2007)	r	230	oil/water emulsion

4-HYDROXYOCTANOIC ACID LACTONE (γ-octalactone, 4-octanolide) *[104-50-7]*

Williams & Ismail (1981)		0.16	sugar-acid base
Wagner & Grosch (1998)		0.12	sunflower oil

(*R*)-(+)-4-HYDROXYOCTANOIC ACID LACTONE ((*R*)-γ-octalactone)

Yamamoto (1999)		0.3	500 ppm ethanol in water?
Brown (2007); Cooke *et al.* (2009)	d	0.238	red wine

(*S*)-(-)-4-HYDROXYOCTANOIC ACID LACTONE ((*S*)-γ-octalactone)

Yamamoto (1999)		0.13	500 ppm ethanol in water?
Brown (2007); Cooke *et al.* (2009)	d	0.135	red wine

5-HYDROXYOCTANOIC ACID LACTONE (δ-octalactone, 5-octanolide) *[698-76-0]*

Etiévant *et al.* (1983)	d	2.4	base white sweet wine
Rothe (1991)		1.1	neutral margarine mass
Ferreira *et al.* (2006)		0.71	ethanol 10% v/v, tartaric acid 2 g/l, pH 3.4
Schmitt & Schieberle (2006)	d	2.49	sunflower oil

5-HYDROXY-2-OCTENOIC ACID LACTONE (δ-octenolactone)

Schmitt & Schieberle (2006)	d	4.73	sunflower oil

15-HYDROXYPENTADECANOIC ACID LACTONE (exaltolide, ω-pentadecalactone, cyclopentadecanolide, thibetolide, muskolactone) *[106-02-5]*

Matsushita *et al.* (1967)		270	liquid paraffin
Yoshida (1972)	d	640	nujiol (a mineral oil)
Yoshida (1972)	r	10,000	nujiol (a mineral oil)
Koelega & Köster (1974)		8,750-87,000	benzyl benzoate
D'Andrea (1975)	d	50 - 920	diethyl phthalate
Kalmus & Seedburgh (1975)		49	paraffin oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Mair *et al.* (1978)		< 385	dibutyl phthalate
Rosen *et al.* (1979)	d	240	mineral oil
Yoshida (1984);Takagi (1989)	d	1,000	nujol
Yoshida (1984);Takagi (1989)	r	1,000	nujol
Jeffcoat & Willis (1986)	d	28,800	diethyl phthalate

α-hydroxytoluene⇒BENZYL ALCOHOL

4-HYDROXYTETRADECANOIC ACID LACTONE (γ-tetradecalactone, 4-tetradecanolide) *[2721-23-5]*

Schlutt *et al.* (2007)	r	12	oil/water emulsion

5-HYDROXYTETRADECANOIC ACID LACTONE (δ-tetradecalactone, 5-tetradecanolide) *[2721-22-4]*

Schlutt *et al.* (2007)	r	29	oil/water emulsion

4-HYDROXYUNDECANOIC ACID LACTONE (γ-undecalactone) *[104-67-6]*

Yoshida (1972)	d	0.006	nujiol (a mineral oil)
Yoshida (1972)	r	100	nujiol (a mineral oil)
Yoshida (1984);Takagi (1989)	d	10	nujol or propylene glycol
Yoshida (1984);Takagi (1989)	r	10	nujol or propylene glycol
Rothe (1991)		9.4	neutral margarine mass

(*R*)-(+)-4-HYDROXYUNDECANOIC ACID LACTONE (*d*-γ-undecalactone)

Yamamoto (1999)		0.004	500 ppm ethanol in water?

(*S*)-(-)-4-HYDROXYUNDECANOIC ACID LACTONE (*l*-γ-undecalactone)

Yamamoto (1999)		0.012	500 ppm ethanol in water?

5-HYDROXYUNDECANOIC ACID LACTONE (δ-undecalactone) *[710-04-3]*

Rothe (1991)		19	neutral margarine mass

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INDOLE *[120-72-9]*

Rosenthal (1927)		9	diethyl phthalate
Thomson (1984)		0.33	aqueous ethanol
Coulter *et al.* (2008)	d	0.023	neutral, dry white wine

iodoform⇒TRIIODOMETHANE
α-ionone⇒4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
β-ionone⇒4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
isoamyl acetate⇒ISOPENTYL ACETATE
isoamyl alcohol⇒3-METHYL-1-BUTANOL
isoamyl butyrate⇒ISOPENTYL BUTANOATE
isoamyl caprate⇒ISOPENTYL DECANOATE
isoamyl caproate⇒ISOPENTYL HEXANOATE
isoamyl caprylate⇒ISOPENTYL OCTANOATE
isoamyl isobutyrate⇒ISOPENTYL 2-METHYLPROPANOATE
isoamyl isovalerate⇒ISOPENTYL 3-METHYLBUTANOATE
isoamyl laurate⇒ISOPENTYL DODECANOATE
isoamyl valerate⇒ISOPENTYL PENTANOATE
isoborneol⇒*exo*-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL
isobutanol⇒2-METHYL-1-PROPANOL

ISOBUTYL ACETATE *[110-19-0]*

Salo *et al.* (1972)	d	3.40	9.4 % (w/w) grain spirit
Ong & Acree (1998)		0.01	artificial saliva and fruit juice matrix
Ferreira *et al.* (2002); Escudero *et al.* (2007)		1.605	10 % hydroalcoholic solution, pH 3.2

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Callejón *et al.* (2008)	d	0.177	7 % acetic acid solution
Pineau *et al.* (2009)	d	2.1	dearomatized red wine

4'-ISOBUTYLACETOPHENONE *[388861-78-8]*

Cerveny *et al.* (1993)		39.9	paraffin oil

isobutyl alcohol⇒2-METHYL-1-PROPANOL
isobutyl butyrate⇒ISOBUTYL BUTANOATE

ISOBUTYL BUTANOATE (isobutyl butyrate, 2-methylpropyl butanoate) *[539-90-2]*

Morales *et al.* (2005)	d	0.1	deodorised olive oil

isobutyl isobutyrate⇒ISOBUTYL 2-METHYLPROPANOATE

5-ISOBUTYL-2-ISOPROPOXY-3-METHYLPYRAZINE

Masuda *et al.* (1981)		1	aqueous ethanol solution

ISOBUTYL 2-HYDROXYPROPANOATE (isobutyl lactate) *[585-24-0]*

Zea *et al.* (2001); Moyano *et al.* (2002)		100	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007)		340	14 % ethanolic solution

isobutyl lactate⇒ISOBUTYL 2-HYDROXYPROPANOATE

5-ISOBUTYL-2-METHOXY-3-METHYLPYRAZINE *[78246-20-5]*

Masuda *et al.* (1981)		0.7	aqueous ethanol solution

2-ISOBUTYL-3-METHOXYPYRAZINE (pepper pyrazine) *[24683-00-9]*

Maga (1990)	r	0.000 002	model wine
Maga (1990)	r	0.000 016	red wine
Allen *et al.* (1991)	d	0.000 001	Sauvignon blanc wine
Czerny (1996); Czerny & Grosch (2000)	d	2 - 8 10^{-3}	cellulose
Wagner & Grosch (1998)		0.000 8	sunflower oil
Stephan & Steinhart (1999)	r	0.0103	refined vegetable oil
Siegmund & Pöllinger-Zierler (2006)	d	0.000 000 6	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.000 0033	apple juice
Pineau *et al.* (2007)	d	0.000 0055	12 % ethanoll/water, 4 g/l tartaric aicd, pH 3.5
Pineau *et al.* (2007)	d	0.000 0065	12 % ethanoll/water, 4 g/l tartaric aicd, pH 3.5 + 50 ng/L β-damascenone
Aceña *et al.* (2011)	d	0.000 022	7% (w/v) acetic acid water solution

5-ISOBUTYL-3-METHYL-2-(METHYLTHIO)PYRAZINE *[78246-17-0]*

Masuda *et al.* (1981)		0.8	aqueous ethanol solution

5-ISOBUTYL-3-METHYL-2-PHENOXYPYRAZINE *[78246-16-9]*

Masuda *et al.* (1981)		1	aqueous ethanol solution

5-ISOBUTYL-3-METHYL-2-(PHENYLTHIO)PYRAZINE *[78246-19-2]*

Masuda *et al.* (1981)		0.7	aqueous ethanol solution

ISOBUTYL 2-METHYLPROPANOATE (isobutyl isobutyrate) *[97-85-8]*

Zea *et al.* (2001)		1.0	14 % (v/v) ethanol ; tartaric acid to pH 3.5

(4*R*)-*cis*-2-ISOBUTYL-4-METHYLTETRAHYDROPYRAN ((4*R*)-*cis*-dihydrorose oxide)

Matsuda et al. (1995); Matsuda & Yamamoto (1999)	d	0.017	aqueous-ethanol solution

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

(4*R*)-*trans*-2-ISOBUTYL-4-METHYLTETRAHYDROPYRAN ((4*R*)-*trans*-dihydrorose oxide)

Matsuda *et al.* (1995);			
Matsuda & Yamamoto (1999)	d	0.150	aqueous-ethanol solution

(4*S*)-*cis*-2-ISOBUTYL-4-METHYLTETRAHYDROPYRAN ((4*S*)-*cis*-dihydrorose oxide)

Matsuda *et al.* (1995);			
Matsuda & Yamamoto (1999)	d	0.450	aqueous-ethanol solution

(4*S*)-*trans*-2-ISOBUTYL-4-METHYLTETRAHYDROPYRAN ((4*S*)-*trans*-dihydrorose oxide)

Matsuda *et al.* (1995);			
Matsuda & Yamamoto (1999)	d	0.160	aqueous-ethanol solution

1-(4-ISOBUTYLPHENYL)ETHANOL

Cerveny *et al.* (1993)		125.9	paraffin oil

1-(4-ISOBUTYLPHENYL)ETHYL ACETATE

Cerveny *et al.* (1993)		75.9	paraffin oil

1-(4-ISOBUTYLPHENYL)ETHYL PROPANOATE

Cerveny *et al.* (1993)		316.3	paraffin oil

ISOBUTYL PROPANOATE (isobutyl propionate) *[540-42-1]*

Pineau *et al.* (2009)	d	3.9	dearomatized red wine

isobutyl propionate⇒ISOBUTYL PROPANOATE

2-ISOBUTYLTHIAZOLE *[18640-74-9]*

Tandon *et al.* (2001)	d	0.098	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	0.102	deodorized tomato homogenate

isobutyraldehyde⇒2-METHYLPROPANAL
isobutyric acid⇒METHYLPROPANOIC ACID
isocaproic acid⇒4-METHYLPENTANOIC ACID
isoeugenol⇒2-METHOXY-4-(1-PROPENYL)PHENOL
(*E*)-isoeugenol⇒(*E*)-2-METHOXY-4-(1-PROPENYL)PHENOL

ISOPENTYL ACETATE (isoamyl acetate, 3-methylbutyl acetate) *[123-92-2]*

Williams (1953)		2	ethanol solution
Furchtgott & Friedman (1960)		55-60	ethylene glycol
Furchtgott & Friedman (1960)		4.3-7.5	ethylene glycol
Kimbrell & Furchtgott (1963)	d	1,300-10,400	ethylene glycol
Sega *et al.* (1967)	d	2.3	lager beer
Salo (1970a,1970b); Salo *et al.* (1972)	d	0.20/0.23	9.4 % (w/w) grain spirit
De Wet (1978)	d	0.16	wine residue medium
Gross-Isseroff *et al.* (1987)	d	330 - 1,040	diethyl phthalate
Gross-Isseroff & Lancet (1988)	d	0.000 1-100	diethyl phthalate
Perry (1989)		1	23 % v. ethanol
Gross-Isseroff *et al.* (1992)	d	1,000	diethyl phthalate
Gross-Isseroff *et al.* (1994)	d	330 - 1,040	diethyl phthalate
Griep *et al.* (1996)		0.14-0.85	diethyl phthalate
Griep *et al.* (1997a)		8.9-195	diethyl phthalate
Griep *et al.* (1997b)		1.3-130	diethyl phthalate
Lee *et al.* (2000)	d	0.1-0.3	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	1-2	3 year old grain whisky: 23 % ethanol
Utsunomiya *et al.* (2004b)	d	0.27	base sake
Utsunomiya *et al.* (2004a)	r	0.52	base sake

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Grossmann *et al.* (2005)	d	0.0096	mineral oil
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		0.03	14 % ethanolic solution
Callejón *et al.* (2008)	d	0.012	7 % acetic acid solution
Poisson & Schieberle (2008)		0.245	water/ethanol 6:4 by vol.
Pineau *et al.* (2009)	d	0.86	dearomatized red wine
Aceña *et al.* (2011)	d	0.0222	7 % (w/v) acetic acid water volume

ISOPENTYL BUTANOATE (isoamyl butyrate) *[106-27-4]*

Vickers *et al.* (2001)		7.8-16	deodorized mineral oil

ISOPENTYL DECANOATE (isoamyl caprate) *[2306-91-4]*

Salo *et al.* (1972)	d	> 5.0	9.4 % (w/w) grain spirit

ISOPENTYL DODECANOATE (isoamyl laurate) *[6309-51-9]*

Zea *et al.* (2001)		100	14 % (v/v) ethanol; tartaric acid to pH 3.5

ISOPENTYL HEXANOATE (isoamyl caproate) *[2198-61-0]*

Salo *et al.* (1972)	d	1.40	9.4 % (w/w) grain spirit

5-ISOPENTYL-2-METHOXY-3-METHYLPYRAZINE

Masuda *et al.* (1981)		1	aqueous ethanol solution

ISOPENTYL 3-METHYLBUTANOATE (isoamyl isovalerate) *[659-70-1]*

Salo *et al.* (1972)	d	> 1.0	9.4 % (w/w) grain spirit

ISOPENTYL 2-METHYLPROPANOATE (isoamyl isobutyrate) *[2050-01-3]*

Salo *et al.* (1972)	d	0.25	9.4 % (w/w) grain spirit

ISOPENTYL OCTANOATE (isoamyl caprylate) *[2035-99-6]*

Salo *et al.* (1972)	d	0.60	9.4 % (w/w) grain spirit
Escudero *et al.* (2004)		0.125	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2

ISOPENTYL PENTANOATE (isoamyl valerate) *[2050-09-1]*

Salo *et al.* (1972)	d	> 0.1	9.4 % (w/w) grain spirit

isophorone⇒3,5,5-TRIMETHYL-2-CYCLOHEXEN-1-ONE
isopropanol⇒2-PROPANOL

8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE (nootkatone) *[4674-50-4]*

MacLeod & Buigues (1964)	d	< 10	32 Baume aqueous sucrose

4-ISOPROPENYL-1-METHYL-1-CYCLOHEXANOL (β-terpineol) *[138-87-4]*

Pokorný *et al.* (1978)		6	orange beverage

(+)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANOL ((+)-dihydrocarveol)

Laska (2004)		74 - 370	diethyl phthalate

(-)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANOL ((-)-dihydrocarveol)

Laska (2004)		15 - 74	diethyl phthalate

(+)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE ((+)-dihydrocarvone)

Laska (2004)		15 - 74	diethyl phthalate

(-)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXANONE ((-)-dihydrocarvone)

Laska (2004)		15 - 74	diethyl phthalate

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE (limonene) *[138-86-3]*

Plotto *et al.* (2004)		8.5 - 13.7	deodorized orange juice matrix
Elss *et al.* (2007)		0.4 - 0.7	10.5 g sucrose + 1.5 g citric acid.H_2O/L tap water
Elss *et al.* (2007)		0.7	apple juice substitute without aroma
García-González *et al.* (2008)	d	0.25	deodorised edible oil
Buhr *et al.* (2010)	d	6.5	whole milk
Buhr *et al.* (2010)	d	14.7	sunflower oil

(*S*)-(-)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE (*S*-(-)-limonene) *[5989-54-8]*

Jagella & Grosch (1998,1999a)	d	2.8	starch
Laska & Teubner (1999)	d	1.6-8	diethyl phthalate

(*R*)-(+)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE (*R*-(+)-limonene) *[5989-27-5]*

Jagella & Grosch (1998,1999a)	d	1.8	starch
Laska & Teubner (1999)	d	1.6-8	diethyl phthalate

5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE (carvone) *[99-49-0]*

Elss *et al.* (2007)		0.17	10.5 g sucrose + 1.5 g citric acid.H_2O/L tap water
Elss *et al.* (2007)		0.4 - 0.6	orange juice substitute without aroma

(*R*)-(-)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE ((*l*)-carvone, *R*-(-)-carvone) *[6485-40-1]*

Gross-Isseroff & Lancet (1988)	d	2,820	diethyl phthalate
Lawless *et al.* (1995)	d	0.005	buffered spring water
Laska & Teubner (1999)	d	0.32-8	diethyl phthalate

(*S*)-(+)-5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE ((*d*)-carvone, *S*-(+)-carvone) *[2244-16-8]*

Laska & Teubner (1999)	d	1.6-40	diethyl phthalate

(+)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXYL ACETATE ((+)-dihydrocarvyl acetate)

Laska (2004)		76 - 380	diethyl phthalate

(-)-5-ISOPROPENYL-2-METHYL-1-CYCLOHEXYL ACETATE ((-)-dihydrocarvyl acetate)

Laska (2004)		15 - 76diethyl phthalate	

ISOPROPYL ACETATE *[108-21-4]*

Granzer *et al.* (1986)		1 - 20	coconut oil
Granzer *et al.* (1986)		2 - 50	liquid cheese
Rüter (1992)	d	13	potato crisps
Rüter (1992)	r	33	potato crisps
Rüter (1992)	d	40	coffee powder
Rüter (1992)	r	92	coffee powder
Rüter (1992)	d	74	'Gummibären'
Rüter (1992)	r	330	'Gummibären'
Rüter (1992)	d	18	milk chocolate
Rüter (1992)	r	43	milk chocolate

isopropyl alcohol⇒2-PROPANOL

4-ISOPROPYLBENZALDEHYDE (cuminaldehyde) *[122-03-2]*

Rosenthal (1927)		177	diethyl phthalate

ISOPROPYL 2-ISOPROPYL-3-THIAZOLINE-4-CARBOXYLATE

Fernandez *et al.* (2002)	d	< 10	ethanol

2-ISOPROPYL-3-METHOXYPYRAZINE *[25773-40-4]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Harder (1975)		0.000 01	diethyl phthalate
Maga (1990)	r	0.000 002	model wine
Maga (1990)	r	0.000 002	red wine
Allen *et al.* (1991)	d	0.000 002	Sauvignon blanc wine
Pfnür (1998b,c)		0.000 07(5)	(sunflower) oil
Stephan & Steinhart (1999)	r	0.0031	refined vegetable oil
Czerny & Grosch (2000); Zimmermann (2001)	d	0.000 1	cellulose
Siegmund & Pöllinger-Zierler (2006)	d	0.000 000 3	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.000 000 6	apple juice
Pickering *et al.* (2007)	d	0.000 000 318	Chardonnay wine
Pickering *et al.* (2007)	d	0.000 001563	Gewürztraminer wine
Pickering *et al.* (2007)	d	0.000 001031	red wine blend
Neta *et al.* (2008)	d	0.000 0023	whole milk
Pickering *et al.* (2008)	d	0.000 001108	Concord grape juice
Pickering *et al.* (2008)	d	0.000 000 738	Niagara grape juice
Aceña *et al.* (2011)	d	0.000 01	7% (w/v) acetic acid water solution

1-ISOPROPYL-4-METHYLBENZENE (*p*-cymene) *[99-87-6]*

Rosenthal (1927)		18	diethyl phthalate
Pokorný *et al.* (1978)		22	orange beverage
Moreno *et al.* (2005)		66	14 % ethanolic solution

4-ISOPROPYL-1-METHYL-1,4-CYCLOHEXADIENE (γ-terpinene) *[99-85-4]*

Plotto *et al.* (2004)		2.39 - 3.26	deodorized orange juice matrix

(*R*)-(-)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE (*R*-(-)-α-phellandrene) *[4221-98-1]*

Jagella & Grosch (1998,1999a)	d	1.4	starch

(*S*)-(+)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE (*S*-(+)-α-phellandrene) *[2243-33-6]*

Jagella & Grosch (1998,1999a)	d	1.05 - 1.1	starch

(1*R*,2*S*,5*R*)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL ((*l*)-menthol) *[2216-51-5]*

Eccles *et al.* (1989)	d	92.6	liquid paraffin
Rosenblatt *et al.* (1998)		180-220	mineral oil

2-ISOPROPYL-5-METHYLCYCLOHEXANONE (menthone) *[10458-14-7]*

Rosen *et al.* (1979)	d	0.15	mineral oil

1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL (1-terpinen-4-ol) *[562-74-3]*

Zea *et al.* (2001)		5	14 % (v/v) ethanol; tartaric acid to pH 3.5

(1*S*)-(-)-1-ISOPROPYL-4-METHYLENEBICYCLO[3.1.0]HEXANE ((-)-sabinene) *[10408-16-9]*

Jagella & Grosch (1999a)	d	50	starch

(1*R*)-(+)-1-ISOPROPYL-4-METHYLENEBICYCLO[3.1.0]HEXANE ((+)-sabinene) *[2009-00-9]*

Jagella & Grosch (1999a)	d	6.3	starch

1-ISOPROPYL-4-METHYL-7-OXABICYCLO[2.2.1]HEPTANE (1,4-cineole) *[470-67-7]*

Bielig *et al.* (1974)		< 1	orange juice

5-ISOPROPYL-2-METHYLPHENOL (carvacrol) *[499-75-2]*

Bitar *et al.* (2008)	d	30.97	sunflower oil

2-ISOPROPYL-5-METHYLPHENOL (thymol) *[89-83-8]*

Bitar *et al.* (2008)	d	124	sunflower oil

3-isopropyl-6-methylpyrocatechol ⇒2,3-DIHYDROXY-4-ISOPROPYL-1-METHYLBENZENE

2-ISOPROPYLTHIAZOLIDINE *[24050-11-1]*

Fernandez *et al.* (2002)	d	< 10	ethanol

2-ISOPROPYL-2-THIAZOLINE

Fernandez *et al.* (2002)	d	< 10	ethanol

isovaleraldehyde⇒3-METHYLBUTANAL
isovaleric acid⇒3-METHYLBUTANOIC ACID

L

lauraldehyde⇒DODECANAL
lauric acid⇒DODECANOIC ACID
leaf alcohol⇒(*Z*)-3-HEXEN-1-OL
leaf aldehyde⇒(*E*)-2-HEXENAL
lenthionine⇒1,2,3,5,6-PENTATHIEPANE
limonene⇒4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
R-(+)-limonene⇒(*R*)-(+)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
S-(-)-limonene⇒(*S*)-(-)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE

LIMONENE HYDROPEROXIDE

Pokorný *et al.* (1978)	19	orange beverage

linalool⇒3,7-DIMETHYL-1,6-OCTADIEN-3-OL
(*l*)-linalool⇒(*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL
R-(-)-linalool⇒(*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL
linalool oxide A⇒*trans*-5-ETHENYLTETRAHYDRO-α,α,5-TRIMETHYL-2-FURANMETHANOL
linalool oxide B⇒*cis*-5-ETHENYLTETRAHYDRO-α,α,5-TRIMETHYL-2-FURANMETHANOL
linalool oxide C⇒*trans*-6-ETHENYLTETRAHYDRO-2,2,6-TRIMETHYL-2*H*-PYRAN-3-OL
linalool oxide D⇒*cis*-6-ETHENYLTETRAHYDRO-2,2,6-TRIMETHYL-2*H*-PYRAN-3-OL
linalyl acetate⇒3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE
linoleic acid⇒(*Z*,*Z*)-9,12-OCTADECADIENOIC ACID

M

maltol⇒3-HYDROXY-2-METHYL-4-PYRONE
maple lactone⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE
massoia lactone⇒5,6-DIHYDRO-6-PENTYL-2*H*-PYRAN-2-ONE

MENTHOL *[89-78-1]*

Proetz (1924)		5.24	liq. petrolatum
Kleinschmidt (1983)	r	100	paraffin oil
Graham *et al.* (1995)	d	1.61	light mineral oil
Frasnelli & Hummel (2005)	r (“trigeminal”)	976 – 62,500	propylene glycol

(*l*)-menthol⇒(1*R*,3*R*,4*S*)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL
menthone⇒2-ISOPROPYL-5-METHYLCYCLOHEXANONE

2-MERCAPTOETHANOL *[60-24-2]*

Rapp *et al.* (1985)	1 - 10	15 vol. % ethanol
Boidron (1995)	0.1	water-ethanol 12 %
Boidron (1995)	0.45	white wine

3-MERCAPTO-3-METHYLBUTANAL (3-methyl-3-sulfanylbutanal)

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Bailly *et al.* (2006)		d	0.000 007	ethanol/water 12:88 (v/v)

3-MERCAPTO-2-METHYL-1-BUTANOL *[227456-33-9]*

Chopra *et al.* (2008)	[female pre-puberty]	d	0.0088	propylene glycol
Chopra *et al.* (2008)	[female puberty]	d	0.0078	propylene glycol
Chopra *et al.* (2008)	[female post-puberty]	d	0.0058	propylene glycol
Chopra *et al.* (2008)	[male pre-puberty]	d	0.020	propylene glycol
Chopra *et al.* (2008)	[male puberty]	d	0.22	propylene glycol
Chopra *et al.* (2008)	[male post-puberty]	d	0.055	propylene glycol

4-mercapto-4-methylpentan-2-one⇒2-MERCAPTO-2-METHYL-4-PENTANONE

2-MERCAPTO-2-METHYL-4-PENTANONE (4-mercapto-4-methylpentan-2-one) *[19872-52-7]*

Tressl *et al.* (1980)			0.000 05	beer
Saxby (1982)			0.000 01	bland foods

mesityl oxide⇒4-METHYL-3-PENTEN-2-ONE
methacrylic acid⇒2-METHYLPROPENOIC ACID

METHANETHIOL (methyl mercaptan) *[74-93-1]*

Brenner *et al.* (1954)			0.006 - 0.012	beer
Kubickova & Grosch (1998); Kubickova (1998)			0.000 06	sunflower oil
Lucas (2000)		d	0.005	aqueous-alcoholic distillate from herbs
Lucas (2000)		r	0.01	aqueous-alcoholic distillate from herbs
Solomon *et al.* (2010)		d	0.0031	neutral white wine
Solomon *et al.* (2010)		d	0.0018	red wine
Aceña *et al.* (2011)		d	0.000 16	7% (w/v) acetic acid water solution

methanoic acid⇒FORMIC ACID

METHANOL (methyl alcohol) *[67-56-1]*

Moreno *et al.* (2005); Chaves *et al.* (2007)			668	14 % ethanolic solution

methional⇒3-(METHYLTHIO)PROPANAL
methionol⇒3-(METHYLTHIO)-1-PROPANOL

2-METHOXYBENZALDEHYDE (*o*-anisaldehyde) *[135-02-4]*

Siegmund & Pöllinger-Zierler (2006)		d	0.174	apple juice
Siegmund & Pöllinger-Zierler (2006)		r	> 0.25	apple juice

3-METHOXYBENZALDEHYDE (*m*-anisaldehyde) *[591-31-1]*

Siegmund & Pöllinger-Zierler (2006)		d	0.181	apple juice
Siegmund & Pöllinger-Zierler (2006)		r	> 0.25	apple juice

4-METHOXYBENZALDEHYDE (*p*-anisaldehyde, anisaldehyde) *[123-11-5]*

Rosenthal (1927)			44	diethyl phthalate
Harder (1975)			1	diethyl phthalate
Lindeman *et al.* (1982)		d	0.021	10 % ethanol
Siegmund & Pöllinger-Zierler (2006)		d	0.139	apple juice
Siegmund & Pöllinger-Zierler (2006)		r	0.226	apple juice

7-METHOXYCOUMARIN *[531-59-9]*

Zeller & Rychlik (2007)		r	9.5	cellulose

2-METHOXY-3,5-DIMETHYLPYRAZINE

Czerny & Grosch (2000)			0.000 006	cellulose

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Simpson *et al.* (2004)		0.000 0021	neutral dry white wine

1-METHOXY-3-METHYL-3-BUTANETHIOL *[94087-83-9]*

Rigaud *et al.* (1986)		0.000 1	black currant liquor
Guth & Grosch (1991)		3-6 10^{-5}	odourless sunflower oil
Reiners & Grosch (1998)	d	0.000 017	refined sunflower oil

2-METHOXY-3-METHYL-5-(2-METHYLBUTYL)PYRAZINE *[78246-21-6]*

Masuda *et al.* (1981)		1	aqueous ethanol solution

2-METHOXY-3-METHYL-5-(2-METHYLPENTYL)PYRAZINE *[78246-14-0]*

Masuda *et al.* (1981)		0.5	aqueous ethanol solution

2-METHOXY-4-METHYLPHENOL (creosol, 4-methylguaiacol) *[93-51-6]*

Wasserman (1966)		0.4	mineral oil
Swan *et al.* (1981)		0.05	20 % ethanol-water
Söllner & Schieberle (2009)		1.485	odorless sunflower oil

2-METHOXY-5-METHYLPHENOL *[1195-09-1]*

Söllner & Schieberle (2009)		0.013	odorless sunflower oil

2-METHOXYNAPHTHALENE (methyl β-naphthyl ether) *[93-04-9]*

Rosenthal (1927)		4	diethyl phthalate

2-METHOXYPHENOL (guaiacol) *[90-05-1]*

Proetz (1924)		3.02	liq. petrolatum
Wasserman (1966)		0.07	mineral oil
Swan *et al.* (1981)		0.03	20 % ethanol-water
Jounela-Eriksson & Lehtonen (1981)		0.003	10 vol. % grain spirit
Perry (1989)		0.06	23 % v. ethanol
Schieberle (1993,1996)	d	0.019-0.05	sunflower oil
Mayer (1996)		0.0084	cellulose
Rychlik & Grosch (1996)	d	0.0042	starch
Pettipher *et al.* (1997)		0.002	juice
Reiners & Grosch (1998)	d	0.016	refined sunflower oil
Ferreira *et al.* (1998)		0.03	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Ong & Acree (1998)		0.002	artificial saliva and fruit juice matrix
Lee *et al.* (2000)	d	0.03-0.09	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	0.7-3	3 year old grain whisky: 23 % ethanol
Orr *et al.* (2000)	r	0.00232	apple juice
Eisele & Semon (2005)	d	0.000 91	apple juice
Morales *et al.* (2005)	d	0.02	deodorised olive oil
Siegmund & Pöllinger-Zierler (2006)	d	0.000 57	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.002	apple juice
Poisson & Schieberle (2008)		0.0092	water/ethanol 6:4 by vol.
Vichi *et al.* (2008)	d	0.01	refined sunflower oil

1-METHOXY-4-(1-PROPENYL)BENZENE (anethole) *[104-46-1]*

Kendall & Neilson (1964)		15	hexadecane
Kendall & Neilson (1964)		0.05	aqueous 0.1 % ethanol

2-METHOXY-4-(1-PROPENYL)PHENOL (isoeugenol) *[97-54-1]*

Rosenthal (1927)		59	diethyl phthalate
Culleré *et al.* (2004)		0.01	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2

(*E*)-2-METHOXY-4-(1-PROPENYL)PHENOL ((*E*)-isoeugenol) *[5932-68-3]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Culleré *et al.* (2004); Gómez-Míguez *et al.* (2007);			
Campo *et al.* (2006); Escudero *et al.* (2007)		0.006	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Schmitt (2005)		0.978	oil

2-METHOXY-4-PROPYLPHENOL (4-propyl-2-methoxyphenol, 4-propylguaiacol) *[2785-87-7]*

Schmitt (2005)		0.157	oil
Campo *et al.* (2006)		0.01	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2

2-methoxy-4-vinylphenol⇒4-ETHENYL-2-METHOXYPHENOL
2-methoxy-5-vinylphenol⇒5-ETHENYL-2-METHOXYPHENOL

METHYL ACETATE *[79-20-9]*

De Wet (1978)	d	470	wine residue medium
Luna *et al.* (2006)	d	2	refined olive oil

4'-METHYLACETOPHENONE *[122-00-9]*

Cerveny *et al.* (1993)		1.99	paraffin oil

methyl alcohol⇒METHANOL

METHYL 2-AMINOBENZOATE (methyl anthranilate) *[134-20-3]*

Rosenthal (1927)		44	diethyl phthalate
Nelson *et al.* (1977)		0.3	white Riesling wine
Escudero *et al.* (2004)		0.011	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2

methyl amyl ketone⇒2-HEPTANONE
methyl anthranilate⇒METHYL 2-AMINOBENZOATE
methyl acrylate⇒METHYL 2-PROPENOATE
methylbenzene⇒TOLUENE

METHYL BENZOATE *[93-58-3]*

Rosenthal (1927)		177	diethyl phthalate

α-methyl-1,3-benzodioxole-5-propanal⇒2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL

2-METHYLBUTANAL *[96-17-3]*

Guadagni *et al.* (1972)	d	0.14	vegetable oil
Czerny (1996)		0.021	cellulose
Wagner & Grosch (1998)		0.010	sunflower oil
Reiners & Grosch (1998)	d	0.0022	refined sunflower oil
Stephan & Steinhart (1999)	r	0.038	refined vegetable oil
Frauendorfer (2003)		0.152	oil
Luna *et al.* (2006)	d	0.0022	refined olive oil
Culleré *et al.* (2007)	d	0.016	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2
Saison *et al.* (2009)	d	0.056	carbonated beer served at 7-8 °C
Aceña *et al.* (2011)	d	0.0843	7 % (w/v) acetic acid water volume

(*S*)-(+)-2-METHYLBUTANAL *[1730-97-8]*

Rychlik & Grosch (1996)	d	0.053	starch

3-METHYLBUTANAL (isovaleraldehyde) *[590-86-3]*

Salo *et al.* (1972)	d	0.12	9.4 % (w/w) grain spirit
Guadagni *et al.* (1972)	d	0.013	vegetable oil
Perry (1989)		0.05	23 % v. ethanol
Preininger & Grosch (1994)		0.0054	sunflower oil
Czerny (1996)		0.029	cellulose
Rychlik & Grosch (1996)	d	0.032	starch

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Kubickova & Grosch (1998)		0.013	sunflower oil
Tandon *et al.* (2001)	d	0.015	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	0.003	deodorized tomato homogenate
Utsunomiya *et al.* (2004a)	d	0.120	base sake
Utsunomiya *et al.* (2004a)	d	0.11 - 0.12	base sake
Utsunomiya *et al.* (2004a)	r	0.31	base sake
Luna *et al.* (2006)	d	0.0054	refined olive oil
Culleré *et al.* (2007)	d	0.0046	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2
García-González *et al.* (2008)	d	0.08	deodorised edible oil
Poisson & Schieberle (2008)		0.0028	water/ethanol 6:4 by vol.
Aceña *et al.* (2011)	d	0.0316	7 % (w/v) acetic acid water volume

METHYL BUTANOATE (methyl butyrate) *[623-42-7]*

Zea *et al.* (2001)		0.1	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		1	14 % ethanolic solution

2-METHYLBUTANOIC ACID (α-methylbutyric acid) *[116-53-0]*

Brennand *et al.* (1989)	r	3.2	pH 2.0 aqeous solution
Guth & Grosch (1993)		0.25	cellulose
Rychlik & Grosch (1996)	d	0.024	starch
Stephan & Steinhart (1999)	r	0.24	refined vegetable oil
Schmitt (2003); Schmitt & Schieberle (2006)	d	0.203	sunflower oil
Averbeck & Schierberle (2011)	d	0.034	citrate buffer (pH 3.8)
Buhr *et al.* (2010)	d	10.3	whole milk
Buhr *et al.* (2010)	d	0.1	sunflower oil
Aceña *et al.* (2011)	d	0.2005	7 % (w/v) acetic acid water volume

3-METHYLBUTANOIC ACID (isovaleric acid, 3-methylbutyric acid) *[503-74-2]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	0.7 - 0.75	9.4 % (w/w) grain spirit
Yoshida (1972)	d	1	nujiol (a mineral oil)
Yoshida (1972)	r	10	nujiol (a mineral oil)
Yoshida (1984);Takagi (1989)	d	1	nujol
Yoshida (1984);Takagi (1989)	r	1	nujol
Amoore & Buttery (1978)	d	0.0662	safflower oil
Brennand *et al.* (1989)	r	0.07	pH 2.0 aqeous solution
Guth & Grosch (1993)		0.25	cellulose
Guth & Grosch (1994)		0.56	0.03 M Na phosphate; pH 5.7
Rychlik & Grosch (1996); Steinhaus & Schieberle (2005)	d	0.055 – 0.006	starch
Ong & Acree (1998)		0.012	artificial saliva and fruit juice matrix
Reiners & Grosch (1998)	d	0.022	refined sunflower oil
Stephan & Steinhart (1999)	r	0.041	refined vegetable oil
Lee *et al.* (2000)	d	0.1-0.2	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	0.3-1	3 year old grain whisky: 23 % ethanol
Moyano *et al.* (2002)		1.5	14 % (v/v) ethanol; tartaric acid to pH 3.5
Utsunomiya *et al.* (2004a)	d	0.4 - 0.41	base sake
Utsunomiya *et al.* (2004a)	r	0.8 - 0.81	base sake
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		3	14 % ethanolic solution
Callejón *et al.* (2008)	d	0.150	7 % acetic acid solution
Lübke *et al.* (2009)	d	1.6- 5	diethyl phthalate
Aceña *et al.* (2011)	d	0.1444	7 % (w/v) acetic acid water volume
Averbeck & Schierberle (2011)	d	0.010	citrate buffer (pH 3.8)

2-METHYL-1-BUTANOL (active amyl alcolhol) *[137-32-6]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Salo *et al.* (1972)	d	32.0	9.4 % (w/w) grain spirit
Morales *et al.* (2005)	d	0.48	deodorised olive oil
Aceña *et al.* (2011)	d	12.2	7 % (w/v) acetic acid water volume

3-METHYL-1-BUTANOL (isoamyl alcohol) *[123-51-3]*

Proetz (1924)		2.75	liq. petrolatum
Williams (1953)		10	ethanol solution
Salo (1970a,1970b); Salo *et al.* (1972)	d	6.5/7.0	9.4 % (w/w) grain spirit
Amerine & Roessler (1976)	r	400	wine
Van der Merwe & Van Wyk (1981)	d	14.5	wine residue medium
Ferreira *et al.* (1998)		40	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Tandon *et al.* (2001)	d	0.296	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	1.296	deodorized tomato homogenate
Moyano *et al.* (2002)		65	14 % (v/v) ethanol; tartaric acid to pH 3.5
Utsunomiya *et al.* (2004a)	d	67 - 68	base sake
Utsunomiya *et al.* (2004a)	r	140	base sake
Morales *et al.* (2005); Luna *et al.* (2006); García-González *et al.* (2008)	d	0.1	deodorised olive/edible oil
Poisson & Schieberle (2008)		56.1	water/ethanol 6:4 by vol.
Aceña *et al.* (2011)	d	3.5	7 % (w/v) acetic acid water volume

(*E*)-2-METHYL-2-BUTENAL (tiglaldehyde, tiglic aldehyde) *[497-03-0]*

Luna *et al.* (2006)	d	0.4	refined olive oil

3-METHYL-2-BUTENE-1-THIOL (prenyl mercaptan) *[5287-45-6]*

Gunst (1977)	r	< 0.000 5	beer
Tressl *et al.* (1980)		0.000 05	beer
Irwin *et al.* (1993)		0.000 007	beer

2-METHYL-3-BUTEN-2-OL *[115-18-4]*

Luna *et al.* (2006); García-González *et al.* (2008)	d	0.48	deodorised/refined olive/edible oil

2-METHYLBUTYL ACETATE (active amyl acetate) *[624-41-9]*

Luna *et al.* (2006)	d	2	refined olive oil

3-methylbutyl acetate⇒ISOPENTYL ACETATE
methyl butyrate⇒METHYL BUTANOATE
α-methylbutyric acid⇒2-METHYLBUTANOIC ACID
3-methylbutyric acid⇒3-METHYLBUTANOIC ACID
methyl cinnamate⇒METHYL 3-PHENYLPROPENOATE

(*R*)-(-)-3-METHYLCYCLOPENTADECANONE ((-)-muscone)

Hiraoka *et al.* (1995)	d	500	liquid paraffin
Yamamoto (1999)		3	aqueous ethanol solution

(*S*)-(+)-3-METHYLCYCLOPENTADECANONE ((*S*)-(+)-muscone)

Yamamoto (1999)		10	aqueous ethanol solution

3-methyl-1,2-cyclopentanedione⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE
methylcyclopentenolone⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE

8-METHYLDECANAL *[86414-41-7]*

Tajima *et al.* (1990)	d	0.010	limonene in water?

METHYLDITHIOETHANE (methylethyl disulphide, 2,3-dithiapentane) *[20333-39-5]*

Adda *et al.* (1978); Cuer *et al.* (1979)	d	0.0060	liquid cheese

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

METHYLDITHIOMETHANE (dimethyl disulphide, DMDS) *[624-92-0]*

Brenner *et al.* (1954)		< 0.002	beer
Leppänen *et al.* (1979)		0.02	10 % w/w ethanol/water
Goniak & Noble (1987)	d	0.029	white wine
García-González *et al.* (2008)	d	0.012	deodorised edible oil

methylene bis(methyl sulphide)⇒BIS(METHYLTHIO)METHANE
(3*R*)-α-methylene-3,7-dimethyl-6-octen-1-al⇒(*R*)-(-)-3,7-DIMETHYL-2-METHYLENE-6-OCTENAL
(3*S*)-α-methylene-3,7-dimethyl-6-octen-1-al⇒(*S*)-(+)-3,7-DIMETHYL-2-METHYLENE-6-OCTENAL
(3*R*)-α-methylene-3,7-dimethyl-6-octen-1-ol⇒(*R*)-(-)-3,7-DIMETHYL-2-METHYLENE-6-OCTEN-1-OL
(3*S*)-α-methylene-3,7-dimethyl-6-octen-1-ol⇒(*S*)-(+)-3,7-DIMETHYL-2-METHYLENE-6-OCTEN-1-OL

3,4-(METHYLENEDIOXY)BENZALDEHYDE (piperonal, heliotropine) *[120-57-0]*

Rosenthal (1927)		44	diethyl phthalate
Jagella & Grosch (1999b)	d	0.12	starch

methyleugenol⇒4-ALLYL-1,2-DIMETHOXYBENZENE
methylethyl disulphide⇒METHYLDITHIOETHANE
methyl ethyl ketone⇒BUTANONE

2-METHYLFURAN *[534-22-5]*

Evans *et al.* (1971)	d	27	cottonseed oil

2-METHYL-3-FURANTHIOL *[28588-74-1]*

Ferreira *et al.* (2002)		0.000 005	10 % hydroalcoholic solution, pH 3.2
Bailly *et al.* (2006)	d	0.000 003	ethanol/water 12:88 (v/v)

5-METHYLFURFURAL *[620-02-0]*

Moreno *et al.* (2005); Chaves *et al.* (2007)		16	14 % ethanolic solution

3-(2-METHYL)FURYLDITHIO-3-(2-METHYL)FURAN (bis(2-methyl-3-furyl) disulphide, 3,3'-dithiobis(2-methylfuran)) *[28588-75-2]*

Liepe (2004)		0.204	oil

4-methylguaiacol⇒2-METHOXY-4-METHYLPHENOL

6-METHYLHEPTANOIC ACID *[929-10-2]*

Brennand *et al.* (1989)	r	0.84	pH 2.0 aqeous solution

6-METHYL-5-HEPTEN-3-OL

Morales *et al.* (2005)	d	2	deodorised olive oil

methylheptenone⇒6-METHYL-5-HEPTEN-2-ONE

6-METHYL-5-HEPTEN-2-ONE (methylheptenone) *[110-93-0]*

Rosenthal (1927)		59	diethyl phthalate
Tandon *et al.* (2001)	d	1.008	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	0.522	deodorized tomato homogenate
Morales *et al.* (2005); Luna *et al.* (2006)	d	1	deodorised olive oil

methyl heptyl ketone⇒2-NONANONE

2-METHYLHEXANOIC ACID *[4536-23-6]*

Brennand *et al.* (1989)	r	55.3	pH 2.0 aqeous solution

4-METHYLHEXANOIC ACID *[1561-11-1]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Brennand *et al.* (1989)	r	7.3	pH 2.0 aqeous solution

(*E*)-3-METHYL-2-HEXENOIC ACID *[35205-70-0]*

Wysocki *et al.* (1993)		1	light, white, mineral oil

methyl hexyl ketone⇒2-OCTANONE

METHYL 4-HYDROXY-3-METHOXYBENZOATE (methyl vanillate) *[3943-74-6]*

Lopez *et al.* (2002)		3	10 % hydroalcoholic solution at pH 3.2
Ferreira *et al.* (2006)		0.195	ethanol 10% v/v, tartaric acid 2 g/l, pH 3.4

3-methyl-4-hydroxyoctanoic acid lactone⇒4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

3-METHYLINDOLE (skatole) *[83-34-1]*

Rosenthal (1927)		4	diethyl phthalate
Yoshida (1972)	d	0.0095	nujiol (a mineral oil)
Yoshida (1972)	r	100	nujiol (a mineral oil)
Yoshida (1984);Takagi (1989)	d	1	nujol or propylene glycol
Yoshida (1984);Takagi (1989)	r	1	nujol or propylene glycol
Thomson (1984)		0.14	aqueous ethanol
Moss *et al.* (1993)		0.08	meat puree mix
Preininger & Grosch (1994)		0.0156	sunflower oil
Sather (1995)		0.13	sunflower oil
Annor-Frempong *et al.* (1997)	d	0.026	sunflower oil and vegetable oil
Heiler (1995); Heiler & Schieberle (1997)		0.0021	buttermilk
Jagella & Grosch (1999b)	d	0.000 23	starch
Font I Furnols *et al.* (2000)	d	0.21	refined sunflower oil
Steinhaus & Schieberle (2005)	r	0.035	starch with 12 white pepper odorants
Lunde *et al.* (2010)		0.1	minced meat

δ-methylionone⇒3-METHYL-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
2-methylisoborneol⇒1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL
methyl isobutyl ketone⇒4-METHYL-2-PENTANONE
methyl mercaptan⇒METHANETHIOL
methyl methacrylate⇒METHYL 2-METHYL-2-PROPENOATE

METHYL 2-METHYLBUTANOATE *[868-57-5]*

Elss *et al.* (2007)		9 - 18 10^{-4}	10.5 g sucrose + 1.5 g citric acid.H_2O/L tap water
Elss *et al.* (2007)		15 - 23 10^{-4}	apple juice substitute without aroma

2-METHYL-3-(METHYLDITHIO)FURAN (methyl 2-methylfur-3-yl disulphide) *[65505-17-1]*

Culleré *et al.* (2008)		0.000 3	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2
Culleré *et al.* (2008)		0.000 25	Tintilla red wine

2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL (helional, α-methyl-1,3-benzodioxole-5-propanal) *[1205-17-0]*

Olsson & Laska (2010)	d	3.704 – 11.111	diethyl phthalate

7-METHYL-3-METHYLENE-1,6-OCTADIENE (myrcene) *[123-35-3]*

Jagella & Grosch (1998,1999a)	d	1.86 - 1.9	starch
Plotto *et al.* (2004)		0.532 - 0.773	deodorized orange juice matrix

methyl 2-methylfur-3-yl disulphide⇒2-METHYL-3-(METHYLDITHIO)FURAN

3-METHYL-5-(2-METHYLPENTYL)-2-(METHYLTHIO)PYRAZINE *[78246-18-1]*

Masuda *et al.* (1981)		1	aqueous ethanol solution

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

METHYL 2-METHYL-2-PROPENOATE (methyl methacrylate) *[80-62-6]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.2 - 10	sunflower oil
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.05 - 1	10 vol. % ethanol
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.05 - 0.5	3 vol. % acetic acid

cis-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN (*cis*-rose oxide) *[876-17-5]*

Ong & Acree (1998)		0.000 1	artificial saliva and fruit juice matrix

(2*S*,4*R*)-*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN ((2*S*,4*R*)-*cis*-rose oxide)

Matsuda *et al.* (1995); Matsuda & Yamamoto (1999); Yamamoto *et al.* (2002)	d	0.000 5	aqueous-ethanol solution

(2*R*,4*R*)-*trans*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN ((2*R*,4*R*)-*trans*-rose oxide)

Matsuda *et al.* (1995); Matsuda & Yamamoto (1999) ; Yamamoto *et al.* (2002)	d	0.160	aqueous-ethanol solution

(2*R*,4*S*)-*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN ((2*R*,4*S*)-*cis*-rose oxide)

Matsuda *et al.* (1995); Matsuda & Yamamoto (1999) ; Yamamoto *et al.* (2002)	d	0.050	aqueous-ethanol solution

(2*S*,4*S*)-*trans*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN ((2*S*,4*S*)-*trans*-rose oxide)

Matsuda *et al.* (1995); Matsuda & Yamamoto (1999) ; Yamamoto *et al.* (2002)	d	0.080	aqueous-ethanol solution

methyl β-naphthyl ether⇒2-METHOXYNAPHTHALENE

8-METHYLNONANAL *[3085-26-5]*

Tajima *et al.* (1990)	d	0.005	limonene in water?

3-METHYL-2,4-NONANEDIONE *[113486-29-6]*

Guth & Grosch (1989)		0.015 - 0.030	freshly refined oil
Guth & Grosch (1990a)		0.0225	sunflower oil
Guth & Grosch (1993)		0.0015	cellulose
Kao *et al.* (1998)		< 0.09	aqueous mineral oil solution

4-METHYLNONANOIC ACID *[45019-28-1]*

Brennand *et al.* (1989)	r	0.65	pH 2.0 aqeous solution

8-METHYLNONANOIC ACID *[5963-14-4]*

Brennand *et al.* (1989)	r	0.66	pH 2.0 aqeous solution

methyl nonyl ketone⇒2-UNDECANONE
(*cis*/*trans*) β-methyl-γ-octalactone⇒(*cis*/*trans*) 4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

6-METHYLOCTANAL *[30689-75-9]*

Tajima *et al.* (1990)	d	0.250	limonene

4-METHYLOCTANOIC ACID *[54947-74-9]*

Brennand *et al.* (1989)	r	0.02	pH 2.0 aqeous solution
Salles *et al.* (2002)	d	0.057	citrate buffer pH 5

6-METHYLOCTANOIC ACID *[504-99-4]*

Brennand *et al.* (1989)	r	4.1	pH 2.0 aqeous solution

methyl octyl ketone⇒2-DECANONE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

2-METHYLPENTANOIC ACID *[97-61-0]*			
Brennand *et al.* (1989)	r	11.8	pH 2.0 aqeous solution
3-METHYLPENTANOIC ACID *[105-43-1]*			
Brennand *et al.* (1989)	r	0.15	pH 2.0 aqeous solution
4-METHYLPENTANOIC ACID (isocaproic acid) *[646-07-1]*			
Brennand *et al.* (1989)	r	0.61	pH 2.0 aqeous solution
2-METHYL-1-PENTANOL *[105-30-6]*			
Zea *et al.* (2001)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
3-METHYL-1-PENTANOL *[589-35-5]*			
Zea *et al.* (2001); Moyano *et al.* (2002)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		50	14 % ethanolic solution
4-METHYL-1-PENTANOL *[626-891-]*			
Zea *et al.* (2001)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		50	14 % ethanolic solution
4-METHYL-2-PENTANONE (methyl isobutyl ketone) *[108-10-1]*			
Luna *et al.* (2006)	d	0.3	refined olive oil
Saison *et al.* (2009)	d	2.56	carbonated beer served at 7-8 °C
4-METHYL-3-PENTEN-2-ONE (mesityl oxide) *[141-79-7]*			
Stephan & Steinhart (1999)	r	0.3	refined vegetable oil
2-METHYLPHENOL (*o*-cresol) *[95-48-7]*			
Swan & Burtles (1978); Swan *et al.* (1981)		0.3	20 % ethanol-water
Jounela-Eriksson & Lehtonen (1981)		0.031	10 vol. % grain spirit
Vichi *et al.* (2008)	d	0.025	refined sunflower oil
3-METHYLPHENOL (*m*-cresol) *[108-39-4]*			
Swan *et al.* (1981)		0.5	20 % ethanol-water
Jounela-Eriksson & Lehtonen (1981)		0.068	10 vol. % grain spirit
Steinhaus & Schieberle (2005)	r	0.38	starch with 12 white pepper odorants
Söllner & Schieberle (2009)		0.19	odorless sunflower oil
4-METHYLPHENOL (*p*-cresol) *[106-44-5]*			
Adda & Jounela-Eriksson (1979)		0.008	9.4 % ethanol-water
Swan *et al.* (1981)		0.06	20 % ethanol-water
Jounela-Eriksson & Lehtonen (1981)		0.010	10 vol. % grain spirit
Jagella & Grosch (1999b)	d	0.13	starch
Frauendorfer (2003)		0.068	oil
Steinhaus & Schieberle (2005)	r	0.33	starch with 12 white pepper odorants
Frauendorfer & Schieberle (2006)	d	0.068	sunflower oil
Vichi *et al.* (2008)	d	0.025	refined sunflower oil
1-(4-METHYLPHENYL)ETHANOL *[536-50-5]*			
Cerveny *et al.* (1993)		42.7	paraffin oil
1-(4-METHYLPHENYL)ETHYL ACETATE			
Cerveny *et al.* (1993)		125.7	paraffin oil
1-(4-METHYLPHENYL)ETHYL BUTANOATE			
Cerveny *et al.* (1993)		417.0	paraffin oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

1-(4-METHYLPHENYL)ETHYL PROPANOATE

Cerveny *et al.* (1993)		513.0	paraffin oil

3-METHYL-1-PHENYL-3-PENTANOL (phenylethylmethylethylcarbinol) *[10415-87-9]*

Weiffenbach & McCarthy (1984)		< 0.01	mineral oil

METHYL 3-PHENYLPROPENOATE (methyl cinnamate) *[103-26-4]*

Rosenthal (1927)		446	diethyl phthalate
Williams & Ismail (1981)		0.12	sugar-acid base

2-METHYLPROPANAL (isobutyraldehyde) *[78-84-2]*

Salo *et al.* (1972)	d	1.30	9.4 % (w/w) grain spirit
Guadagni *et al.* (1972)	d	0.043	vegetable oil
Czerny (1996)		0.018	cellulose
Rychlik & Grosch (1996)	d	0.056	starch
Wagner & Grosch (1998)		0.0034	sunflower oil
Culleré *et al.* (2007)	d	0.006	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2
Poisson & Schieberle (2008)		0.0059	water/ethanol 6:4 by vol.
Saison *et al.* (2009)	d	0.086	carbonated beer served at 7-8 °C

METHYLPROPANOIC ACID (isobutyric acid) *[79-31-2]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	8.1 - 8.2	9.4 % (w/w) grain spirit
Amoore & Buttery (1978)	d	0.755	safflower oil
Brennand *et al.* (1989)	r	5.3	pH 2.0 aqeous solution
Frauendorfer & Schieberle (2006)	d	0.19	sunflower oil
Moreno *et al.* (2005)		20	14 % ethanolic solution
Aceña *et al.* (2011)	d	1.5	7 % (w/v) acetic acid water volume

2-METHYL-1-PROPANOL (isobutyl alcohol, isobutanol) *[78-83-1]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	75.0	9.4 % (w/w) grain spirit
De Wet (1978)	d	25	satur. KHT in 12 % EtOH-H2O
De Wet (1978)	d	227.9	wine residue medium
Moyano *et al.* (2002)		82	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007); Moyano *et al.* (2010)		40	14 % ethanolic solution
García-González *et al.* (2008)	d	1.00	deodorised edible oil
Aceña *et al.* (2011)	d	7.5	7 % (w/v) acetic acid water volume

METHYL 2-PROPENOATE (methyl acrylate) *[96-33-3]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.005 - 0.2	sunflower oil
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.005 - 0.2	10 vol. % ethanol
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.01 - 0.1	3 vol. % acetic acid

2-METHYLPROPENOIC ACID (methacrylic acid) *[79-41-4]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		2 - 100	sunflower oil

2-methylpropyl butanoate⇒ISOBUTYL BUTANOATE
methyl propyl ketone⇒2-PENTANONE

METHYLPYRAZINE *[109-08-0]*

Koehler *et al.* (1971)		27	paraffin oil

METHYL SALICYLATE *[119-36-8]*

Proetz (1924)		2.97	liq. petrolatum
Rosenthal (1927)		446	diethyl phthalate

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Kendall & Neilson (1964)		15	hexadecane
Kendall & Neilson (1964)		0.3	aqueous 0.1 % ethanol
Lindeman *et al.* (1982)	d	0.071	10 % ethanol

3-methyl-3-sulfanylbutanal⇒3-MERCAPTO-3-METHYLBUTANAL
3-(methylsulfanyl)propanal⇒3-(METHYLTHIO)PROPANAL

METHYLTETRATHIOMETHANE (dimethyl tetrasulphide) *[5756-24-1]*

Granvogl & Schieberle (2010)	r	0.0018	sunflower oil

S-METHYL THIOACETATE *[1534-08-3]*

Adda *et al.* (1978); Cuer *et al.* (1979)	d	0.005	liquid cheese

S-METHYL THIOBUTANOATE *[2432-51-1]*

Adda *et al.* (1978); Cuer *et al.* (1979)	d	0.2	liquid cheese

4-(METHYLTHIO)-1-BUTANOL *[20582-85-8]*

Rapp *et al.* (1985)		0.1 - 1	15 vol. % ethanol

METHYLTHIOMETHANE (dimethyl sulphide) *[75-18-3]*

De Mora *et al.* (1987)		0.06	Cabernet Sauvignon wine
Goniak & Noble (1987)	d	0.025	white wine
Kubickova & Grosch (1998)		0.0012	sunflower oil
Lucas (2000)	d	0.18	aqueous-alcoholic distillate from herbs
Lucas (2000)	r	0.23	aqueous-alcoholic distillate from herbs
Karagül-Yüceer *et al.* (2004)	d	0.000 8	skim milk
Utsunomiya *et al.* (2004a)	d	56 - 67 10^{-4}	base sake
Utsunomiya *et al.* (2004a)	r	14 - 15 10^{-3}	base sake
Averbeck & Schierberle (2011)	d	0.032	freshly reconstituted orange juice

S-METHYL THIO-2-METHYLBUTANOATE

Adda *et al.* (1978); Cuer *et al.* (1979)	d	0.1	liquid cheese

(+)-S-METHYL THIO-2-METHYLBUTANOATE

Cuer *et al.* (1979)	d	0.100	liquid cheese

(-)-S-METHYL THIO-2-METHYLBUTANOATE

Adda *et al.* (1978)	d	0.100	liquid cheese

S-METHYL THIO-3-METHYLBUTANOATE

Adda *et al.* (1978); Cuer *et al.* (1979)	d	0.100	liquid cheese

2-[[(methylthio)methyl]thio]ethanol⇒2,4-DITHIA-6-HEXANOL

3-(METHYLTHIO)PROPANAL (methional, 3-(methylsulfanyl)propanal, 4-thiapentanal) *[3268-49-3]*

Wick *et al.* (1967)	d	6.0	beef slurry
Guadagni *et al.* (1972)	d	0.000 2	vegetable oil
Preininger & Grosch (1994);			
Kubickova & Grosch (1998)		0.000 2	sunflower oil
Rychlik & Grosch (1996)	d	0.000 27	starch
Escudero *et al.* (2000)	d	0.000 5	11% ethanol, 5 g/l tartaric acid, 7 g/L glycerin, pH 3.4
Mayer (2000)		0.009	cellulose
Silva Ferreira *et al.* (2002)	r	< 0.03	young, dry white wine
Karagül-Yüceer *et al.* (2004)	d	0.005	skim milk
Isogai *et al.* (2005)	d	0.000 010	base sake

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

S-METHYL THIOPROPANOATE

Adda *et al.* (1978); Cuer *et al.* (1979)	d	0.100	liquid cheese

3-(METHYLTHIO)-1-PROPANOL (methionol) *[505-10-2]*

Moyano *et al.* (2002)		1.5	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Moyano *et al.* (2010)		0.5	14 % ethanolic solution
Ferreira *et al.* (2006)		2.5	ethanol 10% v/v, tartaric acid 2 g/l, pH 3.4

3-METHYLTHIO-1-PROPENE (allyl methyl sulphide) *[10152-76-8]*

Söllner & Schieberle (2009)		0.022	odorless sunflower oil

α-(METHYLTHIO)TOLUENE (benzyl methyl sulphide) *[766-92-7]*

Drake *et al.* (2007)		0.177	vegetable oil

3-METHYL-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE (δ-methylionone) *[127-51-5]*

Harder (1975)		0.01	diethyl phthalate

(1*R*,6*S*)-5-METHYL-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-HEXANOL

Yamamoto (1999)		0.07	500 ppm ethanol in water?

(1*S*,6*R*)-5-METHYL-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-HEXANOL

Yamamoto (1999)		1	500 ppm ethanol in water?

(1'*R*,2*R*)-2-METHYL-4-(2,2,3-TRIMETHYL-1-CYCLOPENTEN-3-YL)-1-BUTANOL ((1'*R*,2*R*)-brahmanol)

Yamamoto (1999)		0.0187	500 ppm ethanol in water?

(1'*R*,2*S*)-2-METHYL-4-(2,2,3-TRIMETHYL-1-CYCLOPENTEN-3-YL)-1-BUTANOL ((1'*R*,2*S*)-brahmanol)

Yamamoto (1999)		0.1765	500 ppm ethanol in water?

(1'*S*,2*R*)-2-METHYL-4-(2,2,3-TRIMETHYL-1-CYCLOPENTEN-3-YL)-1-BUTANOL ((1'*S*,2*R*)-brahmanol)

Yamamoto (1999)		0.0145	500 ppm ethanol in water?

(1'*S*,2*S*)-2-METHYL-4-(2,2,3-TRIMETHYL-1-CYCLOPENTEN-3-YL)-1-BUTANOL ((1'*S*,2*S*)-brahmanol)

Yamamoto (1999)		0.166	500 ppm ethanol in water?

METHYLTRITHIOMETHANE (dimethyl trisulphide, 2,3,4-trithiapentane) *[3658-80-8]*

Leppänen *et al.* (1979)		0.000 1	10 % w/w ethanol/water
Adda & Jounela-Eriksson (1979)		0.000 11	9.4 % ethanol-water
Adda *et al.* (1978); Cuer *et al.* (1979)	d	0.000 1	liquid cheese
Leppänen *et al.* (1983)		0.003	light blended whisky
Leppänen *et al.* (1983)		0.005	heavy blended whisky
Leppänen *et al.* (1983)		0.006	malt whisky
Perry (1989)		0.000 1	23 % v. ethanol
Rychlik & Grosch (1996)	d	0.000 086	starch
Kubickova & Grosch (1998); Kubickova (1998)		0.0025	sunflower oil
Lee *et al.* (2000)	d	0.004-0.03	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	0.02-0.1	3 year old grain whisky: 23 % ethanol
Utsunomiya *et al.* (2004a)	d	0.000 18	base sake
Utsunomiya *et al.* (2004a)	r	47 - 48 10^{-5}	base sake
Wright *et al.* (2006)	d	0.000 8	reconstituted whey protein isolate (10 % solids)
Aceña *et al.* (2011)	d	0.000 035	7% (w/v) acetic acid water solution

methyl vinyl ketone⇒1-BUTEN-3-ONE
MIB⇒1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL

MONOETHYL SUCCINATE (ethyl acid succinate) *[1070-34-4]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Moreno *et al.* (2005)		1,000	14 % ethanolic solution

(-)-muscone⇒(*R*)-(-)-3-METHYLCYCLOPENTADECANONE
(*S*)-(+)-muscone⇒(*S*)-(+)-3-METHYLCYCLOPENTADECANONE
musk 36A⇒3'-ETHYL-5',6',7',8'-TETRAHYDRO-5',5',8',8'-TETRAMETHYL-2'-ACETONAPHTHONE
musk ambrette⇒1-*tert*-BUTYL-2-METHOXY-4-METHYL-3,5-DINITROBENZENE
musk ketone⇒4-*tert*-BUTYL-2,6-DIMETHYL-3,5-DINITROACETOPHENONE
muskolactone⇒15-HYDROXYPENTADECANOIC ACID LACTONE
musk xylene⇒1-*tert*-BUTYL-3,5-DIMETHYL-2,4,6-TRINITROBENZENE
myrcene⇒7-METHYL-3-METHYLENE-1,6-OCTADIENE
myrisitic acid⇒TETRADECANOIC ACID

N

nerol⇒3,7-DIMETHYL-*cis*-2,6-OCTADIEN-1-OL
E-nerolidol⇒ (*E*)-3,7,11-TRIMETHYL-1,6,10-DODECATRIEN-3OL

NITROBENZENE *[98-95-3]*

Proetz (1924)		30.5	liq. petrolatum
Moncrieff (1957)		500	propylene glycol
Henkin & Bartter (1966); Henkin & Hoye (1966); Henkin (1967a,1967b,1968)	d	0.012	mineral oil
Henkin & Bartter (1966); Henkin & Hoye (1966); Henkin (1967b,1968)	r	1.2 - 12	mineral oil
Henkin *et al.* (1968); Hoye *et al.* (1970)	d	0.12	mineral oil
Henkin *et al.* (1968); Hoye *et al.* (1970)	r	1,230	mineral oil
Henkin *et al.* (1971,1972, 1976); Henkin & Smith (1971); Marshall & Henkin (1971); Henkin & Larson (1972); Schechter & Henkin (1974); Delaney *et al.* (1977)	d	0.012 - 0.12	mineral oil
Henkin *et al.* (1971,1972,1976); Henkin & Smith (1971); Marshall & Henkin (1971); Henkin & Larson (1972); Schechter & Henkin (1974); Delaney *et al.* (1977)	r	123 - 1,230	mineral oil
Filla *et al.* (1975); De Michele *et al.* (1976); Campanella *et al.* (1978)	d	0.012 - 12	mineral oil
Ansari (1976)	d	2.4	paraffin oil
Sullivan & Burch (1976); Burch *et al.* (1978)	d	0.44	mineral oil
Sullivan & Burch (1976); Burch *et al.* (1978)	r	6.3	mineral oil
Lawas *et al.* (1979)	d	3.98	mineral oil
Lawas *et al.* (1979)	r	237.1	mineral oil
Bosma *et al.* (1981)	d	1.2	mineral oil
Bosma *et al.* (1981)	r	1,230	mineral oil
Garrett-Laster *et al.* (1984)	d	0.012	mineral oil
Garrett-Laster *et al.* (1984)	r	0.12	mineral oil

γ-nonalactone⇒4-HYDROXYNONANOIC ACID LACTONE
(*R*)-γ-nonalactone⇒(*R*)-4-HYDROXYNONANOIC ACID LACTONE
(*S*)-γ-nonalactone⇒(*S*)-4-HYDROXYNONANOIC ACID LACTONE
δ-nonalactone⇒5-HYDROXYNONANOIC ACID LACTONE

2,4-NONADIENAL *[6750-03-4]*

Morales *et al.* (2005)	d	2.5	deodorised olive oil

(*E*,*E*)-2,4-NONADIENAL *[5910-87-2]*

Meijboom (1964); Belitz & Grosch (1992)		2.5	paraffin oil
Meijboom & Jongenotter (1981)		1.0	paraffin oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Poisson & Schieberle (2008)		0.0026	water/ethanol 6:4 by vol.
Burdack-Freitag (2007)		0.0015	sunflower oil
(*E,Z*)-2,6-NONADIENAL *[557-48-2]*			
Meijboom (1964); Hoffmann & Meijboom (1968); Keppler (1977)		0.01	paraffin oil
Guth & Grosch (1990a)		0.0038	sunflower oil
Guth & Grosch (1993)		0.0045	cellulose
Heiler & Schieberle (1996)	d	0.000 7	buttermilk
Poisson & Schieberle (2008)		0.000 3	water/ethanol 6:4 by vol.
(*E,E*)-2,6-NONADIENAL *[17587-33-6]*			
Meijboom (1964); Hoffmann & Meijboom (1968); Keppler (1977); Meijboom & Jongenotter (1981)		0.21 - 0.25	paraffin oil
1,3-NONADIENE			
Evans *et al.* (1971)	d	12	cottonseed oil
1,8-NONADIENE *[4900-30-5]*			
Evans *et al.* (1971)	d	11	cottonseed oil
2,4-NONADIENE *[34266-16-5]*			
Evans *et al.* (1971)	d	9	cottonseed oil
(*E,Z*)-2,6-NONADIEN-1-OL *[28069-72-9]*			
Heiler & Schieberle (1996,1997)	d	0.0013	buttermilk
Stephan & Steinhart (1999)	r	0.058	refined vegetable oil
NONANAL (nonyl aldehyde) *[124-19-6]*			
Rosenthal (1927)		9	diethyl phthalate
Meijboom (1964); Belitz & Grosch (1992)		13.4 - 13.5	paraffin oil
Wick *et al.* (1967)	d	> 7.6	beef slurry
Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	1.0	vegetable oil
Williams & Ismail (1981)		0.056	sugar-acid base
Tajima *et al.* (1990)	d	5.0	limonene in water?
Aparicio & Luna (2002)		13.5	refined vegetable oil
Plotto *et al.* (2004)		0.232 - 0.312	deodorized orange juice matrix
Culleré *et al.* (2004)		0.015	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Morales *et al.* (2005); García-González *et al.* (2008)	d	0.15	deodorised olive/edible oil
Limpawattana (2007)	d	0.0710	starch
Limpawattana (2007)	d	0.0298	partially deodorized rice at 70° C
Aceña *et al.* (2011)	d	0.0182	7 % (w/v) acetic acid water volume
Culleré *et al.* (2011)	d	0.0025	hydroethanolic solution (13 % v/v), tartaric acid 5 g/L, pH 3.2
Culleré *et al.* (2011)	d	0.01125	citrus-smelling white wine
Culleré *et al.* (2011)	d	0.0075	neutral white wine
Culleré *et al.* (2011)	d	0.0025	floral white wine
NONANE *[111-84-2]*			
Champagne & Nawar (1969)		2,150	mineral oil
Evans *et al.* (1971)	d	650	cottonseed oil
2,4-NONANEDIONE *[6175-23-1]*			
Guth & Grosch (1989)		4 - 8	freshly refined oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

NONANOIC ACID (pelargonic acid) *[112-05-0]*			
Salo *et al.* (1972)	d	> 1.1	9.4 % (w/w) grain spirit
Brennand *et al.* (1989)	r	2.4	pH 2.0 aqeous solution
1-NONANOL (nonyl alcohol) *[143-08-8]*			
De Wet (1978)	d	0.31	wine residue medium
Morales *et al.* (2005); García-González *et al.* (2008)	d	0.28	deodorised olive/edible oil
4-nonanolide⇒4-HYDROXYNONANOIC ACID LACTONE			
5-nonanolide⇒5-HYDROXYNONANOIC ACID LACTONE			
2-NONANONE (methyl heptyl ketone) *[821-55-6]*			
Rothe (1991,1992)		116	cheese 45 % fat dry matter
Luna *et al.* (2006)	d	< 0.1	refined olive oil
García-González *et al.* (2008)	d	0.1	deodorised edible oil
2-NONENAL *[2463-53-8]*			
Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	0.15	vegetable oil
Perry (1989)		0.000 6	23 % v. ethanol
(*E*)-2-NONENAL *[18829-56-6]*			
Meijboom (1964); Meijboom & Jongenotter (1981)		3.2 - 3.5	paraffin oil
Guth & Grosch (1990a)		0.9	sunflower oil
Schieberle (1992)		0.000 6	fresh beer
Grosch *et al.* (1994)		0.045	butter oil
Rychlik & Grosch (1996)	d	0.000 53	starch
Ong & Acree (1998)		0.000 1	artificial saliva and fruit juice matrix
Czerny & Grosch (2000)	d	0.015	cellulose
Culleré *et al.* (2007)	d	0.000 6	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2
Limpawattana (2007)	d	0.0213	starch
Limpawattana (2007)	d	0.0049	partially deodorized rice at 70° C
García-González *et al.* (2008)	d	0.15	deodorised edible oil
Poisson & Schieberle (2008)		0.000 6	water/ethanol 6:4 by vol.
Aceña *et al.* (2011)	d	0.0015	7 % (w/v) acetic acid water volume
(*Z*)-2-NONENAL *[60784-31-8]*			
Guth & Grosch (1990a)		0.0045	sunflower oil
Grosch *et al.* (1994)		0.003	butter oil
Zehentbauer & Grosch (1998)		0.000 036	starch
(*Z*)-3-NONENAL *[31823-43-5]*			
Meijboom & Jongenotter (1981); Belitz & Grosch (1992)		0.25	paraffin oil
(*E*)-4-NONENAL *[2277-16-9]*			
Meijboom & Jongenotter (1981)		2.0	paraffin oil
(*Z*)-4-NONENAL *[2277-15-8]*			
Meijboom & Jongenotter (1981)		0.08	paraffin oil
(*E*)-5-NONENAL			
Meijboom & Jongenotter (1981)		0.45	paraffin oil
(*Z*)-5-NONENAL			
Meijboom & Jongenotter (1981)		0.25	paraffin oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

(*E*)-6-NONENAL *[2277-20-5]*

Keppler *et al.* (1967);			
Meijboom & Jongenotter (1981)		0.0046 - 0.005	paraffin oil

(*Z*)-6-NONENAL *[2277-19-2]*

Meijboom & Jongenotter (1981)		0.04	paraffin oil

(*E*)-7-NONENAL

Meijboom & Jongenotter (1981)		1.0	paraffin oil

(*Z*)-7-NONENAL

Meijboom & Jongenotter (1981)		0.4	paraffin oil

8-NONENAL *[39770-04-2]*

Meijboom & Jongenotter (1981)		1.6	paraffin oil

1-NONENE *[124-11-8]*

Champagne & Nawar (1969)		26	mineral oil
Evans *et al.* (1971)	d	9	cottonseed oil

1-NONEN-3-OL *[21964-44-3]*

Evans *et al.* (1971)	d	1.3	cottonseed oil

2-NONEN-4-OL

Evans *et al.* (1971)	d	12	cottonseed oil

1-NONEN-3-ONE *[24415-26-7]*

Stephan & Steinhart (1999)	r	0.000 7	refined vegetable oil
Pons *et al.* (2011)	d	0.000 008	12% ethanol, 5 g/L tartaric acid, pH 3.5
Pons *et al.* (2011)	d	0.000 03	red wine
Pons *et al.* (2011)	r	3 - 3.5 10^{-5}	dry white wine

nonyl alcohol⇒1-NONANOL
nonyl aldehyde⇒NONANAL

1-NONYNE *[3452-09-3]*

Champagne & Nawar (1969)		18	mineral oil
Evans *et al.* (1971)	d	5	cottonseed oil

nootkatone⇒8-ISOPROPENYL-5,6-DIMETHYLBICYCLO[4.4.0]-1-DECEN-3-ONE
norfuraneol⇒4-HYDROXY-5-METHYL-3(2*H*)-FURANONE

O

oak lactone⇒4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE
E-oak lactone⇒*trans*-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE
Z-oak lactone⇒*cis*-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

(*Z,Z*)-9,12-OCTADECADIENOIC ACID (linoleic acid) *[60-33-3]*

Salo *et al.* (1972)	d	> 1.2	9.4 % (w/w) grain spirit
Chalé-Rush *et al.* (2007)	d	100 - 200	water with 5 % mineral oil + 5 % gum acacia
Chalé-Rush *et al.* (2007)	d	1,500 - 2,000	water with 5 % mineral oil + 5 % gum acacia *

*) retronasal olfaction via odorant presentation containers

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

γ-octadecalactone⇒4-HYDROXYOCTADECANOIC ACID LACTONE
δ-octadecalactone⇒5-HYDROXYOCTADECANOIC ACID LACTONE

OCTADECANOIC ACID (stearic acid) *[57-11-4]*

Salo *et al.* (1972)	d	> 1.6	9.4 % (w/w) grain spirit
Chalé-Rush *et al.* (2007)	d	100	water with 5 % mineral oil + 5 % gum acacia
Chalé-Rush *et al.* (2007)	d	300 - 450	water with 5 % mineral oil + 5 % gum acacia * *) retronasal olfaction via odorant presentation containers

4-octadecanolide⇒4-HYDROXYOCTADECANOIC ACID LACTONE
5-octadecanolide⇒5-HYDROXYOCTADECANOIC ACID LACTONE

(*Z*)-9-OCTADECENOIC ACID (oleic acid) *[112-80-1]*

Salo *et al.* (1972)	d	> 2.2	9.4 % (w/w) grain spirit
Chalé-Rush *et al.* (2007)	d	100 - 200	water with 5 % mineral oil + 5 % gum acacia
Chalé-Rush *et al.* (2007)	d	1,500 - 2,000	water with 5 % mineral oil + 5 % gum acacia * *) retronasal olfaction via odorant presentation containers

(*E*,*E*)-2,4-OCTADIENAL *[30361-28-5]*

Meijboom (1964); Keppler (1977)		1.0	paraffin oil

(*E*,*E*)-2,5-OCTADIENAL

Keppler (1977); Meijboom & Jongenotter (1981)		2.5	paraffin oil

(*E*,*Z*)-2,5-OCTADIENAL *[205380-41-2]*

Meijboom (1964)		15.0	paraffin oil

1,3-OCTADIENE *[1002-33-1]*

Evans *et al.* (1971)	d	20	cottonseed oil

1,4-OCTADIENE *[5675-25-2]*

Evans *et al.* (1971)	d	15	cottonseed oil

2,4-OCTADIENE *[13643-08-8]*

Evans *et al.* (1971)	d	12	cottonseed oil

(*Z*)-1,5-OCTADIEN-3-ONE *[65767-22-8]*

Swoboda & Peers (1977)	d	0.000 44	deodorised groundnut oil
Guth & Grosch (1990a)		0.000 45	sunflower oil
Guth & Grosch (1993)		0.000 05	cellulose
Stephan & Steinhart (1999)	r	0.00013	refined vegetable oil

γ-octalactone⇒4-HYDROXYOCTANOIC ACID LACTONE
(*R*)-γ-octalactone⇒(*R*)-4-HYDROXYOCTANOIC ACID LACTONE
(*S*)-γ-octalactone⇒(*S*)-4-HYDROXYOCTANOIC ACID LACTONE
δ-octalactone⇒5-HYDROXYOCTANOIC ACID LACTONE

OCTANAL (caprylaldehyde) *[124-13-0]*

Rosenthal (1927)		6	diethyl phthalate
Meijboom (1964)		0.32	paraffin oil
Reiners & Grosch (1998)		0.056	refined sunflower oil
Zea *et al.* (2001); Moyano *et al.* (2002)		0.05	14 % (v/v) ethanol; tartaric acid to pH 3.5
Plotto *et al.* (2004)		0.233 - 0.474	deodorized orange juice matrix
Culleré *et al.* (2004)		0.015	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Morales *et al.* (2005); Moreno *et al.* (2005);			

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Moyano *et al.* (2010)		0.64	14 % ethanolic solution
García-González *et al.* (2008)	d	0.32	deodorised olive/edible oil
Limpawattana (2007)	d	0.0398	starch
Limpawattana (2007)	d	0.0087	partially deodorized rice at 70° C
Aceña *et al.* (2011)	d	0.0225	7 % (w/v) acetic acid water volume
Culleré *et al.* (2011)	d	0.0025	hydroethanolic solution (13 % v/v), tartaric acid 5 g/L, pH 3.2
Culleré *et al.* (2011)	d	0.00375	citrus-smelling white wine
Culleré *et al.* (2011)	d	0.00175	neutral white wine
Culleré *et al.* (2011)	d	0.00187	floral white wine

OCTANE *[111-65-9]*

Champagne & Nawar (1969)		1340	mineral oil
Morales *et al.* (2005); García-González *et al.* (2008)	d	0.94	deodorised olive/edible oil

OCTANOIC ACID (caprylic acid) *[124-07-2]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	15.0	9.4 % (w/w) grain spirit
Amoore & Buttery (1978)	d	161	safflower oil
Haslbeck *et al.* (1986)		185 - 215	cream
Haslbeck *et al.* (1986)		> 1,000	30 % coconut oil
Haslbeck *et al.* (1986)		> 1,000	38 % coconut oil aromatized
Ferreira *et al.* (1998)		10	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Morales *et al.* (2005)	d	3	deodorised olive oil
Karagül-Yüceer *et al.* (2004)	d	44.668	skim milk
Utsunomiya *et al.* (2004a)	d	1.7 - 1.8	base sake
Utsunomiya *et al.* (2004a)	r	4 - 4.1	base sake
Moreno *et al.* (2005); Chaves *et al.* (2007)		8.8	14 % ethanolic solution
Rychlik *et al.* (2006)		10.98	fresh yoghurt with 2.03 mg/kg octanoic acid
Callejón *et al.* (2008)	d	0.987	7 % acetic acid solution

1-OCTANOL (octyl alcohol) *[111-87-5]*

Cain (1966)	r	900	diethyl phthalate
Salo *et al.* (1972)	d	1.1	9.4 % (w/w) grain spirit
De Wet (1978)	d	0.82	wine residue medium
Moreno *et al.* (2005)		10	14 % ethanolic solution
García-González *et al.* (2008)	d	0.027	deodorised edible oil

2-OCTANOL *[123-96-6]*

Morales *et al.* (2005)	d	0.1	deodorised olive oil

2-OCTANONE (methyl hexyl ketone) *[111-13-7]*

Morales *et al.* (2005); Luna *et al.* (2006); García-González *et al.* (2008)	d	0.5 - 0.51	deodorised/refined olive/edible oil

3-OCTANONE (ethyl amyl ketone) *[106-68-3]*

Siegmund & Pöllinger-Zierler (2006)	d	0.057	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	> 0.1	apple juice

4-octanolide⇒4-HYDROXYOCTANOIC ACID LACTONE
5-octanolide⇒5-HYDROXYOCTANOIC ACID LACTONE

2-OCTENAL *[2363-89-5]*

Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	0.5	vegetable oil

(*E*)-2-OCTENAL *[2548-87-0]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Meijboom (1964); Meijboom & Jongenotter (1981); Belitz & Grosch (1992)		7.0	paraffin oil
Brewer & Vega (1995)	d	4.20	lean ground beef
Jensen *et al.* (1999)	d	0.6	mashed potatoes
Morales *et al.* (2005); Luna *et al.* (2006); García-González *et al.* (2008)	d	0.004	deodorised//refined olive/edible oil
Culleré *et al.* (2007)	d	0.003	10 % water/ethanol, 5 g/l tartaric acid, pH 3.2

(*Z*)-3-OCTENAL *[78693-34-2]*

Meijboom & Jongenotter (1981)		0.05	paraffin oil

(*E*)-4-OCTENAL

Meijboom & Jongenotter (1981)		0.37	paraffin oil

(*E*)-5-OCTENAL

Meijboom & Jongenotter (1981)		0.15	paraffin oil

(*Z*)-5-OCTENAL *[41547-22-2]*

Meijboom & Jongenotter (1981)		1.0	paraffin oil

(*E*)-6-OCTENAL *[63826-25-5]*

Meijboom & Jongenotter (1981)		0.015	paraffin oil

(*Z*)-6-OCTENAL

Meijboom & Jongenotter (1981)		0.040	paraffin oil

7-OCTENAL *[21573-31-9]*

Meijboom & Jongenotter (1981)		0.51	paraffin oil

1-OCTENE *[111-66-0]*

Champagne & Nawar (1969)		2	mineral oil
Evans *et al.* (1971)	d	2	cottonseed oil

1-OCTEN-3-HYDROPEROXIDE *[72755-76-1]*

Guth & Grosch (1990a)		0.24	sunflower oil

1-OCTEN-3-OL *[3391-86-4]*

Evans *et al.* (1971)	d	0.9	cottonseed oil
Amon *et al.* (1989)	d	0.02	non-aromatic dry white wine
Kubickova & Grosch (1998)		0.034	sunflower oil
Ong & Acree (1998)		0.018	artificial saliva and fruit juice matrix
Morales *et al.* (2005); García-González *et al.* (2008)	d	0.001	deodorised olive/edible oil
Siegmund & Pöllinger-Zierler (2006)	d	0.011	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.031	apple juice

δ-octenolactone⇒5-HYDROXY-2-OCTENOIC ACID LACTONE

1-OCTEN-3-ONE *[4312-99-6]*

Evans *et al.* (1971)	d	0.1	cottonseed oil
Swoboda & Peers (1977)	d	0.077	deodorised groundnut oil
Amon *et al.* (1989)	d	0.000 02	non-aromatic dry white wine
Guth & Grosch (1990a)		0.01	sunflower oil
Guth & Grosch (1993)		0.002	cellulose
Grosch *et al.* (1994)		0.003	butter oil
Rychlik & Grosch (1996)	d	0.000 044	starch

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Kubickova & Grosch (1998)		0.01	sunflower oil
Morales *et al.* (2005) ; García-González *et al.* (2008)	d	0.01	deodorised olive/edible oil
Culleré *et al.* (2006)	d	0.000 015	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Culleré *et al.* (2006)	d	0.000 12	white/red wine with 7-12 ng/L 1-octen-3-one
Aceña *et al.* (2011)	d	0.0068	7 % (w/v) acetic acid water volume
Pons *et al.* (2011)	d	0.000 03	12% ethanol, 5 g/L tartaric acid, pH 3.5
Pons *et al.* (2011)	d	0.000 04	neutral white wine
Pons *et al.* (2011)	d	0.000 07	neutral red wine
Pons *et al.* (2011)	r	0.000 12	neutral white wine

3-OCTEN-2-ONE *[1669-44-9]*

Stephan & Steinhart (1999)	r	0.25	refined vegetable oil

OCTYL ACETATE *[112-41-1]*

De Wet (1978)	d	0.80	wine residue medium
Pineau *et al.* (2009)	d	0.8	dearomatized red wine

octyl alcohol⇒1-OCTANOL

1-OCTYNE *[629-05-0]*

Champagne & Nawar (1969)		8	mineral oil

oenanthic acid⇒HEPTANOIC ACID
oleic acid⇒(*Z*)-9-OCTADECENOIC ACID
4-oxoisophorone⇒2,6,6-TRIMETHYL-2-CYCLOHEXENE-1,4-DIONE
oxymethylfurfural⇒5-(HYDROXYMETHYL)FURFURAL

P

palmitic acid⇒HEXADECANOIC ACID
palmitoleic acid⇒(*Z*)-9-HEXADECENOIC ACID
pantolactone⇒4-HYDROXY(2-HYDROXY-3,3-DIMETHYL)BUTANOIC ACID LACTONE
pelargonic acid⇒NONANOIC ACID

PENTACHLOROPHENOL *[87-86-5]*

Coulter *et al.* (2008)	d	> 0.000 896	neutral, dry white wine

ω -pentadecalactone⇒15-HYDROXYPENTADECANOIC ACID LACTONE

PENTADECANE *[629-62-9]*

Champagne & Nawar (1969)		> 13,000	mineral oil

PENTADECANOIC ACID *[1002-84-2]*

Salo *et al.* (1972)	d	> 0.5	9.4 % (w/w) grain spirit

1-PENTADECENE *[13360-61-7]*

Champagne & Nawar (1969)		2,250	mineral oil

(*E*)-1,3-PENTADIENE *[2004-70-8]*

Horwood *et al.* (1981)		2.5	10 % NaCl

PENTANAL (valeraldehyde) *[110-62-3]*

Meijboom (1964); Belitz & Grosch (1992)		0.24	paraffin oil
Salo *et al.* (1972)	d	0.11	9.4 % (w/w) grain spirit
Brewer & Vega (1995)	d	2.67	lean ground beef

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Morales *et al.* (2005)	d	0.24	deodorised olive oil
PENTANE *[109-66-0]*			
Champagne & Nawar (1969)		340	mineral oil
2,3-PENTANEDIONE *[600-14-6]*			
Salo *et al.* (1972)	d	0.078	9.4 % (w/w) grain spirit
Czerny (1996)		0.097	cellulose
Rychlik & Grosch (1996)	d	0.016	starch
Chaves *et al.* (2007)		1	14 % v/v ethanol/water
Burdack-Freitag (2007)		0.000 3	sunflower oil
PENTANOIC ACID (valeric acid) *[109-52-4]*			
Salo *et al.* (1972)	d	> 0.5	9.4 % (w/w) grain spirit
Amoore & Buttery (1978)	d	1.49	safflower oil
Brennand *et al.* (1989)	r	1.1	pH 2.0 aqeous solution
Brennand *et al.* (1989)	r	6.5	pH 5.2 aqeous solution
Hiraoka *et al.* (1995)	d	0.5	liquid paraffin
Stephan & Steinhart (1999)	r	0.061	refined vegetable oil
Jagella & Grosch (1999b)	d	4.9	starch
Morales *et al.* (2005)	d	0.6	deodorised olive oil
1-PENTANOL (amyl alcohol) *[71-41-0]*			
De Wet (1978)	d	64.43	wine residue medium
Rezek (1987)		0.85	mineral oil
Aparicio & Luna (2002); Luna *et al.* (2006); García-González *et al.* (2008)	d	0.47	refined/deodorised vegetable/olive/edible oil
Morales *et al.* (2005)	d	3	deodorised olive oil
2-PENTANONE (methyl propyl ketone) *[107-87-9]*			
Rothe (1991,1992)		288	cheese 45 % fat dry matter
1,2,3,5,6-PENTATHIEPANE (lenthionine) *[292-46-6]*			
Wada *et al.* (1967)		12.5 - 25	vegetable oil
2-PENTENAL *[764-39-6]*			
Tandon *et al.* (2001)	d	0.243	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	2.265	deodorized tomato homogenate
Morales *et al.* (2005)	d	0.3	deodorised olive oil
(*E*)-2-PENTENAL *[1576-87-0]*			
Meijboom (1964); Keppler (1977); Belitz & Grosch (1992)		2.3 - 2.5	paraffin oil
Aparicio & Luna (2002)		0.3	refined vegetable oil
Plotto *et al.* (2004)		2.33 - 2.77	deodorized orange juice matrix
Luna *et al.* (2006)	d	0.3	refined olive oil
(*Z*)-2-PENTENAL *[1576-86-9]*			
Luna *et al.* (2006)	d	1.5	refined olive oil
4-PENTENAL *[2100-17-6]*			
Keppler (1977)		0.025	paraffin oil
1-PENTENE *[109-67-1]*			
Champagne & Nawar (1969)		92	mineral oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

(*Z*)-2-PENTEN-1-OL *[1576-95-0]*

Luna *et al.* (2006)	d	0.25	refined olive oil

3-PENTEN-2-OL *[1569-50-2]*

Morales *et al.* (2005)	d	0.4	deodorised olive oil

1-PENTEN-3-OL *[616-25-1]*

Luna *et al.* (2006)	d	0.4	refined olive oil

1-PENTEN-3-ONE (ethyl vinyl ketone) *[1629-58-9]*

Evans *et al.* (1971)	d	0.02	cottonseed oil
Guadagni *et al.* (1972); Buttery *et al.* (1973)	d	0.0055	vegetable oil
Reiners & Grosch (1998)	d	0.000 73	refined sunflower oil
Tandon *et al.* (2001)	d	0.138	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	0.398	deodorized tomato homogenate
Aparicio & Luna (2002)		0.05	refined vegetable oil
Luna *et al.* (2006)	d	0.000 7	refined olive oil

PENTYL ACETATE (amyl acetate) *[628-63-7]*

Moncrieff (1957)		20	propylene glycol
Engen (1960)		48-3,100	benzyl benzoate
Sega *et al.* (1967)	d	3.44	lager beer
Koelega & Köster (1974)		80-294	benzyl benzoate
Ansari & Johnson (1975)	d	270	liquid paraffin
Ansari (1976)	d	1.2	paraffin oil
De Wet (1978)	d	0.18	wine residue medium
Mair *et al.* (1978)		< 4.9	dibutyl phthalate
Ward *et al.* (1983)	d	160,000	liquid paraffin
Garrett-Laster *et al.* (1984)	d	0.013	mineral oil
Garrett-Laster *et al.* (1984)	r	0.13	mineral oil
Ophir *et al.* (1986)	d	36-56	diethyl phthalate
Quinn *et al.* (1987)	d	500	liquid paraffin
Ophir *et al.* (1988)	d	3.3-10.5	diethyl phthalate
Wysocki *et al.* (1989)	d	0.55-4.3	light mineral oil
Pierce *et al.* (1993)	r	14-27	light mineral oil
Jacob *et al.* (2006)	d	0.51	silicone oil
Wise *et al.* (2008)	d	0.78 - 1.9	light mineral oil
Olsson & Laska (2010)	d	41 - 123	diethyl phthalate

6-pentyl-5,6-dihydro-2*H*-pyran-2-one⇒5,6-DIHYDRO-6-PENTYL-2*H*-PYRAN-2-ONE

2-PENTYLFURAN *[3777-69-3]*

Evans *et al.* (1971); Belitz & Grosch (1992)	d	2	cottonseed oil
García-González *et al.* (2008)	d	0.1	deodorised edible oil

PENTYLPYRAZINE *[6303-75-9]*

Koehler *et al.* (1971)		9	paraffin oil

2-PENTYLPYRIDINE *[2294-76-0]*

Schieberle (1993,1996)	d	0.005	sunflower oil

PENTYL SALICYLATE (amyl salicylate) *[2050-08-0]*

Rosenthal (1927)		88	diethyl phthalate

1-PENTYNE *[627-19-0]*

Champagne & Nawar (1969)		180	mineral oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Evans *et al.* (1971)	d	0.7	cottonseed oil

pemenone⇒*cis*-4-(4'-*tert*-BUTYLCYCLOHEXYL)-4-METHYL-2-PENTANONE
pepper pyrazine⇒2-ISOBUTYL-3-METHOXYPYRAZINE
R-(-)-α-phellandrene⇒(*R*)-(-)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE
S-(+)-α-phellandrene⇒(*S*)-(+)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE
phenethyl acetate⇒2-PHENYLETHYL ACETATE
phenethyl alcohol⇒2-PHENYLETHANOL
phenethyl octanoate⇒2-PHENYLETHYL OCTANOATE

PHENOL *[108-95-2]*

Proetz (1924)		19.21	liq. petrolatum
Adams & Crabtree (1961)		96	liquid paraffin
Fordyce (1961)		38 - 96	liquid paraffin
Joyner (1963)		40	mineral oil
Matsushita *et al.* (1967)		15	liquid paraffin
Yoshida (1972)	d	9.1	nujiol (a mineral oil)
Yoshida (1972)	r	1,000	nujiol (a mineral oil)
Funasaka *et al.* (1975)		0.026	muscle of silver carp
Swan & Burtles (1978); Swan *et al.* (1981)		40	20 % ethanol-water
Jounela-Eriksson & Lehtonen (1981)		7.1	10 vol. % grain spirit
Yoshida (1984);Takagi (1989)	d	100	nujol or propylene glycol
Yoshida (1984);Takagi (1989)	r	1,000	nujol or propylene glycol
Vichi *et al.* (2008)	d	0.1	refined sunflower oil

PHENOXYBENZENE (diphenyl ether) *[101-84-8]*

Rosenthal (1927)		885	diethyl phthalate
Gross-Isseroff & Lancet (1988)	d	0.000 1-3,200	diethyl phthalate

phenylacetaldehyde⇒2-PHENYLETHANAL

PHENYLACETIC ACID *[103-82-2]*

Kubota *et al.* (1989)		0.1 - 0.5	4.5 % acetic acid solution
Ong & Acree (1998)		2	artificial saliva and fruit juice matrix
Kerscher (2000a)		0.186	oil
Ferreira *et al.* (2002)		2.5	10 % hydroalcoholic solution, pH 3.2
Carunchia Whetstine *et al.* (2005)	d	0.017	water with pH 5.5. buffer
Schmitt (2003); Schmitt & Schieberle (2006)	d	0.36	sunflower oil
Averbeck & Schierberle (2009, 2011)	d	0.150	citrate buffer (pH 3.8)

2-PHENYLETHANAL (phenylacetaldehyde) *[122-78-1]*

Wick *et al.* (1967)	d	0.94	beef slurry
Guadagni *et al.* (1972)	d	0.022	vegetable oil
Etiévant *et al.* (1983)	d	1.1	base white sweet wine
Pfnür (1998a)		0.025	oil
Fickert (2001)		0.028	starch
Silva Ferreira *et al.* (2002)	r	0.025	young, dry white wine
Culleré *et al.* (2004, 2007); Campo *et al.* (2006); Gómez-Míguez *et al.* (2007); Escudero *et al.* (2007)		0.001	10 % water/ethanol, tartaric acid 5 g/l, pH 3.2
Carunchia Whetstine *et al.* (2005)	d	0.000 8	water with pH 5.5 buffer
Schirack *et al.* (2006)	d	0.154	vegetable oil
Chetschik *et al.* (2010)		0.083	sunflower oil

2-PHENYLETHANETHIOL (2-phenylethyl mercaptan) *[4410-9-5]*

Schieberle (1996,1998)	d	$5 - 10^{-5}$	sunflower oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

1-PHENYLETHANOL (styralyl alcohol) *[98-85-1]*

Tandon *et al.* (2001)	d	1.38	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	10.794	deodorized tomato homogenate

2-PHENYLETHANOL (phenethyl alcohol, phenyl ethyl alcohol, β-phenylethyl alcohol) *[60-12-8]*

Rosenthal (1927)		6	diethyl phthalate
Engen (1960)		48-760	benzyl benzoate
Sega *et al.* (1967)	d	47.9	lager beer
Salo (1970a,1970b); Salo *et al.* (1972)	d	7.5	9.4 % (w/w) grain spirit
Yoshida (1972)	d	1.9	nujiol (a mineral oil)
Yoshida (1972)	r	100	nujiol (a mineral oil)
Perry *et al.* (1980)	d	3.1	propylene glycol
Thumfart *et al.* (1980)		250 - 1,000	diethyl phthalate
Williams & Ismail (1981)		1.4	sugar-acid base
Lindeman *et al.* (1982)	d	0.9	10 % ethanol
Ghorbanian *et al.* (1983)	d	< 32	propylene glycol
Doty *et al.* (1984)	d	0.3 - 100,000	propylene glycol
Yoshida (1984);Takagi (1989)	d	10	nujol or propylene glycol
Yoshida (1984);Takagi (1989)	r	100	nujol or propylene glycol
Doty *et al.* (1986)	d	< 0.03 - 23	propylene glycol
Deems & Doty (1987)	d	1-1,000	mineral oil
Doty *et al.* (1987)	d	1.2	propylene glycol
Doty *et al.* (1988a)	d	35 - 54	propylene glycol
Doty *et al.* (1988b)	d	10 - 100	propylene glycol
Roger *et al.* (1988)	d	7.6	curd with 20 % milk fat
Lewitt *et al.* (1989)	d	0.06	glycerine
Doty (1991)	d	0.032-42	mineral oil
Doty *et al.* (1991)		0.01-0.1	propylene glycol?
Hummel *et al.* (1991)	d	31-123	propylene glycol
Guth & Grosch (1993)		0.125	cellulose
Smith *et al.* (1993)	d	0.1 - 1	light mineral oil
Pierce *et al.* (1996)		0.1-1	propylene glycol
Pierce *et al.* (1996)		0.01-0.32	light mineral oil
Moberg & Doty (1997)	d	0.1 - 0.63	propylene glycol
Betchen & Doty (1998)		3.2-32	light mineral oil
Ferreira *et al.* (1998)		14	12 % v/v ethanol, 5 g/l tartaric acid, pH 3.5
Kubickova & Grosch (1998);			
Reiners & Grosch (1998)	d	0.211	(refined) sunflower oil
Ong & Acree (1998)		2	artificial saliva and fruit juice matrix
Caccappolo *et al.* (2000)		488-977	glycerol
Lee *et al.* (2000)	d	2-8	3 year old grain whisky: 23 % ethanol
Lee *et al.* (2000)	r	80-246	3 year old grain whisky: 23 % ethanol
Kareken *et al.* (2001)		3.3	light mineral oil
Moyano *et al.* (2002)		210	14 % (v/v) ethanol; tartaric acid to pH 3.5
Postolache *et al.* (2002)	d	0.015 - 0.59	light mineral oil
Dalton *et al.* (2003)		0.003 - 0.01	glycerol
Tsukatani *et al.* (2003)	d	2.0	liquid paraffin
Tsukatani *et al.* (2003)	d	20.4	mineral oil
Tsukatani *et al.* (2003)	d	644	propylene glycol
Tsukatani *et al.* (2003)	d	644	dipropylene glycol
Wudarski & Doty (2004)	d	0.245	light mineral oil; glass sniff bottles
Wudarski & Doty (2004)	d	0.741	light mineral oil; plastic squeeze bottles
Utsunomiya *et al.* (2004a)	d	29	base sake
Utsunomiya *et al.* (2004a)	r	56 - 57	base sake
Carunchia Whetstine *et al.* (2005)	d	0.176	water with pH 5.5. buffer

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Moreno *et al.* (2005); Chaves *et al.* (2007);			
Moyano *et al.* (2010)		10	14 % ethanolic solution
Djordjevic *et al.* (2008)	d	10 - 100	diethyl phthalate
Poisson & Schieberle (2008)		2.6	water/ethanol 6:4 by vol.
Aceña *et al.* (2011)	d	1.4	7 % (w/v) acetic acid water volume

2-PHENYLETHYL ACETATE (phenethyl acetate) *[103-45-7]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	0.65 - 0.70	9.4 % (w/w) grain spirit
De Wet (1978)	d	1.80	wine residue medium
Roger *et al.* (1988)	d	19.8	curd with 20 % milk fat
Kubickova & Grosch (1998)		0.137	sunflower oil
Carunchia Whetstine *et al.* (2005)	d	0.061	water with pH 5.5. buffer
Moreno *et al.* (2005);Chaves *et al.* (2007);			
Moyano *et al.* (2010)		0.25	14 % ethanolic solution
Schmitt & Schieberle (2006)	d	0.233	sunflower oil
Callejón *et al.* (2008)	d	0.088	7 % acetic acid solution
Poisson & Schieberle (2008)		0.108	water/ethanol 6:4 by vol.
Aceña *et al.* (2011)	d	0.0978	7 % (w/v) acetic acid water volume

phenyl ethyl alcohol⇒2-PHENYLETHANOL
β-phenylethyl alcohol⇒2-PHENYLETHANOL
2-phenylethyl mercaptan⇒2-PHENYLETHANETHIOL
phenylethylmethylethylcarbinol⇒3-METHYL-1-PHENYL-3-PENTANOL

2-PHENYLETHYL PROPANOATE *[122-70-3]*

Roger *et al.* (1988)	d	18	curd with 20 % milk fat

2-PHENYLETHYL OCTANOATE (phenethyl octanoate) *[5457-70-5]*

Zea *et al.* (2001)	500	14 % (v/v) ethanol; tartaric acid to pH3.5
Moreno *et al.* (2005); Chaves *et al.* (2007)	10	14 % ethanolic solution

phenylmethanol⇒BENZYL ALCOHOL

3-PHENYLPROPANOIC ACID (hydrocinnamic acid) *[501-52-0]*

Ong & Acree (1998)	5	artificial saliva and fruit juice matrix

3-PHENYL-2-PROPENAL (cinnamaldehyde) *[104-55-2]*

Rosenthal (1927)	177	diethyl phthalate

3-PHENYL-2-PROPEN-1-OL (cinnamic alcohol, cinnamlyl alcohol) *[104-54-1]*

Rosenthal (1927)	295	diethyl phthalate

(*E*)-3-PHENYL-2-PROPEN-1-YL ACETATE ((*E*)-cinnamyl acetate) *[21040-45-9]*

Pino & Queris (2011)	0.002	11% ethanol solution

3-PHENYL-2-PROPEN-1-YL BUTANOATE (cinnamyl butyrate) *[103-61-7]*

Mair *et al.* (1978)	< 159	dibutyl phthalate

3-PHENYLPROPYL ACETATE *[122-72-5]*

Pino & Queris (2011)	0.1	11% ethanol solution

α-pinene⇒2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
(-)-α-pinene⇒(*S*)-(-)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
(+)-α-pinene⇒(*R*)-(+)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
β-pinene⇒6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE
(-)-β-pinene⇒(1*S*)-(-)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE
(+)-β-pinene⇒(1*R*)-(+)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

piperonal⇒3,4-METHYLENEDIOXYBENZALDEHYDE
prenyl mercaptan⇒3-METHYL-2-BUTENE-1-THIOL

PROPANAL (propionaldehyde) *[123-38-6]*

Meijboom (1964)		3.6	paraffin oil
Salo *et al.* (1972)	d	2.00	9.4 % (w/w) grain spirit
Herrmann & Abd El Salam (1980a)		9.84	sunflower oil
Czerny (1996)		0.086	cellulose
Reiners & Grosch (1998)	d	0.0094	refined sunflower oil

PROPANOIC ACID (propionic acid) *[79-09-4]*

Salo (1970a,1970b); Salo *et al.* (1972)	d	20	9.4 % (w/w) grain spirit
Amoore & Buttery (1978)	d	3.84	safflower oil
Morales *et al.* (2005)	d	0.72	deodorised olive oil

1-PROPANOL (propyl alcohol) *[71-23-8]*

Williams (1953)		250	ethanol solution
Cain (1966)	r	14,190	diethyl phthalate
Salo *et al.* (1972)	d	> 720	9.4 % (w/w) grain spirit
De Wet (1978)	d	25	satur. KHT in 12 % EtOH-H2O
De Wet (1978)	d	306.1	wine residue medium
Richard & Bizzini (1981)		> 15,600	diethyl phthalate
Moyano *et al.* (2002)		314	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005); Chaves *et al.* (2007)		830	14 % ethanolic solution

2-PROPANOL (isopropyl alcohol, isopropanol) *[67-63-0]*

Granzer *et al.* (1986)		50 - 500	coconut oil
Rüter (1992)	d	260	potato crisps
Rüter (1992)	r	720	potato crisps
Rüter (1992)	d	270	coffee powder
Rüter (1992)	r	710	coffee powder
Rüter (1992)	d	75	'Gummibären'
Rüter (1992)	r	370	'Gummibären'
Rüter (1992)	d	200	milk chocolate
Rüter (1992)	r	620	milk chocolate

PROPANONE (acetone) *[67-64-1]*

Forrai *et al.* (1970)	*bimodal distribution*	r	312-1,768	silicone oil
Granzer *et al.* (1986)			100 - 1,000	coconut oil
Granzer *et al.* (1986)			50 - 1,000	liquid cheese
Rüter (1992)		d	190	potato crisps
Rüter (1992)		r	330	potato crisps
Rüter (1992)		d	43	coffee powder
Rüter (1992)		r	960	coffee powder
Rüter (1992)		d	75	'Gummibären'
Rüter (1992)		r	420	'Gummibären'
Rüter (1992)		d	16	milk chocolate
Rüter (1992)		r	57	milk chocolate
Tandon *et al.* (2001)		d	2.549	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)		d	17.433	deodorized tomato homogenate

2-propanoyl-1-pyrroline⇒2-PROPIONYL-1-PYRROLINE

2-PROPENAL (acrolein) *[107-02-8]*

Balavoine (1948c)		40	glycerol

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

PROPENOIC ACID (acrylic acid) *[79-10-7]*

Piringer & Granzer (1984); Granzer *et al.* (1986)		0.5 - 10	sunflower oil
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.05 - 2	10 vol. % ethanol
Piringer & Granzer (1984); Granzer *et al.* (1986)		0.05 - 1	3 vol. % acetic acid

2-PROPENYLDITHIO-2-PROPENE (diallyl disulphide) *[2179-57-9]*

Yoshida (1972)	d	10	nujiol (a mineral oil)
Yoshida (1972)	r	1,000	nujiol (a mineral oil)

2-PROPENYL HEXANOATE (allyl caproate) *[123-68-2]*

Moncrieff (1957)		0.2	propylene glycol

2-PROPENYLTHIO-2-PROPENE (diallyl sulphide) *[592-88-1]*

Yoshida (1984);Takagi (1989)	d	100	nujol or diethyl phthalate
Yoshida (1984);Takagi (1989)	r	100	nujol or diethyl phthalate

propionaldehyde⇒PROPANAL
propionic acid⇒PROPANOIC ACID

2-PROPIONYL-1-PYRROLINE (2-propanoyl-1-pyrroline) *[133447-37-7]*

Burdack-Freitag (2007)		0.000 1	sunflower oil

PROPYL ACETATE *[109-60-4]*

De Wet (1978)	d	4.74	wine residue medium
Moreno *et al.* (2005)		65	14 % ethanolic solution
Callejón *et al.* (2008)	d	0.6.708	7 % acetic acid solution

4'-PROPYLACETOPHENONE *[2932-65-2]*

Cerveny *et al.* (1993)		151.5	paraffin oil

propyl alcohol⇒1-PROPANOL

PROPYL BUTANOATE (propyl butyrate) *[105-66-8]*

Zea *et al.* (2001); Moyano *et al.* (2002)		0.1	14 % (v/v) ethanol; tartaric acid to pH 3.5
Morales *et al.* (2005)	d	0.15	deodorised olive oil

propyl butyrate⇒PROPYL BUTANOATE

2-PROPYLFURAN *[4229-91-8]*

Evans *et al.* (1971)	d	6	cottonseed oil

4-propylguaiacol⇒2-METHOXY-4-PROPYLPHENOL
4-propyl-2-methoxyphenol⇒2-METHOXY-4-PROPYLPHENOL

1-(4-PROPYLPHENYL)ETHANOL

Cerveny *et al.* (1993)		52.5	paraffin oil

1-(4-PROPYLPHENYL)ETHYL ACETATE

Cerveny *et al.* (1993)		165.8	paraffin oil

1-(4-PROPYLPHENYL)ETHYL BUTANOATE

Cerveny *et al.* (1993)		263.6	paraffin oil

1-(4-PROPYLPHENYL)ETHYL PROPANOATE

Cerveny *et al.* (1993)		282.2	paraffin oil

prunolide⇒4-HYDROXYNONANOIC ACID LACTONE

trans-pyran linalool oxide⇒*trans*-6-ETHENYLTETRAHYDRO-2,2,6-TRIMETHYL-2*H*-PYRAN-3-OL
cis-pyran linalool oxide⇒*cis*-6-ETHENYLTETRAHYDRO-2,2,6-TRIMETHYL-2*H*-PYRAN-3-OL

PYRIDINE (azine) *[110-86-1]*

Koelega & Köster (1974)		25 - 34	benzyl benzoate
Hyde *et al.* (1981)		3	mineral oil
Pollack *et al.* (1982)		1.9 - 3.8	light mineral oil
Amoore & Ollman (1983)		3.8	mineral oil
Wysocki & Beauchamp (1984)	d	0.92-3.7	mineral oil
Satya-Murti & Crisostomo (1988)		12	mineral oil
Koss *et al.* (1987,1988)	d	7.3	mineral oil
Dorries *et al.* (1989)		1 - 2	mineral oil
Wysocki *et al.* (1989)	d	0.92-7.4	light mineral oil
Caccappolo *et al.* (2000)		1.9-3.8	formulary grade mineral oil

R

cis-rose oxide⇒ *cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
(2*S*,4*R*)-*cis*-rose oxide⇒(2*S*,4*R*)-*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
(2*R*,4*R*)-*trans*-rose oxide⇒(2*R*,4*R*)-*trans*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
(2*R*,4*S*)-*cis*-rose oxide⇒(2*R*,4*S*)-*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
(2*S*,4*S*)-*trans*-rose oxide⇒(2*S*,4*S*)-*trans*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN

ROTUNDONE *[18374-76-0]*

Wood *et al.* (2008, 2010)	d	0.000 016	red wine

S

(-)-sabinene⇒(1*S*)-(-)-1-ISOPROPYL-4-METHYLENEBICYCLO[3.1.0]HEXANE
(+)-sabinene⇒(1*R*)-(+)-1-ISOPROPYL-4-METHYLENEBICYCLO[3.1.0]HEXANE
safrole⇒4-ALLYL-1,2-(METHYLENEDIOXY)BENZENE
salicylaldehyde⇒2-HYDROXYBENZALDEHYDE
skatole⇒3-METHYLINDOLE
sorbic aldehyde⇒(*E*,*E*)-2,4-HEXADIENAL
sotolon⇒4,5-DIMETHYL-3-HYDROXY-2(5*H*)-FURANONE
stearic acid⇒OCTADECANOIC ACID
styralyl alcohol⇒1-PHENYLETHANOL
styrene⇒ETHENYLBENZENE
sugarone⇒2-ETHYL-4-HYDROXY-5-METHYL-3(2*H*)-FURANONE
sulfurol⇒5-(2-HYDROXYETHYL)-4-METHYLTHIAZOLE

SULPHUR DIOXIDE *[7446-09-5]*

Brenner *et al.* (1955)	r	50 - 100	club soda
Amerine & Roessler (1976)		5 - 100	wine
Sudraud & Sauvet (1985)	r	2	white wine

syringol⇒2,6-DIMETHOXYPHENOL

T

γ-terpinene⇒4-ISOPROPYL-1-METHYL-1,4-CYCLOHEXADIENE
1-terpinen-4-ol⇒1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL
α-terpineol⇒4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE
(4*R*)-(+)-α-terpineol⇒(*R*)-(+)-4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE
β-terpineol⇒4-ISOPROPENYL-1-METHYL-1-CYCLOHEXANOL
α-terpinol⇒4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

2,3,4,6-TETRACHLOROANISOLE *[938-22-7]*

Nijssen (1988)	d	0.000 014	20 % ethanol

2,3,4,6-TETRACHLOROPHENOL *[58-90-2]*

Coulter *et al.* (2008)	d	> 0.000 896	neutral, dry white wine

γ-tetradecalactone⇒4-HYDROXYTETRADECANOIC ACID LACTONE
δ-tetradecalactone⇒5-HYDROXYTETRADECANOIC ACID LACTONE

TETRADECANE *[629-59-4]*

Champagne & Nawar (1969)		> 13,000	mineral oil

TETRADECANOIC ACID (myristic acid) *[544-63-8]*

Salo *et al.* (1972)	d	> 12	9.4 % (w/w) grain spirit

1-TETRADECANOL *[112-72-1]*

Salo *et al.* (1972)	d	> 5.0	9.4 % (w/w) grain spirit

4-tetradecanolide⇒4-HYDROXYTETRADECANOIC ACID LACTONE
5-tetradecanolide⇒5-HYDROXYTETRADECANOIC ACID LACTONE

1-TETRADECENE *[1120-36-1]*

Champagne & Nawar (1969)		1,680	mineral oil

1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL (2-methylisoborneol, MIB) *[2371-42-8]*

Persson & York (1978)	r	0.000 070	cooked pike
Persson (1980)	r	0.000 085	cooked pike
Persson (1980)	r	0.000 095	cooked bream
Persson (1980)	r	0.000 075	cooked pikeperch
Persson (1980)	r	0.000 55	cooked rainbow trout
Ito *et al.* (1988)	d	7-11 10^{-6}	water with 0.6 mg/l chlorine
Ito *et al.* (1988)	d	11-14 10^{-6}	water with 1.5 mg/l chlorine
Ito *et al.* (1988)	r	27-44 10^{-6}	water with 0.6 mg/l chlorine
Ito *et al.* (1988)	r	45-48 10^{-6}	water with 1.5 mg/l chlorine
Tomita *et al.* (1988)		0.000 015	water with citral or with leaf alcohol (both 0.01 mg/l)
Amon *et al.* (1989)	d	0.000 03	non-aromatic dry white wine
Siegmund & Pöllinger-Zierler (2006)	d	0.000 0017	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.000 033	apple juice

TETRAMETHYLPYRAZINE *[1124-11-4]*

Koehler *et al.* (1971)		38	paraffin oil

4-thiapentanal⇒3-(METHYLTHIO)PROPANAL
thibetolide⇒15-HYDROXYPENTADECANOIC ACID LACTONE

THIOPHENE *[110-02-1]*

Henkin & Bartter (1966); Henkin & Hoye (1966); Henkin (1967a,1967b,1968)	d	0.000 84	mineral oil
Henkin & Bartter (1966); Henkin & Hoye (1966); Henkin (1967b,1968)	r	8.4 - 84	mineral oil
Henkin *et al.* (1968); Hoye *et al.* (1970)	d	0.0084	mineral oil
Henkin *et al.* (1968); Hoye *et al.* (1970)	r	840	mineral oil
Henkin & Smith (1971); Henkin & Larson (1972)	d	0.084	mineral oil

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Henkin & Smith (1971); Henkin & Larson (1972)	r	84	mineral oil
Henkin *et al.* (1971,1972); Marshall & Henkin (1971)	d	0.0084	mineral oil
Henkin *et al.* (1971,1972); Marshall & Henkin (1971)	r	84	mineral oil
Filla *et al.* (1975); De Michele *et al.* (1976); Campanella *et al.* (1978)	d	0.084 - 0.84	mineral oil
Henkin *et al.* (1976); Delaney *et al.* (1977); Bosma *et al.* (1981)	d	0.084	mineral oil
Henkin *et al.* (1976); Delaney *et al.* (1977); Bosma *et al.* (1981)	r	84	mineral oil
Sullivan & Burch (1976); Burch *et al.* (1978)	d	0.014	mineral oil
Sullivan & Burch (1976); Burch *et al.* (1978)	r	0.41	mineral oil
Perry *et al.* (1980)	d	0.000 000 17	mineral oil
Garrett-Laster *et al.* (1984)	d	0.0084	mineral oil
Garrett-Laster *et al.* (1984)	r	0.67	mineral oil

thymol⇒2-ISOPROPYL-5-METHYLPHENOL
tiglaldehyde⇒(*E*)-2-METHYL-2-BUTENAL
tiglic aldehyde⇒(*E*)-2-METHYL-2-BUTENAL

TOLUENE (methylbenzene) *[108-88-3]*

Granzer *et al.* (1986)		100 - 500	coconut oil
Granzer *et al.* (1986)		50 - 300	biscuits
Rüter (1992)	d	54	potato crisps
Rüter (1992)	r	150	potato crisps
Rüter (1992)	d	67	coffee powder
Rüter (1992)	r	230	coffee powder
Rüter (1992)	d	10	'Gummibären'
Rüter (1992)	r	38	'Gummibären'
Rüter (1992)	d	9	milk chocolate
Rüter (1992)	r	22	milk chocolate
Schirack *et al.* (2006)	d	94.66	vegetable oil
García-González *et al.* (2008)	d	0.33	deodorised edible oil

tonalide⇒7-ACETYL-1,1,3,4,4,6-HEXAMETHYL-1,2,3,4-TETRAHYDRONAPHTHALENE

2,4,6-TRIBROMOANISOLE *[607-99-8]*

Whitfield *et al.* (1997)	d	0.000 002	sultanas
Chatonnet *et al.* (2004)	d	0.000 0034	12 % ethanol, tartaric acid 5 g/l, pH 3.65
Chatonnet *et al.* (2004)	d	0.000 0079	standard red wine

TRIBROMOMETHANE (bromoform) *[75-25-2]*

Matsushita *et al.* (1967)		530	liquid paraffin

2,4,6-TRICHLOROANISOLE *[87-40-1]*

Nijssen (1988)	d	0.000 0056	20 % ethanol
Nijssen (1988)	d	0.000 036	40 % ethanol
Nijssen (1988)	d	0.007	edible oil
Amon *et al.* (1989)	d	0.000 004	non-aromatic dry white wine
Cantagrel & Vidal (1990)	d	3-15 10^{-4}	cognac
Spadone *et al.* (1990)		0.000 008	coffee
Utsunomiya *et al.* (2004a)	d	73 - 75 10^{-5}	base sake
Utsunomiya *et al.* (2004a)	r	17 -18 10^{-4}	base sake

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

2,4,6-TRICHLOROPHENOL *[88-06-2]*

Coulter *et al.* (2008)	d	> 0.000 896	neutral, dry white wine

TRICYCLO[3.3.1.1(3.7)]DECANE (adamantane) *[281-23-2]*

Pelosi & Pisanelli (1981)	d	2,100	mineral oil

TRIDECANOIC ACID *[638-53-9]*

Salo *et al.* (1972)	d	> 0.5	9.4 % (w/w) grain spirit

(*E,Z,Z*)-2,4,7-TRIDECATRIENAL *[13552-96-0]*

Lin *et al.* (2001)		0.18	oil

TRIIODOMETHANE (iodoform) *[75-47-8]*

Proetz (1924)		2.5	liq. petrolatum

TRIMETHYLAMINE *[75-50-3]*

Von Gunten *et al.* (1976)	d	1	milk
Griffiths *et al.* (1979)	r	0.8	stirred eggs

1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL (fenchyl alcohol) *[1632-73-1]*

Siegmund & Pöllinger-Zierler (2006)	d	0.0021	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.0032	apple juice

endo-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL (borneol) *[507-70-0]*

Rosenthal (1927)		18	diethyl phthalate

(1*S*)-(-)-*endo*-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL ((l)-borneol) *[464-45-9]*

Pelosi & Pisanelli (1981)	d	91	mineral oil
Siegmund & Pöllinger-Zierler (2006)	d	0.048	apple juice
Siegmund & Pöllinger-Zierler (2006)	r	0.067	apple juice

exo-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL (isoborneol) *[124-76-5]*

Rosenthal (1927)		18	diethyl phthalate
Ong & Acree (1998)		0.01	artificial saliva and fruit juice matrix

1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE (camphor) *[76-22-2]*

Rosenthal (1927)		88	diethyl phthalate
Matsushita *et al.* (1967)		120	liquid paraffin
Yoshida (1972)	d	640	nujiol (a mineral oil)
Yoshida (1972)	r	10,000	nujiol (a mineral oil)
Kleinschmidt (1983)	r	100	paraffin oil
Ophir *et al.* (1986)	d	1,200-6,500	diethyl phthalate
Yoshida (1984);Takagi (1989)	d	1,000	nujol
Yoshida (1984);Takagi (1989)	r	1,000	nujol

2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE (α-pinene) *[80-56-8]*

Pelosi & Pisanelli (1981)	d	200	mineral oil
Plotto *et al.* (2004)		1.65 - 2.12	deodorized orange juice matrix
Söllner & Schieberle (2009)		0.274	odorless sunflower oil

(*S*)-(-)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE ((-)-α-pinene) *[7785-26-4]*

Jagella & Grosch (1999a)	d	3.4	starch
Laska & Teubner (1999)	d	0.32-1.6	diethyl phthalate

(*R*)-(+)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE ((+)-α-pinene) *[7785-70-8]*

Jagella & Grosch (1999a)	d	2.1	starch

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Laska & Teubner (1999)	d	0.064-0.32	diethyl phthalate

endo-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTYL ACETATE (bornyl acetate) *[76-49-3]*

Rosenthal (1927)		885	diethyl phthalate

1-(2,6,6-TRIMETHYL-1,3-CYCLOHEXADIEN-1-YL)-(*E*)-2-BUTEN-1-ONE ((*E*)-damascenone, β-damascenone) *[23726-93-4]*

Williams & Ismail (1981)		0.30	sugar-acid base
Masuda & Nishimura (1980)		0.01	40 % ethanol
Etiévant *et al.* (1983)		1.6	base white sweet wine
Perry (1989)		0.000 04	23 % v. ethanol
Guth & Grosch (1993)		0.000 2	cellulose
Mayer (1996, 2000)		0.000 15	cellulose
Ong & Acree (1998)		0.000 01	artificial saliva and fruit juice matrix
Reiners & Grosch (1998)	d	0.011	refined sunflower oil
Stephan & Steinhart (1999)	r	0.0017	refined vegetable oil
Plotto *et al.* (2006)	d	0.000 282	deodorized orange juice (perceivers)
Plotto *et al.* (2006)	d	0.195	deodorized orange juice (nonperceivers)
Plotto *et al.* (2006)	r	0.000 638	deodorized orange juice (perceivers)
Plotto *et al.* (2006)	r	0.153	deodorized orange juice (nonperceivers)
Pineau *et al.* (2007); Barbe *et al.* (2008)	d	0.000 05	12 % ethanol/water, 4 g/l tartaric aicd, pH 3.5
Pineau *et al.* (2007); Barbe *et al.* (2008)	d	0.000 14	model white wine
Pineau *et al.* (2007); Barbe *et al.* (2008)	d	0.0021	model red wine 1
Pineau *et al.* (2007); Barbe *et al.* (2008)	d	0.000 85	model red wine 2
Pineau *et al.* (2007); Barbe *et al.* (2008)	d	0.007	red wine
Poisson & Schieberle (2008)		0.000 1	water/ethanol 6:4 by vol.
Saison *et al.* (2009)	d	0.203	carbonated beer served at 7-8 °C
Aceña *et al.* (2011)	d	0.000 19	7 % (w/v) acetic acid water volume

2,6,6-TRIMETHYL-2-CYCLOHEXENE-1,4-DIONE (4-oxoisophorone) *[1125-21-9]*

Rogerson *et al.* (2001)	r	0.025	12 % ethanol, tartaric acid 5g/l, pH 3.5
Rogerson *et al.* (2001)	r	0.025	20 % ethanol, tartaric acid 5g/l, pH 3.5

2,4,4-TRIMETHYL-2-CYCLOHEXEN-1-ONE *[13395-71-6]*

Hochstetler & Schmitt (1976)	d	300 – 1,000	diethyl phthalate

3,5,5-TRIMETHYL-2-CYCLOHEXEN-1-ONE (isophorone) *[78-59-1]*

Stephan & Steinhart (1999)	r	0.068	refined vegetable oil

4,4,6-TRIMETHYL-2-CYCLOHEXEN-1-ONE *[13395-73-8]*

Hochstetler & Schmitt (1976)	d	30 – 100	diethyl phthalate

4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE (α-ionone) *[127-41-3]*

Moncrieff (1957)		500	propylene gycol
Plotto *et al.* (2006)	d	0.0433	deodorized orange juice
Plotto *et al.* (2006)	r	0.0956	deodorized orange juice

4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE (β-ionone) *[14901-07-6]*

Etiévant *et al.* (1983)	d	0.0045	base white sweet wine
Tandon *et al.* (2001)	d	0.197	ethanol 100 ppm/methanol 500 ppm
Tandon *et al.* (2001)	d	0.431	deodorized tomato homogenate
Moyano *et al.* (2002)		0.005	14 % (v/v) ethanol; tartaric acid to pH 3.5
Zimmermann (2001)	r	0.000 87	starch
Plotto *et al.* (2006)	d	0.00267	deodorized orange juice (perceivers)
Plotto *et al.* (2006)	d	2.63	deodorized orange juice (nonperceivers)
Plotto *et al.* (2006)	r	0.0165	deodorized orange juice (perceivers)

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Plotto *et al.* (2006)	r	2.25	deodorized orange juice (nonperceivers)

(1*R*,2'*RS*,6*S*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-BUTANOL

Yamamoto *et al.* (2004)	0.4	500 ppm ethanol in water

(1*R*,2'*RS*,6*S*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-BUTANOL

Yamamoto *et al.* (2004)	1	500 ppm ethanol in water

(1*R*,6*S*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-1-BUTANONE

Yamamoto *et al.* (2004)	0.01	500 ppm ethanol in water

(1*S*,6*R*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-1-BUTANONE

Yamamoto *et al.* (2004)	0.025	500 ppm ethanol in water

(*E*)-(1*R*,6*S*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-2-BUTEN-1-ONE ((1*R*,6*S*)-(*E*)-dihydrodamascone)

Yamamoto *et al.* (2004)	0.005	500 ppm ethanol in water

(*E*)-(1*S*,6*R*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-2-BUTEN-1-ONE ((1*S*,6*R*)-(*E*)-dihydrodamascone)

Yamamoto *et al.* (2004)	0.000 75	500 ppm ethanol in water

(*E*)-(1*R*,6*S*)-4-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-BUTEN-2-ONE

Yamamoto *et al.* (2004)	0.0025	500 ppm ethanol in water

(*E*)-(1*S*,6*R*)-4-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-BUTEN-2-ONE

Yamamoto *et al.* (2004)	0.01	500 ppm ethanol in water

(1*R*,6*S*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-HEXANOL

Yamamoto (1999)	0.004	500 ppm ethanol in water?

(1*S*,6*R*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-HEXANOL

Yamamoto (1999)	0.13	500 ppm ethanol in water?

(1*R*,6*S*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-5-HEXEN-3-OL

Yamamoto (1999)	0.01	500 ppm ethanol in water?

(1*S*,6*R*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-5-HEXEN-3-OL

Yamamoto (1999)	0.3	500 ppm ethanol in water?

(1*R*,6*S*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-PENTANOL

Yamamoto (1999)	0.056	500 ppm ethanol in water?

(1*S*,6*R*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-PENTANOL

Yamamoto (1999)	0.89	500 ppm ethanol in water?

(1*R*,3'*S*,6*S*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-PENTANOL

Yamamoto (1999)	0.033	500 ppm ethanol in water?

(1*S*,3'*S*,6*R*)-1-(2,2,6-TRIMETHYLCYCLOHEXYL)-3-PENTANOL

Yamamoto (1999)	0.6	500 ppm ethanol in water?

3,7,11-TRIMETHYL-2,6,10-DODECATRIEN-1-OL (farnesol) *[4602-84-0]*

Harder (1975)	10	diethyl phthalate
Zea *et al.* (2001); Moyano *et al.* (2002)	1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
Chaves *et al.* (2007)	5	14 % v/v ethanol/water

(*E*)-3,7,11-TRIMETHYL-1,6,10-DODECATRIEN-3-OL (*E*-nerolidol) *[40716-66-3]*

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Zea *et al.* (2001); Moyano *et al.* (2002)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5

(*Z*)-3,7,11-TRIMETHYL-1,6,10-DODECATRIEN-3-OL (*Z*-nerolidol) *[3790-78-1]*

Zea *et al.* (2001)		1.0	14 % (v/v) ethanol; tartaric acid to pH 3.5
Moreno *et al.* (2005)		64	14 % ethanolic solution

1,3,3-TRIMETHYL-2-OXABICYCLO[2.2.2]OCTANE (1,8-cineole, eucalyptol) *[470-82-6]*

Rosenthal (1927)		0.1	diethyl phthalate
Rosen *et al.* (1979)	d	0.015	mineral oil
Latrasse *et al.* (1982)		0.21	liqueur de cassis; low odour intensity
Latrasse *et al.* (1982)		0.38	liqueur de cassis; high odour intensity
Knupfer & Spiegel (1986)	d	1	propylene glycol
Marin *et al.* (1991)		22 - 47	Concord grape juice and 1 % ethanol
Lawless *et al.* (1995)	d	0.0025	buffered spring water
Jagella & Grosch (1999a)	d	0.084	starch
Herve *et al.* (2003)	d	0.0011	Californian Merlot wine
Herve *et al.* (2003)	r	0.0032	Californian Merlot wine
Farina *et al.* (2005)	d	0.0013	Tannat wine
Ferreira *et al.* (2006)		0.55	ethanol 10% v/v, tartaric acid 2 g/l, pH 3.4

(*E*)-1-(2,3,6-TRIMETHYLPHENYL)-1,3-BUTADIENE

Janusz *et al.* (2003)		0.000 04	neutral dry white wine
Janusz *et al.* (2003)		0.000 43	10% aqueous ethanol

TRIMETHYLPYRAZINE *[14667-55-1]*

Koehler *et al.* (1971)		27	paraffin oil
Frauendorfer & Schieberle (2006)	d	0.29	sunflower oil

2,3,4-trithiapentane⇒METHYLTRITHIOMETHANE

U

(*E*,*E*)-2,4-UNDECADIENAL *[30361-29-6]*

Keppler (1977)		2	paraffin oil
Karagül-Yüceer *et al.* (2004)	d	0.001	skim milk
Aceña *et al.* (2011)	d	0.000 4356	7 % (w/v) acetic acid water volume

γ-undecalactone⇒4-HYDROXYUNDECANOIC ACID LACTONE
d-γ-undecalactone⇒(*R*)-(+)-4-HYDROXYUNDECANOIC ACID LACTONE
l-γ-undecalactone⇒(*S*)-(-)-4-HYDROXYUNDECANOIC ACID LACTONE
δ-undecalactone⇒5-HYDROXYUNDECANOIC ACID LACTONE

UNDECANAL *[112-44-7]*

Meijboom (1964)		6.8	paraffin oil
Tajima *et al.* (1990)	d	5.0	limonene in water?

UNDECANE *[1120-21-4]*

Champagne & Nawar (1969)		5,750	mineral oil

UNDECANOIC ACID *[112-37-8]*

Salo *et al.* (1972)	d	> 0.5	9.4 % (w/w) grain spirit
Brennand *et al.* (1989)	r	0.10	pH 2.0 aqeous solution

2-UNDECANONE (methyl nonyl ketone) *[112-12-9]*

Matsushita *et al.* (1967)		500	liquid paraffin

OTHER MEDIA detection (d) and recognition (r) odour threshold values in mg/kg

Rothe (1991,1992)		444	cheese 45 % fat dry matter
Kubickova & Grosch (1998)		3.4	sunflower oil
(*E*)-2-UNDECENAL *[53448-07-0]*			
Meijboom (1964)		150.0	paraffin oil
1-UNDECENE *[821-95-4]*			
Champagne & Nawar (1969)		570	mineral oil
10-UNDECENOIC ACID *[112-38-9]*			
Brennand *et al.* (1989)	r	2.3	pH 2.0 aqeous solution
1-UNDECYNE *[2243-98-3]*			
Champagne & Nawar (1969)		110	mineral oil

V

valeraldehyde⇒PENTANAL
valeric acid⇒PENTANOIC ACID
vanillic acid⇒4-HYDROXY-3-METHOXYBENZOIC ACID
vanillin⇒4-HYDROXY-3-METHOXYBENZALDEHYDE
vinylbenzene⇒ ETHENYLBENZENE
2-vinylfuran⇒2-ETHENYLFURAN
p-vinylguaiacol⇒4-ETHENYL-2-METHOXYPHENOL
4-vinylguaiacol⇒4-ETHENYL-2-METHOXYPHENOL
4-vinylphenol⇒4-ETHENYLPHENOL

W

whisky lactone⇒4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE
Z-whisky lactone⇒*cis*-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

X

m-xylene⇒1,3-DIMETHYLBENZENE
2,3-xylenol⇒2,3-DIMETHYLPHENOL

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